

**Application of Legendre Polynomials to Construct the
Approximation Technique for the Solution of Coupled System of
Generalized Heat Conduction Model with Boundary Constrains.**



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This thesis is submitted in partial fulfillment
of the requirements for award of the degree of
Masters of Science in Mathematics

ATTOCK CAMPUS
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Dedicated to My Beloved Parents, Loving Brothers

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My Respected Teachers

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ABSTRACT

This thesis presents a numerical method for solving an integer-order coupled system related to generalized heat conduction with a system of boundary constraints. The model is written as a coupled pair of linear heat-type partial differential equations for two unknown functions, where each equation contains second-order spatial derivatives and coupling terms, along with linear reaction terms and given source functions. Since the problem is coupled and boundary conditions must be satisfied properly, a direct and accurate numerical approach is needed.

In this work, the unknown solutions are approximated using shifted Legendre polynomials on a finite domain. The main idea is to represent the required integration and differentiation operations through operational matrices. In this way, the coupled differential system, together with the boundary conditions, is converted into a system of algebraic equations for the unknown expansion coefficients. After forming this algebraic system, it is solved numerically and the approximate solutions are obtained for different scale levels M .

The complete algorithm is implemented in MATLAB. Several test problems with known exact solutions are solved to check the accuracy and reliability of the proposed method. Comparisons of exact and approximate solutions are given in tables and figures, along with absolute error plots and maximum error graphs. The results show that the method converges rapidly as the scale level increases. For a representative test problem, the absolute error is around 10^{-5} at $M = 5$ and it reduces to about 10^{-15} at $M = 11$. Overall, the proposed shifted Legendre operational matrix technique is simple to implement, computationally efficient, and it provides highly accurate results for smooth coupled heat-type models under standard boundary conditions.

Keywords: coupled heat equation, generalized heat conduction, shifted Legendre polynomials, operational matrix, spectral approximation, numerical solution, MATLAB.

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1 INTRODUCTION

1.1 Background and Motivation

Heat transfer is a basic and very important topic in engineering and applied sciences. It comes in power systems, electronics cooling, manufacturing, thermal protection, and many other areas. In many real systems, heat does not move alone. It is often linked with another temperature field, another layer of material, or another physical effect. Because of this, coupled heat-type models are used in many applications and studies (Kovács & Ván, 2015; Z. Shomali, Kovács, Ván, Kudinov, & Ghazanfarian, 2022).

In the classical (integer-order) setting, the heat equation is commonly derived from Fourier's law. This model is simple and it works well for many practical cases. Still, research shows that Fourier-type models can face limitations in fast thermal processes because the classical model leads to an infinite speed paradox for thermal signal propagation (Z. Shomali et al., 2022; Zhmakin, 2023). Due to this reason, generalized or non-Fourier heat conduction models have also been studied, especially in heat pulse experiments and in situations where lagging or wave-like effects become noticeable (Kovács & Ván, 2015; Ván et al., 2017a).

In this thesis, the focus is on a **classical (integer-order)** coupled system related to generalized heat conduction, with boundary constraints. The goal is to compute accurate numerical approximations using shifted Legendre polynomials and an operational matrix based approach. Research shows that polynomial spectral methods can give high accuracy for smooth solutions, often with fewer unknowns than many grid-based methods (Trefethen, 2000). Also, operational matrix techniques help in converting differential models into algebraic systems, which makes implementation more systematic (Abd-Elhameed, Youssri, & Doha, 2015; Khalil & Khan, 2014).

So the motivation is simple: we want a method that is accurate, stable, and practical to implement in MATLAB for a coupled heat-type system, while keeping the algorithm clear and repeatable (Khalil & Khan, 2014; Mei, Wu, & Lin, 2022).

1.2 Considered Model

The main problem studied in this thesis is a coupled system of two heat-type partial differential equations with two unknown functions $u(x, t)$ and $v(x, t)$. A general form of

the coupled model can be written as:

$$\frac{\partial u}{\partial t} = \alpha_1 \frac{\partial^2 u}{\partial x^2} + \alpha_2 \frac{\partial^2 v}{\partial x^2} + a_0 u + a_1 v + F(x, t), \quad (1.1)$$

$$\frac{\partial v}{\partial t} = \alpha_3 \frac{\partial^2 u}{\partial x^2} + \alpha_4 \frac{\partial^2 v}{\partial x^2} + b_0 u + b_1 v + G(x, t). \quad (1.2)$$

Initial and boundary conditions are given by,

$$u(x, 0) = g_1(x) \quad v(x, 0) = g_2(x)$$

,

$$u(0, t) = f_1(t) \quad v(0, t) = f_3(t)$$

,

$$u(1, t) = f_2(t) \quad v(1, t) = f_4(t)$$

Coupled systems like this are useful because they allow interaction between two fields, and this helps in modelling multi-field or multi-layer heat transfer situations (Kovács & Ván, 2015; Z. Shomali et al., 2022). In generalized heat conduction literature, different extensions (including lagging heat models) are used to describe behaviour that the classical Fourier model does not capture well in some cases (Z. Shomali et al., 2022; Zhmakin, 2023).

1.3 Why shifted Legendre polynomials and operational matrices?

Shifted Legendre polynomials are useful when the computational domain is $[0, 1]$ (or can be scaled to $[0, 1]$). They also keep an orthogonal structure, which helps in getting stable approximations (Mei et al., 2022; Trefethen, 2000). In spectral methods, the idea is to approximate the unknown solution as a truncated series in basis functions, and then compute the coefficients.

Operational matrices are important because they turn integration and differentiation operations into matrix multiplications. So, instead of repeatedly differentiating the polynomial expansions by hand, we use pre-built matrices. Research shows that operational matrix techniques based on shifted Legendre polynomials reduce boundary value and differential systems into algebraic equations in a neat way (Abd-Elhameed et al., 2015; Khalil & Khan, 2014).

1.4 Problem statement

The problem addressed in this thesis can be stated in a direct way.

We consider a classical coupled generalized heat conduction system of the form (1.1)–(1.2), along with appropriate initial and boundary conditions. The task is to develop a numerical scheme that:

- approximates $u(x, t)$ and $v(x, t)$ accurately on the given domain,
- handles the coupling terms in a stable way,
- converts the coupled PDE system into a solvable algebraic system using shifted Legendre operational matrices,
- and produces reliable results with decreasing error as the approximation level is increased (Abd-Elhameed et al., 2015; Trefethen, 2000).

1.5 Objectives of the thesis

The main objectives of this thesis are:

1. To study the classical coupled heat-type system and write it in a form suitable for spectral approximation (Kovács & Ván, 2015; Z. Shomali et al., 2022).
2. To use shifted Legendre polynomials for approximating the unknown functions on the selected domain (Mei et al., 2022).
3. To apply operational matrices for integration/differentiation so that the coupled PDE system is reduced to algebraic equations (Abd-Elhameed et al., 2015; Khalil & Khan, 2014).
4. To implement the method in MATLAB and analyse the accuracy using error measures while increasing the approximation level (Trefethen, 2000).

1.6 Scope and limitations

Scope: This thesis focuses on a classical (integer-order) coupled generalized heat conduction system on a finite domain. The numerical solution is based on shifted Legendre polynomials and operational matrices. The aim is to obtain accurate approximations for $u(x, t)$ and $v(x, t)$ under boundary constraints (Khalil & Khan, 2014; Mei et al., 2022).

Limitations: Some limitations are also natural.

- Spectral polynomial methods work best when the solution is smooth. If the solution has sharp layers or discontinuities, the accuracy may reduce unless special treatment is used (Trefethen, 2000).
- Increasing the approximation level increases the size of the algebraic system, so the computation cost can grow for very large expansions (Abd-Elhameed et al., 2015).
- This work is mainly for a one-dimensional space variable setting and does not target complex geometries directly. For complex geometries, other methods like FEM are often preferred (Trefethen, 2000).

1.7 Comparison and Workflow

Table 1.1: Simple comparison of common numerical approaches for heat-type PDEs.

Method	Basic idea	Common strength	Common issue
Finite Difference (FDM)	Discretize derivatives on grid points	Easy to implement	Often needs fine grid for high accuracy
Finite Element (FEM)	Weak form + piecewise basis (mesh based)	Works well on complex domains	More setup and meshing effort
Spectral / Polynomial	Global polynomial approximation	Very high accuracy for smooth solutions	Less suitable for non-smooth solutions

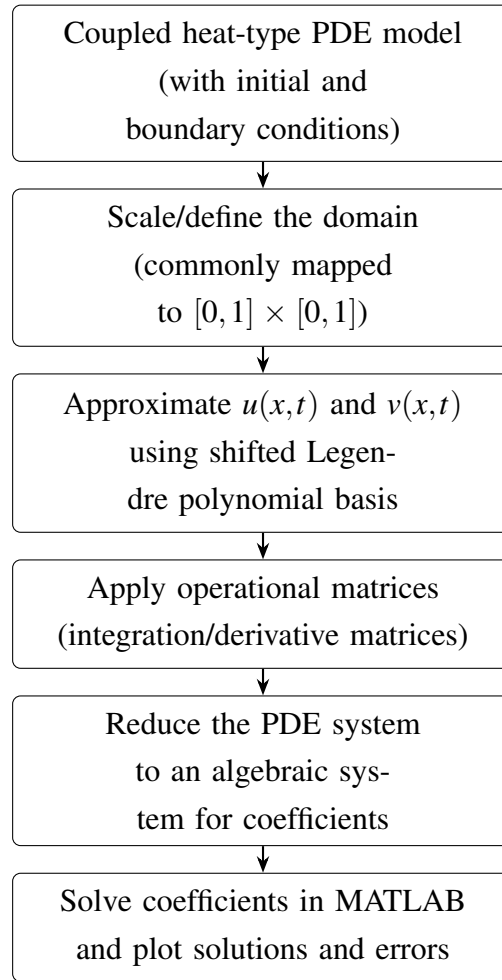


Figure 1.1: Workflow of the proposed shifted Legendre operational-matrix based method.

1.8 Thesis organization

This thesis is organized as follows. Chapter 1 gives the background, motivation, the problem statement, objectives, scope and limitations, and the overall structure. Chapter 2 presents preliminaries and related literature. Chapter 3 covers the mathematical tools needed for the method, including shifted Legendre polynomials and operational matrices. Chapter 4 develops the proposed method for different types of problems and shows how the coupled system is converted into algebraic form. Chapter 5 presents numerical results, comparisons, and error analysis. Finally, Chapter 6 concludes the thesis and suggests future directions.

2 LITERATURE REVIEW

2.1 Background

Heat conduction is one of the most common transport processes studied in engineering and applied mathematics. In many basic problems, Fourier's law and the classical heat equation are used because they are simple and they usually give acceptable results at macroscopic scales (Fourier, 1822). Still, when the material has strong heterogeneity, internal interfaces, porosity, or when heating happens very fast (like pulse heating), the classical Fourier model may not match experimental trends well. Research shows that in such cases, the temperature response may look like a delayed or wave-like propagation rather than a smooth diffusion response (Fehér & Kovács, 2021; M. Shomali et al., 2022; Ván et al., 2017b).

In recent years, many studies focused on so-called non-Fourier or generalized heat conduction models. The motivation is mostly physical: the model should allow finite speed of heat propagation and should capture extra mechanisms that become visible in experiments and in micro/meso-scale problems (Ramos, Kovács, Freitas, & Almeida Júnior, 2023; M. Shomali et al., 2022; Tóth, Kovács, & Ván, 2023). Some works also consider bioheat transfer and thermal therapies, where lagging behaviour and model selection become important because treatment planning depends on temperature prediction (M. Shomali et al., 2022). In this way, the literature connects real physical observations with mathematical modelling and numerical computation.

2.2 Non-Fourier heat conduction models

2.2.1 Cattaneo-type and extended models

A key limitation of Fourier's law is that it leads to infinite propagation speed in the mathematical model. For many engineering cases this is fine, but it becomes questionable in very short time scales. A standard historical correction is the Cattaneo-type modification, where a relaxation time is introduced in the constitutive relation. This idea leads to hyperbolic-type heat equations and it gives finite propagation speed in the simplified setting. Many modern papers discuss these models again, mainly to compare them with experiments and to understand when the extra parameters are really needed (Fehér & Kovács, 2021; Ramos, Kovács, et al., 2023).

The Guyer–Krumhansl (GK) model is another important extension and it can be

derived in the context of phonon hydrodynamics and kinetic theory (Guyer & Krumhansl, 1966). The GK equation includes additional spatial terms and it can be seen as a higher-order extension of relaxation-type models. Research shows that GK-type behaviour can appear not only at low temperatures but also in room-temperature heat pulse experiments when the material structure is complex (Fehér & Kovács, 2021; Ván et al., 2017b). Because of this, the GK model is often treated as a promising candidate for modelling heterogeneous media in a more realistic way (Ramos, Kovács, et al., 2023; Tóth, Kovács, & Ván, 2024).

2.2.2 Dual-phase-lag model and moving heat sources

Another widely used framework is the dual-phase-lag (DPL) model, where two different lag times are used (one for temperature gradient and one for heat flux). This model is often applied in fast heating and laser-type problems. A recent example is the analytical and numerical study of a two-dimensional plate subjected to a moving point heat source, where the heat source follows curved trajectories and the lags change the peak temperature and its location (Chen, Hu, & Xu, 2025). In many such studies, the authors focus on how the lag parameters influence stability, peak shifting, and transient profiles. This helps in understanding the practical meaning of phase lags rather than only treating them as fitting parameters.

2.2.3 Experimental and Mathematical Analysis

Experimental reports are an important part of this area. For example, research shows deviation from the Fourier law in room-temperature heat pulse experiments and explains it using generalized heat conduction ideas (Both et al., 2016). Also, GK-type heat conduction at room temperature has been discussed in detail with experimental observations and theoretical fitting (Ván et al., 2017b). On the mathematical side, analysis work has also grown. For example, well-posedness, stability, and structural properties of the GK equation have been investigated, and the results are supported by numerical demonstrations (Ramos, Kovács, et al., 2023). Another study revisits the GK equation with focus on non-negativity and maximum principle issues, which is important because negative temperature predictions are physically unacceptable in many applications (Ramos, Miranda, Freitas, & Kovács, 2023).

Overall, the literature shows that non-Fourier models are not only theoretical. They connect to real measurements and practical systems. At the same time, they introduce extra parameters and extra mathematical difficulty, so careful numerical treatment becomes necessary (M. Shomali et al., 2022; Tóth et al., 2023, 2024).

2.3 Coupled Heat Conduction systems

In many generalized frameworks, the governing equations naturally become coupled. A common example is when temperature and heat flux (or related auxiliary variables) are treated together. This coupling is one reason why wave-like features and lagging behaviour can appear. Such coupled descriptions are also useful when boundary conditions are specified in terms of flux, or when the material response is better described by additional internal variables (Ramos, Kovács, et al., 2023; Tóth et al., 2023). In computational studies, the coupled form may be handled either by reducing it to a single higher-order PDE or by keeping it as a system and solving it directly. The second approach often gives more control on the physical meaning of each variable, but the algebra and implementation become heavier.

The literature also shows that coupled formulations arise in many application directions such as porous or layered structures, thermal damping in micro-scale systems, and fast transient heating. For example, GK-based modelling has been used in refined thermodynamic theories and mixed formulations, where temperature and flux-type variables are approximated together and boundary conditions are imposed in a weak form (Tóth et al., 2023, 2024). These works motivate numerical methods that can handle coupled systems efficiently with good accuracy.

2.4 Numerical methods

2.4.1 Traditional discretization: FDM, FEM, BEM, and meshless ideas

For the classical heat equation and related parabolic systems, finite difference and finite element methods are still widely used. Their main advantage is that they are flexible for complex geometries and boundary conditions. However, for generalized models (with relaxation or higher-order spatial terms), standard schemes may suffer from stability restrictions and may require careful parameter tuning. That is why recent papers also propose improved discretizations and space–time strategies.

A recent work proposed a space–time generalized finite difference method for transient heat conduction problems, aiming to balance accuracy in space and time and to reduce error mismatch between the two directions (Lei, Wang, Liu, Gu, & Fan, 2020). In another direction, boundary element methods have also been used for sets of parabolic differential equations for heat-conducting problems, mainly because BEM reduces the dimension and can be efficient in some settings (Asghar et al., 2023). In applied engineering computation, practical algorithms are also proposed with emphasis

on implementation and performance. For example, a subdivision collocation algorithm for the heat conduction equation was recently presented and tested with different benchmark cases (Ejaz et al., 2024).

2.4.2 High-order and spectral-type methods

Spectral methods and related polynomial-based methods are popular when the domain is regular and high accuracy is needed. The main idea is to approximate the solution using global basis functions (often orthogonal polynomials). This can give fast convergence for smooth solutions. For time-dependent PDEs, classical spectral approaches often use spectral discretization in space and low-order discretization in time, but research shows that this can create imbalance between spatial and temporal errors. That is why space–time spectral methods are increasingly discussed (Kaur, Lui, Nataj, & Liyanage, 2025).

A recent paper presented an efficient space–time Legendre rational spectral method for parabolic problems on unbounded domains and reported good accuracy with suitable basis choices (Yu, 2021). Another strong line of work uses Legendre spectral collocation in space and time for PDEs, forming a global space–time system (Lui, 2017). Such global systems can become large, so acceleration and solver strategies become important. For this, Anderson acceleration and iterative improvement ideas have been applied to space–time spectral discretizations of nonlinear PDEs and the numerical evidence showed improved convergence in several cases (Nataj & He, 2024b). A more recent work also combined Anderson acceleration with Newton-type iterations for spectral discretizations and studied the numerical performance (Nataj & He, 2024a).

Also, there are works that propose new Galerkin-spectral methods using discrete Legendre polynomials for parabolic differential equations, focusing on efficiency and stability in implementation (Rezazadeh & Darehmira, 2024). In a different application direction, arXiv preprints discuss parabolic PDEs in climate-related modelling and show how numerical parabolic solvers appear in that context too (Płocińczak et al., 2021). Although the application is different, such studies still highlight the importance of stable and accurate solvers for parabolic-type equations.

2.5 Shifted Legendre polynomials, operational matrices, and collocation frameworks

When the physical domain is $[0, 1]$ or can be mapped to $[0, 1]$, shifted Legendre polynomials are convenient because they keep orthogonality on the shifted interval and allow compact matrix forms. The operational matrix idea is especially useful: differentiation

and integration can be represented as matrices acting on coefficient vectors. This converts differential equations into algebraic systems, which are easier to compute once the matrices are formed.

Operational matrix methods have been used in many problem types: ODEs, integral equations, and PDEs. A collocation method based on the differentiation operational matrix for shifted Legendre polynomials was presented recently, and the paper reports good performance on model problems (Abbassi, 2024). Another work developed operational matrices based on two-dimensional shifted Legendre polynomials and applied them to partial integro-differential equations, showing that the 2D basis can reduce complexity when the solution is smooth (Singh & Singh, 2016). Also, collocation methods using almost operational matrices for two-dimensional shifted Legendre polynomials have been investigated for PDE-type settings (Patel, Darbari, & Saha, 2017).

2.5.1 Computational considerations

A practical observation is that global space–time methods can give high accuracy but can produce very large linear systems. So, solver design and stabilization matter. Survey-style work on space–time spectral methods discusses this point clearly and highlights the trade-off between full spectral convergence and system size (Kaur et al., 2025). In the same direction, Anderson acceleration is one of the tools to improve iterative convergence for the nonlinear cases (Nataj & He, 2024a, 2024b). This is relevant when the governing equations are coupled or nonlinear, which is common in generalized heat conduction frameworks.

In a related setting, mixed finite element and hp-refinement strategies have been developed for GK-type models to get robust and accurate solutions for both temperature and flux variables (Tóth et al., 2023, 2024). These studies show that even with strong mathematical models, the numerical method decides whether the model can be used reliably in practice.

2.6 Summary and Gaps

From the discussion above, it is clear that (i) generalized heat conduction models are actively studied due to experimental deviations from Fourier predictions and due to the need for finite-speed heat transport descriptions, (ii) coupled formulations naturally arise and they are useful for modelling temperature and flux together, and (iii) modern numerical methods are moving towards high-order accuracy, space–time treatment, and efficient algebraic solvers (Fehér & Kovács, 2021; Kaur et al., 2025; Ramos, Kovács, et al., 2023; M. Shomali et al., 2022; Ván et al., 2017b).

At the same time, there is still a strong need for methods that are accurate, not too difficult to implement, and suitable for coupled systems in a clean algebraic form. Polynomial operational matrix approaches based on shifted Legendre polynomials look useful here because they can transform the coupled PDE system into a computable matrix system in a direct way (Abbassi, 2024; Patel et al., 2017; Singh & Singh, 2016). This motivates the use of shifted Legendre operational matrices and collocation-type discretization for the integer-order coupled generalized heat conduction models studied in this thesis.

2.7 Comparison and Proposed Method Map

Table 2.1 summarizes a few recent directions in generalized heat conduction and numerical treatment. The main point is not to claim that one method is always best, but to show what is commonly done and where a shifted-Legendre operational-matrix approach fits.

Table 2.1: Recent trends in generalized heat conduction and typical numerical approaches.

Model / focus		Typical setting	Common numerical idea	Example refs.
GK-type Fourier conduction	non-heat	Heat pulse and heterogeneous materials; coupled temperature–flux view-point	FDM stability analysis; mixed FEM and hp-refinement; model analysis	(Fehér & Kovács, 2021; Ramos, Kovács, et al., 2023; Tóth et al., 2023, 2024; Ván et al., 2017b)
DPL heat conduction		Fast transient heating; laser/moving source problems	Green’s functions + series; time integration with parameter studies	(Chen et al., 2025; M. Shomali et al., 2022)
Classical heat equation (benchmark)		Validation and baseline comparisons	Space–time generalized FDM; collocation-type engineering solvers	(Ejaz et al., 2024; Lei et al., 2020)
High-order spectral (space–time)		Regular geometries; smooth solutions; large global systems	Legendre space–time collocation; acceleration for nonlinear systems	(Lui, 2017; Nataj & He, 2024b; Rezazadeh & Darehmiraki, 2024; Yu, 2021)
Shifted Legendre operational matrix		Mapped domains like $[0, 1]$; algebraic conversion of PDEs	Differentiation/integration operational matrices + collocation	(Abbassi, 2024; Patel et al., 2017; Singh & Singh, 2016)

Figure 2.1 gives a simple visual map of how the literature connects models and numerical choices, and where the present thesis method sits.

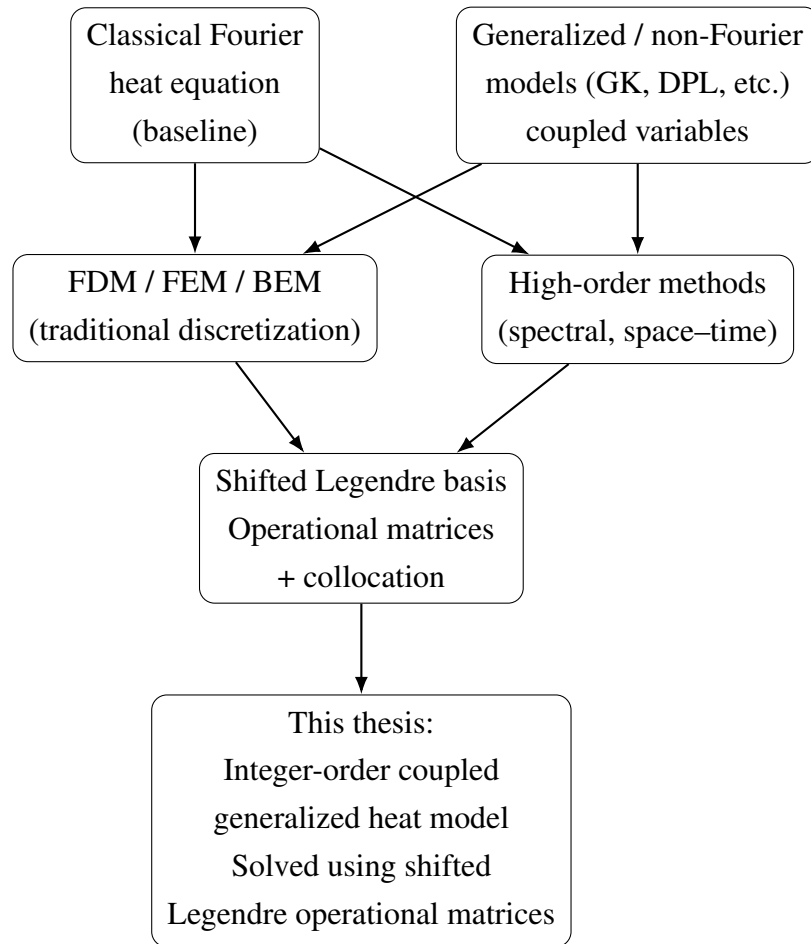


Figure 2.1: A simple method map from models to numerical approaches and the position of this thesis approach.

3 PRELIMINARIES

In this chapter, we discussed the pre-requisites for the development of Proposed scheme in detail. This include the revision of theory of Legendre Polynomials and their Shifted version. We developed the one-dimensional and two dimensional approximation theory based on Legendre Polynomials. We also discussed different properties of Shifted Legendre Polynomials. Next, we developed the required operational matrices of integration and differentiation.

3.1 Legendre Polynomials

The standard Legendre's Polynomials are defined on $[-1,1]$ given by the recurrence relation(Khalil, Hashim, Khan, & Ghaffari, 2021; Tadoummant, Khalil, Echarggaoui, Aljohani, & Mlaiki, 2025).

$$\begin{aligned} \mathbf{L}_{i+1}(z) &= \frac{2i+1}{i+1} z \mathbf{L}_i(z) - \frac{i}{i+1} \mathbf{L}_{i-1}(z), \quad i = 1, 2, \dots, \\ \mathbf{L}_0(z) &= 0, \quad \mathbf{L}_1(z) = z. \end{aligned} \quad (3.1)$$

3.1.1 Shifted Legendre Polynomials

We will develop the proposed theory on the domain $[0, 1] \times [0, 1]$. For this purpose, we use transformation $x = \frac{z+1}{2}$ to shift the standard Legendre Polynomials from $[-1,1]$ to $[0,1]$. The closed form for the Shifted Legendre Polynomials on the domain $[0,1]$ is given by

$$\begin{aligned} \mathbf{P}_i(x) &= \sum_{k=0}^i (-1)^{i+k} \frac{(i+k)!}{(i-k)!(k!)^2} x^k, \quad i = 0, 1, 2, 3, \dots, \\ \mathbf{P}_i(0) &= (-1)^i, \quad \mathbf{P}_i(1) = 1. \end{aligned} \quad (3.2)$$

We calculate the expressions for the first few polynomials to check the validity of (3.2). Putting $i = 0, 1, 2, 3, 4$ in (3.2) we get first five Shifted Legendre Polynomials $\mathbf{P}_0(x)$, $\mathbf{P}_1(x)$, $\mathbf{P}_2(x)$, $\mathbf{P}_3(x)$, and $\mathbf{P}_4(x)$.

For $\mathbf{P}_0(x)$:

$$\begin{aligned} \mathbf{P}_0(x) &= -1^{(0+0)} \frac{(0+0)!}{(0-0)!(0)!} x^0, \\ \mathbf{P}_0(x) &= -(1)^0(1) \end{aligned}$$

$$\boxed{\mathbf{P}_0(x) = 1} \quad (3.3)$$

For $\mathbf{P}_1(x)$:

$$\begin{aligned} \mathbf{P}_1(x) &= -1^{(1+0)} \frac{(1+0)!}{(1-0)!(0)!} x^0 + -1^{(1+1)} \frac{(1+1)!}{(1-1)!(1!)^2} x^1 \\ \mathbf{P}_1(x) &= (-1)^1 + (-1)^2(2x) \end{aligned}$$

$$\boxed{\mathbf{P}_1(x) = 2x - 1} \quad (3.4)$$

For $\mathbf{P}_2(x)$:

$$\begin{aligned} \mathbf{P}_2(x) &= -1^{(2+0)} \frac{(2+0)!}{(2-0)!(0)!} x^0 + -1^{(2+1)} \frac{(2+1)!}{(2-1)!(1!)^2} x^1 + (-1)^{2+2} (2+2)! \frac{(2-2)!}{(2!)^2} x^2 \\ \mathbf{P}_2(x) &= 1 + (-1)(3!)x + (1) \frac{(1)(4!)}{4} x^2 \end{aligned}$$

$$\boxed{\mathbf{P}_2(x) = 1 - 6x + 6x^2} \quad (3.5)$$

For $\mathbf{P}_3(x)$

$$\begin{aligned} \mathbf{P}_3(x) &= (-1)^{3+0} \frac{(3+0)!}{(3-0)!(0!)^2} x^0 + (-1)^{3+1} \frac{(3+1)!}{(3-1)!(1!)^2} x^1 + (-1)^{3+2} \frac{(3+2)!}{(3-2)!(2!)^2} x^2 \\ &\quad + (-1)^{3+3} \frac{(3+3)!}{(3-3)!(3!)^2} x^3 \\ \mathbf{P}_3(x) &= -1 + \frac{(1)(4!)}{2} x + (-1) \frac{(5!)}{4} x^2 + (1) \frac{6!}{36} x^3 \end{aligned}$$

$$\boxed{\mathbf{P}_3(x) = 20x^3 - 30x^2 + 12x - 1} \quad (3.6)$$

For $\mathbf{P}_4(x)$

$$\begin{aligned} \mathbf{P}_4(x) &= (-1)^{4+0} \frac{(4+0)!}{(4-0)!(0!)^2} x^0 + (-1)^{4+1} \frac{(4+1)!}{(4-1)!(1!)^2} x^1 + (-1)^{4-2} \frac{(4+2)!}{(4-2)!(2!)^2} x^2 \\ &\quad + (-1)^{4+3} \frac{(4+3)!}{(4-3)!(3!)^2} x^3 + (-1)^{4+4} \frac{(4+4)!}{(4-4)!(4!)^2} x^4 \\ \mathbf{P}_4(x) &= 1 + \frac{(-1)(5!)}{6} x + \frac{1(6!)}{2(4)} x^2 + (-1) \frac{(7!)}{36} x^3 + (1) \frac{8!}{576} x^4 \end{aligned}$$

$$\boxed{\mathbf{P}_4(x) = 70x^4 - 140x^3 + 90x^2 - 20x + 1} \quad (3.7)$$

These five Shifted Legendre Polynomials are shown graphically in the Figure 3.1.

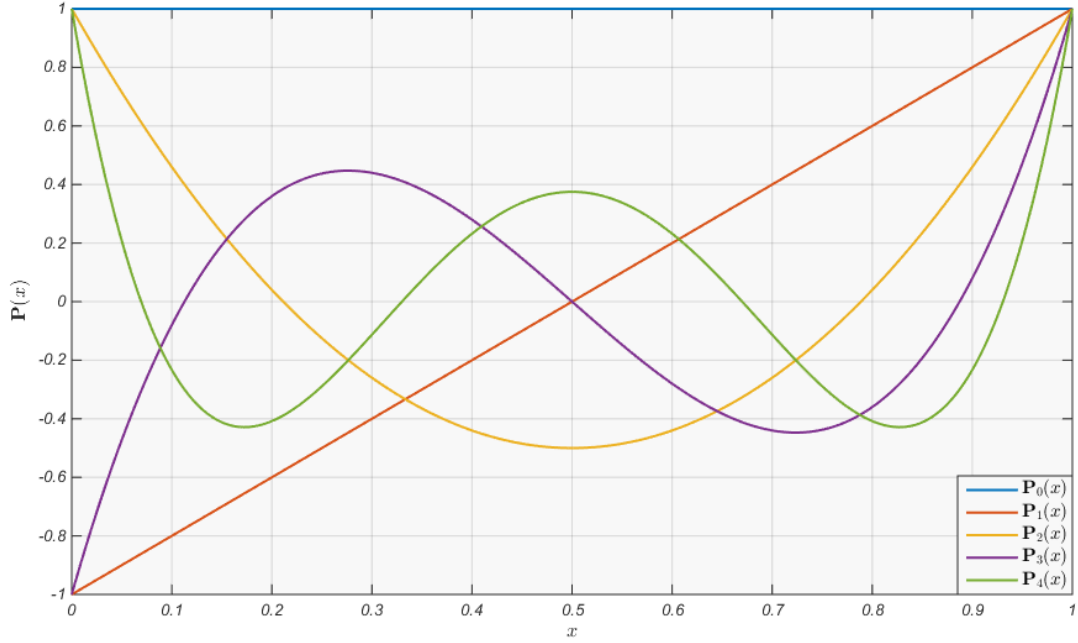


Figure 3.1: First Five Shifted Legendre Polynomials

3.1.2 Orthogonality Condition

Shifted Legendre Polynomials defined in (3.2) are orthogonal (Khalil et al., 2021) and orthogonality condition for these polynomials is given by

$$\int_0^1 \mathbf{P}_i(x) \mathbf{P}_j(x) dx = \begin{cases} \frac{1}{2i+1}, & i = j, \\ 0, & i \neq j. \end{cases} \quad (3.8)$$

To verify the orthogonality condition numerically, we check the orthogonality for the first few polynomials defined in (3.3), (3.4), (3.5), (3.6), and (3.7).

For $\mathbf{P}_0(x)$:

We check the orthogonality of $\mathbf{P}_0(x)$ with other polynomials upto $\mathbf{P}_4(x)$

$$\begin{aligned} \int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_0(x) dx &= \int_0^1 \mathbf{P}_0^2(x) dx \\ \int_0^1 \mathbf{P}_0^2(x) dx &= \int_0^1 1 dx \\ &= x \Big|_0^1 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_0^2(x) dx = 1} \quad (3.9)$$

$$\begin{aligned} \int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_1(x) dx &= \int_0^1 (1)(2x-1) dx \\ &= (x^2 - x)|_0^1 \\ &= 1 - 1 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_1(x) dx = 0} \quad (3.10)$$

$$\begin{aligned} \int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_2(x) dx &= \int_0^1 (1)(6x^2 - 6x + 1) dx \\ &= (2x^3 - 3x^2 + x)|_0^1 \\ &= 2 - 3 + 1 \\ &= 0 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_2(x) dx = 0} \quad (3.11)$$

$$\begin{aligned} \int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_3(x) dx &= \int_0^1 (1)(20x^3 - 30x^2 + 12x - 1) dx \\ &= (5x^4 - 10x^3 + 6x^2 - x)|_0^1 \\ &= 5 - 10 + 6 - 1 \\ &= 0 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_3(x) dx = 0} \quad (3.12)$$

$$\begin{aligned} \int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_4(x) dx &= \int_0^1 (1)(70x^4 - 140x^3 + 90x^2 - 20x + 1) dx \\ &= (14x^5 - 25x^4 + 30x^3 - 10x^2 + x)|_0^1 \\ &= 45 - 45 \\ &= 0 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_0(x) \cdot \mathbf{P}_4(x) dx = 0} \quad (3.13)$$

For $\mathbf{P}_1(x)$:

$$\begin{aligned} \int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_1(x) dx &= \int_0^1 \mathbf{P}_1^2(x) dx \\ \int_0^1 \mathbf{P}_1^2(x) dx &= \int_0^1 (2x-1)^2 dx = \int_0^1 (4x^2 - 4x + 1) dx \\ &= \left(\frac{4x^3}{3} - 2x^2 + x \right) \Big|_0^1 \\ &= \frac{1}{3} \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_1(x) dx = \frac{1}{3}} \quad (3.14)$$

$$\begin{aligned} \int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_2(x) dx &= \int_0^1 (2x-1)(6x^2 - 6x + 1) dx \\ &= \int_0^1 (12x^3 - 18x^2 + 8x - 1) dx \\ &= (3x^4 - 6x^3 + 4x^2 - x) \Big|_0^1 \\ &= 3 - 6 + 4 - 1 = 7 - 7 \\ &= 0 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_2(x) dx = 0} \quad (3.15)$$

$$\begin{aligned} \int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_3(x) dx &= \int_0^1 (2x-1)(20x^3 - 30x^2 + 12x - 1) dx \\ &= \int_0^1 (40x^4 - 80x^3 + 54x^2 - 14x + 1) dx \\ &= (8x^5 - 20x^4 + 18x^3 - 7x^2 + x) \Big|_0^1 \\ &= 8 - 20 + 18 - 7 + 1 = 27 - 27 \\ &= 0 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_3(x) dx = 0} \quad (3.16)$$

$$\begin{aligned}
\int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_4(x) dx &= \int_0^1 (2x-1)(70x^4 - 140x^3 + 90x^2 - 20x + 1) dx \\
&= \int_0^1 (140x^5 - 350x^4 + 320x^3 - 130x^2 + 22x - 1) dx \\
&= \left(\frac{70}{3}x^6 - 70x^5 + 80x^4 - \frac{130}{3}x^3 + 11x^2 - x \right) \Big|_0^1 \\
&= 10 + 11 - 1 - 20 = 21 - 21 \\
&= 0
\end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_1(x) \cdot \mathbf{P}_4(x) dx = 0} \quad (3.17)$$

Similarly we check the orthogonality of the other Shifted Legendre Polynomials,

For $\mathbf{P}_2(x)$:

$$\begin{aligned}
\int_0^1 \mathbf{P}_2^2(x) dx &= \int_0^1 (6x^2 - 6x + 1)^2 dx \\
&= \int_0^1 (36x^4 - 72x^3 + 48x^2 - 12x + 1) dx \\
&= \left(\frac{36x^5}{5} - 18x^4 + 16x^3 - 6x^2 + x \right) \Big|_0^1 \\
&= \frac{36}{5} - 7 \\
\int_0^1 \mathbf{P}_2^2(x) dx &= \frac{1}{5}
\end{aligned}$$

For $\mathbf{P}_3(x)$:

$$\begin{aligned}
\int_0^1 \mathbf{P}_3^2(x) dx &= \int_0^1 (20x^3 - 30x^2 + 12x - 1)^2 dx \\
&= \int_0^1 (400x^6 - 1200x^5 + 1380x^4 - 760x^3 + 204x^2 - 24x + 1) dx \\
&= \left(\frac{400x^7}{7} - \frac{1200x^6}{6} + \frac{138x^5}{5} - \frac{760x^4}{4} + \frac{204x^3}{3} - \frac{24x^2}{2} + x \right) \Big|_0^1 \\
&= \frac{400}{7} - \frac{399}{7}
\end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_3^2(x) dx = \frac{1}{7}} \quad (3.18)$$

For $\mathbf{P}_4(x)$:

$$\begin{aligned}
 \int_0^1 \mathbf{P}_4^2(x) dx &= \int_0^1 (70x^4 - 140x^3 + 90x^2 - 20x + 1)^2 dx \\
 &= \int_0^1 (4900x^8 - 19600x^7 + 32200x^6 - 28000x^5 + 13840x^4 - 3880x^3 + 580x^2 \\
 &\quad - 40x + 1) dx \\
 &= \left(\frac{4900x^9}{9} - 2450x^8 + 4600x^7 - \frac{14000x^6}{3} + 2768x^5 - 970x^4 + \frac{580x^3}{3} - 20x^2 \right. \\
 &\quad \left. + x \right) \Big|_0^1 \\
 &= \frac{1}{9}
 \end{aligned}$$

$$\boxed{\int_0^1 \mathbf{P}_4^2(x) dx = \frac{1}{9}} \quad (3.19)$$

We checked the orthogonality for the first five Shifted Legendre Polynomials defined in (3.3), (3.4), (3.5), (3.6) and (3.7). We got the following matrix for the orthogonality.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{9} \end{bmatrix}$$

3.2 Approximation using Shifted Legendre Polynomials

After understanding the Shifted Legendre Polynomials and their different properties, now we move to the next stage to approximate a function.

3.2.1 One-Dimensional Approximation

For one dimensional approximations, any continuous function $\psi(x)$ defined over the interval $[0, 1]$ can be approximated by Shifted Legendre polynomials as

$$\psi(x) \simeq \sum_{a=0}^m \mathbf{C}_a \mathbf{P}_a(x), \quad (3.20)$$

where $\mathbf{P}_a$ denotes the Shifted Legendre Polynomials of degree a and $\mathbf{C}_a$ represents the expansion co-efficients and is defined as

$$\mathbf{C}_a = (2a + 1) \int_0^1 \psi(x) \mathbf{P}_a(x) dx. \quad (3.21)$$

$$\psi(x) = \mathbf{C}_0 \mathbf{P}_0(x) + \mathbf{C}_1 \mathbf{P}_1(x) + \mathbf{C}_2 \mathbf{P}_2(x) + \mathbf{C}_3 \mathbf{P}_3(x) + \cdots + \mathbf{C}_m \mathbf{P}_m(x) \quad (3.22)$$

Equation (3.22) can be written in vector form also and is given as

$$\begin{aligned} \psi(x) &= [\mathbf{C}_0, \mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \dots, \mathbf{C}_m] \begin{bmatrix} \mathbf{P}_0 \\ \mathbf{P}_1 \\ \mathbf{P}_2 \\ \vdots \\ \mathbf{P}_m \end{bmatrix} \\ \mathbf{V}_M &= [\mathbf{C}_0, \mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \dots, \mathbf{C}_m] \end{aligned} \quad (3.23)$$

$$\hat{\mathbf{P}}_M = \begin{bmatrix} \mathbf{P}_0 \\ \mathbf{P}_1 \\ \mathbf{P}_2 \\ \vdots \\ \mathbf{P}_m \end{bmatrix} \quad (3.24)$$

$$\psi(x) \simeq \mathbf{V}_M \hat{\mathbf{P}}_M(x). \quad (3.25)$$

where $M = m + 1$, $\mathbf{V}_M$ is the $1 \times M$ row vector of Shifted Legendre coefficients and $\hat{\mathbf{P}}$ denotes the $M \times 1$ column vector of Shifted Legendre Polynomials.

Example: To explain the procedure that how we can approximate a continuous function on the domain $[0,1]$ using the Shifted Legendre Polynomials, we approximate the function $\psi(x) = \sin(2\pi)x$ at $M = 4$.

For approximating this function, we calculate $\mathbf{C}_0$, $\mathbf{C}_1$, $\mathbf{C}_2$, $\mathbf{C}_3$, $\mathbf{C}_4$ and $\mathbf{C}_5$.

We calculate $\mathbf{C}_0$

$$\begin{aligned}
 \mathbf{C}_0 &= \int_0^1 (\sin(2\pi)x) \mathbf{P}_0(x) dx \\
 &= \int_0^1 \sin(2\pi x) (1) dx \mathbf{C}_0 \\
 &= \int_0^1 \sin(2\pi x) dx \\
 &= -\frac{1}{2\pi} (\cos(2\pi) - \cos(0)) \\
 &= -\frac{1}{2\pi} (1 - 1) \\
 &= 0 \\
 \mathbf{C}_0 &= 0 \tag{3.26}
 \end{aligned}$$

now we calculate $\mathbf{C}_1$

$$\begin{aligned}
 \mathbf{C}_1 &= 3 \int_0^1 \psi(x) \mathbf{P}_1(x) dx \\
 \mathbf{C}_1 &= 3 \int_0^1 \sin(2\pi x) (2x - 1) dx \\
 &= 3 \left(2 \int_0^1 x \sin(2\pi x) dx - \int_0^1 \sin(2\pi x) dx \right) \\
 &= -3 \left(\frac{1}{\pi} \cos(2\pi) - 3 \left(-\frac{0}{\pi} \cos(0) \right) \right) + 3 \left(\frac{1}{2\pi^2} (\sin(2\pi) - \sin(0)) \right) \\
 &= -\frac{3}{\pi} \cdot 1 + \frac{1}{2\pi^2} (0 - 0) \\
 &= -\frac{3}{\pi} + 0 \\
 \mathbf{C}_1 &= -\frac{3}{\pi} \approx -0.95493 \tag{3.27}
 \end{aligned}$$

now we calculate $\mathbf{C}_2$

$$\begin{aligned}
 \mathbf{C}_2 &= 5 \int_0^1 \psi(x) \mathbf{P}_2(x) dx \\
 \mathbf{C}_2 &= 5 \int_0^1 \sin(2\pi x) (6x^2 - 6x + 1) dx \\
 &= 5 \left(6 \left[-\frac{x^2}{2\pi} \cos(2\pi x) \right]_0^1 + \frac{1}{\pi} \int_0^1 x \cos(2\pi x) dx - 6 \left(-\frac{1}{2\pi} \right) + 0 \right) \\
 &= 5 \left(6 \left(-\frac{1}{2\pi} \right) + 0 - 6 \left(-\frac{1}{2\pi} \right) \right) \\
 &= 5(0)
 \end{aligned}$$

$$\mathbf{C}_2 = 0 \quad (3.28)$$

similarly we calculate $\mathbf{C}_3, \mathbf{C}_4$ and $\mathbf{C}_5$

$$\mathbf{C}_3 = \frac{7(15 - \pi^2)}{\pi^3} \approx 1.15770191219441 \quad (3.29)$$

$$\mathbf{C}_4 = 0 \quad (3.30)$$

$$\mathbf{C}_5 = 11 \left(-\frac{1}{\pi} + \frac{105}{\pi^3} - \frac{945}{\pi^5} \right) \approx -0.2192895472 \quad (3.31)$$

now we put the results obtained from the (3.26),(3.27),(3.28),(3.29),(3.30)(3.31) in equation (3.22)

3.2.2 Operational Matrix Of Derivatives

Theorem 3.2.1. *Let $\hat{\mathbf{n}}(x)$ represent the vector function specified in (3.24). The derivative operator of integer order σ acting on $\hat{\mathbf{n}}(x)$ admits the representation:*

$$\bar{\mathbf{D}}^\sigma(\hat{\mathbf{n}}(x)) \approx \mathbf{D}_{M \times M}^\sigma \hat{\mathbf{n}}(x) \quad (3.32)$$

where $\mathbf{D}_{M \times M}^\sigma$ serves as the differentiation operational matrix of order σ , expressed as:

$$\mathbf{D}_{M \times M}^\sigma = \begin{bmatrix} \Psi_{0,0} & \Psi_{0,1} & \cdots & \Psi_{0,j} & \cdots & \Psi_{0,m} \\ \Psi_{1,0} & \Psi_{1,1} & \cdots & \Psi_{1,j} & \cdots & \Psi_{1,m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Psi_{i,0} & \Psi_{i,1} & \cdots & \Psi_{i,j} & \cdots & \Psi_{i,m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Psi_{m,0} & \Psi_{m,1} & \cdots & \Psi_{m,j} & \cdots & \Psi_{m,m} \end{bmatrix} \quad (3.33)$$

Each element $\Psi_{i,j}$ is determined through:

$$\Psi_{i,j} = w_j \sum_{s=\lceil \sigma \rceil}^i \frac{(-1)^{i+s} (s+1)! \cdot s!}{(i-s)!(s!)^2 (s-\sigma)!} \quad (3.34)$$

Theorem 3.2.2. *Given the bivariate function vector $\hat{\mathbf{n}}(x,t)$ from (3.24), we derive the differentiation of integer order σ with respect to t :*

$$\bar{\mathbf{D}}^{\sigma,t}(\hat{\mathbf{n}}(x,t)) \approx \mathbf{D}_{M^2 \times M^2}^{\sigma,t} \hat{\mathbf{n}}(x,t) \quad (3.35)$$

The matrix $\mathbf{D}_{M^2 \times M^2}^{\sigma, t}$ characterizes the σ -order differentiation operator in the temporal direction:

$$\mathbf{D}_{M^2 \times M^2}^{\sigma, t} = \begin{bmatrix} \Xi_{0,0,s} & \Xi_{0,1} & \cdots & \Xi_{0,q,s} & \cdots & \Xi_{0,M^2,s} \\ \Xi_{1,0,s} & \Xi_{1,1,s} & \cdots & \Xi_{1,2,s} & \cdots & \Xi_{1,M^2,s} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Xi_{r,0,s} & \Xi_{r,1,s} & \cdots & \Xi_{r,q,s} & \cdots & \Xi_{r,M^2,s} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Xi_{M^2,0,s} & \Xi_{M^2,1,s} & \cdots & \Xi_{M^2,q,s} & \cdots & \Xi_{M^2,M^2,s} \end{bmatrix} \quad (3.36)$$

with $r = Mi + j + 1$, $q = Mc + h + 1$, and $i, j, c, h \in \{0, 1, 2, 3, \dots, m\}$.

Elements are computed as:

$$\Xi_{r,q,s} = \mathbf{E}_{i,j,h,c,s} = \delta_{i,c}(2j+1) \sum_{s=\lceil \sigma \rceil}^c \sum_{n=0}^j \frac{(-1)^{j+n+h+s}(j+n)!(h+s)!}{(j-n)!(n!)^2(s+n-\sigma+1)(h-s)!s!(s-\sigma)!} \quad (3.37)$$

where $\mathbf{E}_{i,j,h,c,s} = 0$ if $h < \sigma$.

Theorem 3.2.3. For the function vector $\hat{\mathbf{n}}(x, t)$ described in (3.24), differentiation of integer order σ in the spatial variable x is formulated as:

$$\bar{\mathbf{D}}^{\sigma, x}(\hat{\mathbf{n}}(x, t)) \approx \mathbf{D}_{M^2 \times M^2}^{\sigma, x} \hat{\mathbf{n}}(x, t) \quad (3.38)$$

The operator $\mathbf{D}_{M^2 \times M^2}^{\sigma, x}$ embodies the σ -order differentiation matrix in the x -direction:

$$\mathbf{D}_{M^2 \times M^2}^{\sigma, x} = \begin{bmatrix} \Omega_{0,0,s} & \Omega_{0,1} & \cdots & \Omega_{0,q,s} & \cdots & \Omega_{0,M^2,s} \\ \Omega_{1,0,s} & \Omega_{1,1,s} & \cdots & \Omega_{1,2,s} & \cdots & \Omega_{1,M^2,s} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Omega_{r,0,s} & \Omega_{r,1,s} & \cdots & \Omega_{r,q,s} & \cdots & \Omega_{r,M^2,s} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \Omega_{M^2,0,s} & \Omega_{M^2,1,s} & \cdots & \Omega_{M^2,q,s} & \cdots & \Omega_{M^2,M^2,s} \end{bmatrix} \quad (3.39)$$

where indexing follows $r = Mi + j + 1$ and $q = Mc + h + 1$, with $i, j, c, h \in \{0, 1, 2, 3, \dots, m\}$.

The matrix entries satisfy:

$$\Omega_{r,q,s} = \mathbf{E}_{i,j,h,c,s} = \delta_{i,h}(2j+1) \sum_{s=\lceil \sigma \rceil}^c \sum_{n=0}^j \frac{(-1)^{j+n+c+s}(j+n)!(c+s)!}{(j-n)!(n!)^2(s+n-\sigma+1)(c-s)!s!(s-\sigma)!} \quad (3.40)$$

with the condition $\mathbf{E}_{i,j,h,c,s} = 0$ for $c < \sigma$.

3.2.3 Operational Matrix Of Integration

Theorem 3.2.4. Let $\hat{\mathbf{n}}(x)$ be the function vector defined in (3.24), then the integration of order α is given by:

$$\mathcal{J}^\alpha(\hat{\mathbf{n}}(x)) \simeq \mathbf{O}_{M \times M}^\alpha \hat{\mathbf{n}}(x) \quad (3.41)$$

where $\mathbf{O}_{M \times M}^\alpha$ is the operational matrix of integration defined as:

$$\mathbf{O}_{M \times M}^\alpha = \begin{bmatrix} \hat{O}_{0,0} & \hat{O}_{0,1} & \cdots & \hat{O}_{0,j} & \cdots & \hat{O}_{0,m} \\ \hat{O}_{1,0} & \hat{O}_{1,1} & \cdots & \hat{O}_{1,j} & \cdots & \hat{O}_{1,m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{O}_{i,0} & \hat{O}_{i,1} & \cdots & \hat{O}_{i,j} & \cdots & \hat{O}_{i,m} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{O}_{m,0} & \hat{O}_{m,1} & \cdots & \hat{O}_{m,j} & \cdots & \hat{O}_{m,m} \end{bmatrix} \quad (3.42)$$

The elements of the operational matrix are defined as:

$$\hat{O}_{i,j} = (2j+1) \sum_{k=0}^i \frac{(-1)^{i+k}(i+k)!}{(i-k)!(k!)^2} \cdot \frac{k!}{(k+\alpha)!} \quad (3.43)$$

Theorem 3.2.5. Let $\hat{\mathbf{n}}(x,t)$ be the function vector defined in (3.24), then the integration of order α with respect to x is generalized as:

$$\mathcal{J}^{\alpha,x}(\hat{\mathbf{n}}(x,t)) \simeq \mathbf{O}_{M^2 \times M^2}^{\alpha,x} \hat{\Lambda}(x,t) \quad (3.44)$$

where $\mathbf{O}_{M^2 \times M^2}^{\alpha,x}$ is the operational matrix of integration.

$$\mathbf{O}_{M^2 \times M^2}^{\alpha,x} = \begin{bmatrix} \hat{O}_{0,0} & \hat{O}_{0,1} & \cdots & \hat{O}_{0,p} & \cdots & \hat{O}_{0,M^2} \\ \hat{O}_{1,0} & \hat{O}_{1,1} & \cdots & \hat{O}_{1,2} & \cdots & \hat{O}_{1,M^2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{O}_{d,0} & \hat{O}_{d,1} & \cdots & \hat{O}_{d,p} & \cdots & \hat{O}_{d,M^2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{O}_{M^2,0} & \hat{O}_{M^2,1} & \cdots & \hat{O}_{M^2,p} & \cdots & \hat{O}_{M^2,M^2} \end{bmatrix} \quad (3.45)$$

The entries of the matrix are defined as:

$$\hat{O}_{d,p} = C_{i,j,b,a} \quad (3.46)$$

$$C_{i,j,b,a} = \sum_{k=0}^a \delta_{i,a}(2j+1) \sum_{l=0}^j \frac{(-1)^{j+l+a+k}(j+l)!(a+k)!}{(j-l)!(l!)^2(k+l+\alpha+1)(a-k)!k!(k+\alpha)!} \quad (3.47)$$

where

$$\delta_{i,a} = \begin{cases} 1 & \text{if } i = a \\ 0 & \text{if } i \neq a \end{cases} \quad (3.48)$$

Condition:

$$C_{i,j,b,a} = 0 \quad \text{if } a < \alpha \quad (3.49)$$

Theorem 3.2.6. Let $\hat{\mathbf{n}}(x, t)$ be the vector function defined in (3.24). Then, the integration of integer order α with respect to t can be expressed as:

$$\mathcal{J}_t^\alpha(\hat{\mathbf{n}}(x, t)) \approx \mathbf{O}_t^\alpha \hat{\mathbf{n}}(x, t) \quad (3.50)$$

where $\mathbf{O}_t^\alpha$ denotes the operational integration matrix corresponding to the variable t .

$$\mathbf{O}_t^\alpha = \begin{bmatrix} \bar{O}_{0,0} & \bar{O}_{0,1} & \cdots & \bar{O}_{0,q} & \cdots & \bar{O}_{0,N} \\ \bar{O}_{1,0} & \bar{O}_{1,1} & \cdots & \bar{O}_{1,q} & \cdots & \bar{O}_{1,N} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{O}_{r,0} & \bar{O}_{r,1} & \cdots & \bar{O}_{r,q} & \cdots & \bar{O}_{r,N} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{O}_{N,0} & \bar{O}_{N,1} & \cdots & \bar{O}_{N,q} & \cdots & \bar{O}_{N,N} \end{bmatrix} \quad (3.51)$$

The elements of this matrix are given by:

$$\bar{O}_{r,q} = \Phi_{m,n,s,\ell} \quad (3.52)$$

$$\Phi_{m,n,s,\ell} = \sum_{\kappa=0}^{\ell} \delta_{m,\ell}(2n+1) \sum_{\eta=0}^n \frac{(-1)^{n+\eta+\ell+\kappa}(n+\eta)!(\ell+\kappa)!}{(n-\eta)!(\eta!)^2(\kappa+\eta+\alpha+1)(\ell-\kappa)!\kappa!(\kappa+\alpha)!} \quad (3.53)$$

where

$$\delta_{m,\ell} = \begin{cases} 1 & \text{if } m = \ell \\ 0 & \text{if } m \neq \ell \end{cases} \quad (3.54)$$

Constraint:

$$\Phi_{m,n,s,\ell} = 0 \quad \text{whenever } \ell < \alpha \quad (3.55)$$

Lemma 3.2.1. Assume $F_\mu(x) = \beta x^\mu$ represents a polynomial, $\hat{\mathbf{n}}(x, t)$ denotes a function vector, and $\zeta \in \mathbb{R}$. Under these conditions, we obtain:

$$\beta \xi^\mu \hat{\mathbf{n}}(\zeta, t) = \mathbf{A}^{(\beta, \mu, \zeta)} \hat{\mathbf{n}}(x, t) \quad (3.56)$$

The transformation matrix $\mathbf{A}^{(\beta, \mu, \zeta)}$ possesses the structure:

$$\mathbf{A}^{(\beta, \mu, \zeta)} = \begin{pmatrix} \Phi_{(1,1)} & \Phi_{(1,2)} & \cdots & \Phi_{(1,N^2)} \\ \Phi_{(2,1)} & \Phi_{(2,2)} & \cdots & \Phi_{(2,N^2)} \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_{(N^2,1)} & \Phi_{(N^2,2)} & \cdots & \Phi_{(N^2,N^2)} \end{pmatrix} \quad (3.57)$$

wherein $\Phi_{(u,v)}$ relates to $\Sigma_{(v,p,q)}$ through the index mapping $u = Nq + p + 1, v = Nq + h + 1$, where $v, h, p, q = 0, 1, 2, \dots, N$. The element $\Sigma_{(v,p,q)}$ is expressed as:

$$\Sigma_{(v,p,q)} = \sum_{w=0}^{\mu} \frac{(v+w)!(-1)^{v+w}}{(w!)^2(v-w)!} \zeta^w \sum_{g=0}^N \sum_{f=0}^N \Upsilon_{p(q,g)}. \quad (3.58)$$

4 DEVELOPMENT OF PROPOSED METHOD

In this chapter, we will develop our proposed scheme for the Solution of Coupled System of Generalized Heat Conduction Model with Boundary Constraints. First, we developed the scheme for the ordinary differential equations and their coupled systems. Then, we extended this solution methodology to Partial Differential Equations (PDEs) and their coupled system to achieve our main aim.

4.1 Ordinary Differential Equations(ODEs)

4.1.1 First-order ODEs

Consider the following first order differential equation

$$\frac{d}{dx}u(x) = a_0u(x) + f(x) \quad (4.1)$$

with initial condition

$$u(0) = u_0$$

To solve this general first order differential equation, we first express the highest order derivative term $\frac{d}{dx}u(x)$ in terms of the shifted legendre polynomials.

$$\frac{d}{dx}u(x) = \mathbf{V}_M \hat{\mathbf{M}}_M(x) \quad (4.2)$$

We apply the first order integration to (4.2) from 0 to x and get

$$\begin{aligned} \int_0^x \frac{d}{dx}u(x)dx &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x) \\ u(x) \Big|_0^x &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x) \\ u(x) - u(0) &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x) \\ u(x) &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x) + u_0 \end{aligned} \quad (4.3)$$

We approximate u_0 using the Shifted Legendre Polynomials as $u_0 = G_1 \hat{\mathbf{M}}_M(x)$ and get the following expression for the solution $u(x)$.

$$\boxed{u(x) = \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x) + G_1 \hat{\mathbf{M}}_M(x)} \quad (4.4)$$

In equation (4.4), the co-efficient vector $\mathbf{V}_M$ is unknown to be determined. To find this, we will put the equations (4.2) and (4.4) in the (4.1).

$$\mathbf{V}_M \hat{\mathbf{M}}_M(x) = a_0(\mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x) + G_1 \hat{\mathbf{M}}_M(x)) + G_2 \hat{\mathbf{M}}_M(x)$$

Rearranging

$$\mathbf{V}_M \hat{\mathbf{M}}_M(x) = \hat{\mathbf{M}}_M(x) (a_0(\mathbf{V}_M \mathbf{O}_{M \times M} + G_1) + G_2)$$

Cancelling out the common term, we get

$$\mathbf{V}_M - a_0 \mathbf{V}_M \mathbf{O}_{M \times M} = a_0 G_1 + G_2$$

$$\mathbf{V}_M (1 - a_0 \mathbf{O}_{M \times M}) = a_0 G_1 + G_2$$

$$\mathbf{V}_M (I - a_0 \mathbf{O}_{M \times M}) = a_0 G_1 + G_2$$

$$\boxed{\mathbf{V}_M = (a_0 G_1 + G_2)(I - a_0 \mathbf{O}_{M \times M})^{-1}} \quad (4.5)$$

We use the result obtained in (4.5) into (4.4) to get the solution $u(x)$ obtained through the proposed method.

4.1.2 Second-order ODEs

Consider the following second order differential equation,

$$\frac{d^2}{dx^2} u(x) = a_0 \frac{d}{dx} (u(x)) + a_1 u(x) + f \quad (4.6)$$

with initial conditions given by,

$$u(0) = u_0,$$

$$\frac{d}{dx} u(0) = u_1$$

To solve this general second order differential equation, we will express the highest order derivative term $\frac{d^2}{dx^2} u(x)$ in terms of Shifted Legendre Polynomials,

$$\frac{d^2}{dx^2} (u(x)) = \mathbf{V}_M \hat{\mathbf{M}}_M(x) \quad (4.7)$$

We Integrate (4.7) from 0 to x ,

$$\int_0^x \frac{d^2}{dx^2} u(x) dx = \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x)$$

$$\left. \frac{d}{dx} u(x) \right|_0^x = \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x)$$

$$\frac{d}{dx} u(x) - \frac{d}{dx} u_0 = \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{M}}_M(x)$$

Now again applying integration from 0 to x we get,

$$\begin{aligned}\int_0^x \frac{d}{dx} u(x) dx - \int_0^x \frac{d}{dx} u_0 dx &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{u}}_M(x) \\ u(x) \Big|_0^x - u_1 x \Big|_0^x &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{u}}_M(x) \\ u(x) - u_0 - u_1 x &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{u}}_M(x) \\ u(x) &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{u}}_M(x) + (u_1(x) + u_0)\end{aligned}$$

We approximate $u_1(x) + u_0(x)$ using the Shifted Legendre Polynomials as $u_1(x) + u_0 = G_1 \hat{\mathbf{u}}_M(x)$ and get the following expression for $u(x)$.

$$\boxed{u(x) = \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{u}}_M(x) + G_1 \hat{\mathbf{u}}_M(x)} \quad (4.8)$$

Applying derivative to the equation (4.8) we get the following expression,

$$\frac{d}{dx} u(x) = \mathbf{V}_M \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} \hat{\mathbf{u}} + G_1 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x) \quad (4.9)$$

In equation (4.8) the $\mathbf{V}_M$ is unknown coefficient vector, we put the equations (4.7), (4.8) and (4.9) in equation (4.6) to find $\mathbf{V}_M$,

$$\mathbf{V}_M \hat{\mathbf{u}}(x) = a_0 (\mathbf{V}_M \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x) + G_1 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x)) + a_1 (\mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{u}}(x) + G_1 \hat{\mathbf{u}}(x)) + G_2 \hat{\mathbf{u}}(x)$$

Rearranging

$$\mathbf{V}_M \hat{\mathbf{u}}(x) = \hat{\mathbf{u}}(x) (a_0 (\mathbf{V}_M \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} + G_1 \mathbf{D}_{M \times M}) + a_1 (\mathbf{V}_M \mathbf{O}_{M \times M}^2 + G_1) + G_2 \hat{\mathbf{u}}(x))$$

Cancelling the common terms, we get the following expression

$$\mathbf{V}_M = a_0 \mathbf{V}_M \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} + a_0 G_1 \mathbf{D}_{M \times M} + a_1 \mathbf{V}_M \mathbf{O}_{M \times M}^2 + a_1 G_1 + G_2$$

$$\mathbf{V}_M - a_0 \mathbf{V}_M \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} - a_1 \mathbf{V}_M \mathbf{O}_{M \times M}^2 = a_0 G_1 \mathbf{D}_{M \times M} + a_1 G_1 + G_2$$

$$\mathbf{V}_M (1 - (a_0 \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} + a_1 \mathbf{O}_{M \times M}^2)) = a_0 G_1 \mathbf{D}_{M \times M} + a_1 G_1 + G_2$$

$$\mathbf{V}_M (I - (a_0 \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} + a_1 \mathbf{O}_{M \times M}^2)) = a_0 G_1 \mathbf{D}_{M \times M} + a_1 G_1 + G_2$$

$$\boxed{\mathbf{V}_M = (a_0 G_1 \mathbf{D}_{M \times M} + a_1 G_1 + G_2) (I - (a_0 \mathbf{O}_{M \times M}^2 \mathbf{D}_{M \times M} + a_1 \mathbf{O}_{M \times M}^2))^{-1}} \quad (4.10)$$

we use the result obtained in the (4.10) into (4.8) to get solution $u(x)$ obtained through our proposed method.

4.1.3 Third-order ODEs

Consider the following 3rd Order Differential equation,

$$\frac{d^3}{dx^3}u(x) = a_0 \frac{d^2}{dx^2}u(x) + a_1 \frac{d}{dx}u(x) + a_2 u(x) + f \quad (4.11)$$

with initial conditions given by,

$$\begin{aligned} u(0) &= u_0, \\ \frac{d}{dx}u(x) &= u_1, \\ \frac{d^2}{dx^2}u(x) &= u_2. \end{aligned}$$

To solve this general third order differential equation, we will express the highest order derivative term $\frac{d^3}{dx^3}u(x)$ in terms of Shifted Legendre Polynomials,

$$\frac{d^3}{dx^3}u(x) = \mathbf{V}_M \hat{\mathbf{\Pi}}(x) \quad (4.12)$$

where $\mathbf{V}_M$ is the unknown row vector of Shifted Legendre Polynomials, and $\hat{\mathbf{\Pi}}(x)$ is the function vector of Shifted Legendre Polynomials. We integrate(4.12) from 0 to x we get

$$\begin{aligned} \int_0^x \frac{d^3}{dx^3}u(x) &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{\Pi}}(x) \\ \left. \frac{d^2}{dx^2}u(x) \right|_0^x &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{\Pi}}(x) \\ \frac{d^2}{dx^2}u(x) - \frac{d^2}{dx^2}u(0) &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{\Pi}}(x) \\ \frac{d^2}{dx^2}u(x) - u_2 &= \mathbf{V}_M \mathbf{O}_{M \times M} \hat{\mathbf{\Pi}}(x) \end{aligned}$$

Now again we apply Integration from 0 to x we get

$$\begin{aligned} \int_0^x \frac{d^2}{dx^2}u(x) - \int_0^x u_2 &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{\Pi}}(x) \\ \left. \frac{d}{dx}u(x) \right|_0^x - u_2 x \Big|_0^x &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{\Pi}}(x) \\ \frac{d}{dx}u(x) - \frac{d}{dx}u(0) - u_2 x - 0 &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{\Pi}}(x) \\ \frac{d}{dx}u(x) - u_1 - u_2 x &= \mathbf{V}_M \mathbf{O}_{M \times M}^2 \hat{\mathbf{\Pi}}(x) \end{aligned}$$

Again applying Integration from 0 to x we get

$$\begin{aligned}
 \int_0^x \frac{d}{dx} u(x) dx - \int_0^x u_1 dx - \int_0^x u_2 x dx &= \mathbf{V}_M \mathbf{O}_{M \times M}^3 \hat{\mathbf{u}}(x) \\
 u(x) \Big|_0^x - u_1 x \Big|_0^x - u_2 \frac{x^2}{2} \Big|_0^x &= \mathbf{V}_M \mathbf{O}_{M \times M}^3 \hat{\mathbf{u}}(x) \\
 u(x) - u_0 - u_1 x - u_2 \frac{x^2}{2} &= \mathbf{V}_M \mathbf{O}_{M \times M}^3 \hat{\mathbf{u}}(x) \\
 u(x) &= \mathbf{V}_M \mathbf{O}_{M \times M}^3 \hat{\mathbf{u}}(x) + u_0 + u_1 x + u_2 \frac{x^2}{2}
 \end{aligned} \tag{4.13}$$

We approximate $u_0 + u_1 x + u_2 \frac{x^2}{2}$ using Shifted Legendre Polynomials as $u_0 + u_1 x + u_2 \frac{x^2}{2} = G_1 \hat{\mathbf{u}}(x)$ and get the following solution for the solution $u(x)$ the equation (4.13) becomes,

$$\boxed{u(x) = \mathbf{V}_M \mathbf{O}_{M \times M}^3 \hat{\mathbf{u}}(x) + G_1 \hat{\mathbf{u}}(x)} \tag{4.14}$$

To find $\frac{d}{dx} u(x)$ we take derivative of equation (4.14),

$$\frac{d}{dx} u(x) = \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x) + G_1 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x) \tag{4.15}$$

Again taking derivative of the equation (4.15)

$$\frac{d^2}{dx^2} u(x) = \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 \hat{\mathbf{u}}(x) + G_1 \mathbf{D}_{M \times M}^2 \hat{\mathbf{u}}(x) \tag{4.16}$$

In (4.14) $\mathbf{V}_M$ is the unknown coefficient vector, we have all the required results to find $\mathbf{V}_M$, so we put (4.14), (4.15), (4.16) and (4.12) in the equation (4.11)

$$\begin{aligned}
 \mathbf{V}_M \hat{\mathbf{u}}(x) &= a_0 (\mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 \hat{\mathbf{u}}(x) + G_1 \mathbf{D}_{M \times M}^2 \hat{\mathbf{u}}(x)) + a_1 (\mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x) \\
 &\quad + G_1 \mathbf{D}_{M \times M} \hat{\mathbf{u}}(x)) + a_2 (\mathbf{V}_M \mathbf{O}_{M \times M}^3 \hat{\mathbf{u}}(x) + G_1 \hat{\mathbf{u}}(x)) + G_2 \hat{\mathbf{u}}(x)
 \end{aligned}$$

Rearranging the above equation,

$$\begin{aligned}
 \mathbf{V}_M \hat{\mathbf{u}}(x) &= \hat{\mathbf{u}}(x) (a_0 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_1 G_1 \mathbf{D}_{M \times M} \\
 &\quad + a_2 \mathbf{V}_M \mathbf{O}_{M \times M}^3 + a_2 G_1 + G_2)
 \end{aligned}$$

Cancelling the common terms, we get the following expression

$$\begin{aligned}
 \mathbf{V}_M &= a_0 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_1 G_1 \mathbf{D}_{M \times M} \\
 &\quad + a_2 \mathbf{V}_M \mathbf{O}_{M \times M}^3 + a_2 G_1 + G_2
 \end{aligned}$$

$$\begin{aligned}
 \mathbf{V}_M &= a_0 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_1 G_1 \mathbf{D}_{M \times M} + a_2 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \\
 &\quad + a_2 G_1 + G_2
 \end{aligned}$$

$$\mathbf{V}_M - a_0 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 - a_1 \mathbf{V}_M \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} - a_2 \mathbf{V}_M \mathbf{O}_{M \times M}^3 = a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 G_1 \mathbf{D}_{M \times M} + a_2 G_1 + G_2$$

$$\mathbf{V}_M (1 - (a_0 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_2 \mathbf{O}_{M \times M}^3)) = a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 G_1 \mathbf{D}_{M \times M} + a_2 G_1 + G_2$$

$$\mathbf{V}_M (I - (a_0 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_2 \mathbf{O}_{M \times M}^3)) = a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 G_1 \mathbf{D}_{M \times M} + a_2 G_1 + G_2$$

$$\mathbf{V}_M = (I - (a_0 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_2 \mathbf{O}_{M \times M}^3))^{-1} (a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 G_1 \mathbf{D}_{M \times M} + a_2 G_1 + G_2) \quad (4.17)$$

$$\boxed{\mathbf{V}_M = (I - (a_0 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M}^2 + a_1 \mathbf{O}_{M \times M}^3 \mathbf{D}_{M \times M} + a_2 \mathbf{O}_{M \times M}^3))^{-1} (a_0 G_1 \mathbf{D}_{M \times M}^2 + a_1 G_1 \mathbf{D}_{M \times M} + a_2 G_1 + G_2)} \quad (4.18)$$

we use the results from (4.18) into (4.14) to get solution $u(x)$ obtained through our proposed method.

4.2 Coupled System Of ODEs

4.2.1 First-order Coupled Systems

Consider the following system of equations of first order,

$$\frac{d}{dx} u(x) = a_0 u(x) + a_1 v(x) + f(x) \quad (4.19)$$

$$\frac{d}{dx} v(x) = b_0 u(x) + b_1 v(x) + g(x) \quad (4.20)$$

The initial conditions are given by,

$$u(0) = u_0$$

and

$$v(0) = v_0$$

We express the highest order terms in terms of Shifted Legendre Polynomials as,

$$\frac{d}{dx}u(x) = \hat{\mathbf{K}}\hat{\mathbf{u}}(x) \quad (4.21)$$

where $\hat{\mathbf{K}}$ is the unknown Shifted Legendre coefficient vector.

$$\frac{d}{dx}v(x) = \hat{\mathbf{L}}\hat{\mathbf{v}}(x) \quad (4.22)$$

where $\hat{\mathbf{L}}$ is the unknown Shifted Legendre coefficient vector.

Integrating (4.21) from 0 to x ,

$$\int_0^x \frac{d}{dx}u(x)dx = \hat{\mathbf{K}}_M \mathbf{O}\hat{\mathbf{u}}_M(x)$$

$$u(x)\Big|_0^x = \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_M(x)$$

$$u(x) - u_0 = \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_M(x)$$

$$u(x) = \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_M(x) + u_0$$

we express $u(0)$ in terms of Shifted Legendre Polynomials,

$$u_0 = G_1 \hat{\mathbf{u}}_M(x)$$

where G_1 is the Shifted Legendre coefficient vector.

we approximate the source term $f(x)$ in terms of shifted Legendre Polynomials as,

$$f(x) = H_1 \hat{\mathbf{u}}_M(x)$$

where H_1 is the Shifted Legendre coefficient for the source term $f(x)$.

$$u(x) = \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_M(x) + G_1 \hat{\mathbf{u}}_M(x) \quad (4.23)$$

Integrating (4.22) from 0 to x ,

$$\int_0^x \frac{d}{dx}v(x)dx = \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{v}}_M(x)$$

$$\begin{aligned}
v(x) \Big|_0^x &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) \\
v(x) - v(0) &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) \\
v(x) &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) + v_0
\end{aligned}$$

we express $v(0)$ in terms of Shifted Legendre Polynomials,

$$v_0 = G_2\hat{\mathbf{\Pi}}_M(x)$$

where G_2 is the Shifted Legendre coefficient vector. we approximate the source term $g(x)$ in terms of Shifted Legendre Polynomials as,

$$g(x) = H_2\hat{\mathbf{\Pi}}_M(x)$$

$H_2(M \times M)$ is the Shifted Legendre coefficient vector of source term $g(x)$.

$$v(x) = \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) + G_2\hat{\mathbf{\Pi}}_M(x) \quad (4.24)$$

Now we put the calculated results in (4.19) and (4.20)

$$\hat{\mathbf{K}}\hat{\mathbf{\Pi}}_M(x) = a_0\hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) + a_0G_1\hat{\mathbf{\Pi}}_M(x) + a_1\hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) + a_1G_2\hat{\mathbf{\Pi}}_M(x) + H_1\hat{\mathbf{\Pi}}_M(x)$$

$$\hat{\mathbf{K}}\hat{\mathbf{\Pi}}_M(x) = \hat{\mathbf{\Pi}}_M(x) (a_0\hat{\mathbf{K}}\mathbf{O} + a_0G_1 + a_1\hat{\mathbf{L}}\mathbf{O} + a_1G_2 + H_1)$$

cancelling common terms we get the following expression

$$\boxed{\hat{\mathbf{K}} = a_0\hat{\mathbf{K}}\mathbf{O} + a_0G_1 + a_1\hat{\mathbf{L}}\mathbf{O} + a_1G_2 + H_1} \quad (4.25)$$

Putting the calculated results in the (4.20) we get the following expression.

$$\hat{\mathbf{L}}\hat{\mathbf{\Pi}}_M(x) = b_0\hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) + b_0G_1\hat{\mathbf{\Pi}}_M(x) + b_1\hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}}_M(x) + b_1G_2 + H_2\hat{\mathbf{\Pi}}_M(x)\hat{\mathbf{\Pi}}_M(x)$$

$$\hat{\mathbf{L}}\hat{\mathbf{\Pi}}_M(x) = \hat{\mathbf{\Pi}}_M(x) (b_0\hat{\mathbf{K}}\mathbf{O} + b_0G_1 + b_1\hat{\mathbf{L}}\mathbf{O} + b_1G_2 + H_2)$$

cancelling common terms we get the following expression

$$\boxed{\hat{\mathbf{L}} = b_0\hat{\mathbf{K}}\mathbf{O} + b_0G_1 + b_1\hat{\mathbf{L}}\mathbf{O} + b_1G_2 + H_2} \quad (4.26)$$

we express (4.25) and (4.26) in matrix form

$$\begin{bmatrix} \hat{\mathbf{K}} \\ \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} a_0 \hat{\mathbf{K}} \mathbf{O} \\ b_1 \hat{\mathbf{L}} \mathbf{O} \end{bmatrix} + \begin{bmatrix} a_1 \hat{\mathbf{L}} \mathbf{O}^\alpha \\ b_0 \hat{\mathbf{K}} \mathbf{O}^\alpha \end{bmatrix} + \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 \\ b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix} \quad (4.27)$$

Taking transpose of the (4.27)

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} a_0 \hat{\mathbf{K}} \mathbf{O} & b_1 \hat{\mathbf{L}} \mathbf{O} \end{bmatrix} + \begin{bmatrix} a_1 \hat{\mathbf{L}} \mathbf{O} & b_0 \hat{\mathbf{K}} \mathbf{O} \end{bmatrix} + \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 & b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} a_0 \mathbf{O} & 0 \\ 0 & b_1 \mathbf{O} \end{bmatrix} + \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} 0 & b_0 \mathbf{O} \\ a_1 \mathbf{O} & 0 \end{bmatrix} + \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 & b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} a_0 \mathbf{O} & b_0 \mathbf{O} \\ a_1 \mathbf{O} & b_1 \mathbf{O} \end{bmatrix} + \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 & b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} - \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} a_0 \mathbf{O} & b_0 \mathbf{O} \\ a_1 \mathbf{O} & b_1 \mathbf{O} \end{bmatrix} = \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 & b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \left(\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} - \begin{bmatrix} a_0 \mathbf{O} & b_0 \mathbf{O} \\ a_1 \mathbf{O} & b_1 \mathbf{O} \end{bmatrix} \right) = \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 & b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} I - a_0 \mathbf{O} & -b_0 \mathbf{O} \\ -a_1 \mathbf{O} & I - b_1 \mathbf{O} \end{bmatrix}^{-1} \begin{bmatrix} a_0 G_1 + a_1 G_2 + H_1 & b_0 G_1 + b_1 G_2 + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} I - a_0 \mathbf{O} & -b_0 \mathbf{O} \\ -a_1 \mathbf{O} & I - b_1 \mathbf{O} \end{bmatrix}^{-1} \quad (4.28)$$

$$\blacksquare = \begin{bmatrix} I - a_0 \mathbf{O} & -b_0 \mathbf{O} \\ -a_1 \mathbf{O} & I - b_1 \mathbf{O} \end{bmatrix}^{-1}$$

$$\blacksquare^{-1} = \frac{1}{\det \blacksquare} \text{adj}(\blacksquare)$$

$$\begin{aligned}
\text{adj}(\blacksquare) &= \begin{bmatrix} I - b_1 \mathbf{O} & b_0 \mathbf{O} \\ a_1 \mathbf{O} & I - a_0 \mathbf{O} \end{bmatrix} \\
\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} &= \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} \frac{I - b_1 \mathbf{O}}{\det \blacksquare} & \frac{b_0 \mathbf{O}}{\det \blacksquare} \\ \frac{a_1 \mathbf{O}}{\det \blacksquare} & \frac{I - a_0 \mathbf{O}}{\det \blacksquare} \end{bmatrix} \\
\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} &= \begin{bmatrix} \frac{A(I - b_1 \mathbf{O}) + B(a_1 \mathbf{O})}{\det \blacksquare} & \frac{A(b_0 \mathbf{O}) + B(I - a_0 \mathbf{O})}{\det \blacksquare} \end{bmatrix} \\
\hat{\mathbf{K}} &= \frac{A(I - b_1 \mathbf{O}) + B(a_1 \mathbf{O})}{\det \blacksquare} \\
\hat{\mathbf{L}} &= \frac{A(b_0 \mathbf{O}) + B(I - a_0 \mathbf{O})}{\det \blacksquare}
\end{aligned}$$

substituting the obtained results in (4.23) and (4.26) we get the solution $u(x)$ and $v(x)$ by using our proposed method.

4.2.2 Second-order Coupled Systems

Consider the following coupled system of equations of second order,

$$\frac{d^2}{dx^2}u(x) = a_0u(x) + a_1\frac{d}{dx}u(x) + a_2v(x) + a_3\frac{d}{dx}v(x) + f(x) \quad (4.29)$$

$$\frac{d^2}{dx^2}v(x) = a_0u(x) + a_1\frac{d}{dx}u(x) + a_2v(x) + a_3\frac{d}{dx}v(x) + g(x) \quad (4.30)$$

the initial conditions are given by,

$$u(0) = u_0, \quad \frac{d}{dx}u(0) = u_1$$

and

$$v(0) = v_0, \quad \frac{d}{dx}v(0) = v_1$$

we express the highest order term in terms of Shifted Legendre Polynomials,

$$\frac{d^2}{dx^2}u(x) = \hat{\mathbf{K}}\hat{\mathbf{u}}_M(x) \quad (4.31)$$

$$\frac{d^2}{dx^2}v(x) = \hat{\mathbf{L}}\hat{\mathbf{v}}_M(x) \quad (4.32)$$

Integrating (4.31) from 0 to x ,

$$\begin{aligned}\int_0^x \frac{d^2}{dx^2} u(x) dx &= \hat{\mathbf{K}} \mathbf{O} \hat{\mathbf{m}}_M(x) \\ \frac{d}{dx} u(x) \Big|_0^x &= \hat{\mathbf{K}} \mathbf{O} \hat{\mathbf{m}}_M(x) \\ \frac{d}{dx} u(x) - \frac{d}{dx} u(0) &= \hat{\mathbf{K}} \mathbf{O} \hat{\mathbf{m}}_M(x) \\ \frac{d}{dx} u(x) - u_0 &= \hat{\mathbf{K}} \mathbf{O} \hat{\mathbf{m}}_M(x)\end{aligned}$$

Again integrating from 0 to x ,

$$\begin{aligned}\int_0^x \left(\frac{d}{dx} u(x) - u_0 \right) dx &= \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{m}}_M(x) \\ u(x) \Big|_0^x - u_0 x \Big|_0^x &= \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{m}}_M(x) \\ u(x) - u(0) - u_1 x &= \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{m}}_M(x) \\ u(x) - u_0 - u_1 x &= \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{m}}_M(x) \\ u(x) &= \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{m}}_M(x) + u_0 + u_1 x\end{aligned}$$

$$u(x) = \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{m}}_M(x) + G_1 \hat{\mathbf{m}}_M(x) \quad (4.33)$$

where $\hat{\mathbf{K}}$ is the unknown Shifted Legendre coefficient vector, $\mathbf{O}^2$ is the second order operational matrix of integration. G_1 is the Shifted Legendre coefficient vector for $u_0 + u_1 x$.

Integrating (4.32) from 0 to x ,

$$\begin{aligned}\int_0^x \frac{d^2}{dx^2} v(x) dx &= \hat{\mathbf{L}} \mathbf{O} \hat{\mathbf{m}}_M(x) \\ \frac{d}{dx} v(x) \Big|_0^x &= \hat{\mathbf{L}} \mathbf{O} \hat{\mathbf{m}}_M(x) \\ \frac{d}{dx} v(x) - \frac{d}{dx} v(0) &= \hat{\mathbf{L}} \mathbf{O} \hat{\mathbf{m}}_M(x) \\ \frac{d}{dx} v(x) - v_0 &= \hat{\mathbf{L}} \mathbf{O} \hat{\mathbf{m}}_M(x)\end{aligned}$$

Again integrating from 0 to x ,

$$\begin{aligned}
\int_0^x \left(\frac{d}{dx} v(x) - v_0 \right) dx &= \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}}_M(x) \\
v(x) \Big|_0^x - v_0 x \Big|_0^x &= \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}}_M(x) \\
v(x) - v(0) - v_1 x &= \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}}_M(x) \\
v(x) - v_0 - v_1 x &= \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}}_M(x) \\
v(x) &= \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}}_M(x) + v_0 + v_1 x \\
v(x) &= \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}}_M(x) + G_2 \hat{\mathbf{u}}_M(x)
\end{aligned} \tag{4.34}$$

where $\hat{\mathbf{L}}$ is the unknown Shifted Legendre coefficient vector, $\mathbf{O}^2$ is the second order operational matrix of integration. G_2 is the Shifted Legendre coefficient vector for $v_0 + v_1 x$.

$$\frac{d}{dx} u(x) = \hat{\mathbf{K}} \mathbf{O}^2 \mathbf{D} \hat{\mathbf{u}}_M(x) + G_1 \mathbf{D} \hat{\mathbf{u}}_M(x) \tag{4.35}$$

$$\frac{d}{dx} v(x) = \hat{\mathbf{L}} \mathbf{O}^2 \mathbf{D} \hat{\mathbf{u}}_M(x) + G_2 \mathbf{D} \hat{\mathbf{u}}_M(x) \tag{4.36}$$

Now we substitute the results from (4.31), (4.32), (4.33), (4.34), (4.35) and (4.36) in (4.29) and (4.30).

$$\begin{aligned}
\hat{\mathbf{K}} \hat{\mathbf{u}}_M(x) &= a_0 (\hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{u}} + G_1 \hat{\mathbf{u}}) + a_1 (\hat{\mathbf{K}} \mathbf{O}^2 \mathbf{D} \hat{\mathbf{u}} + G_1 \mathbf{D} \hat{\mathbf{u}}) + a_2 (\hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}} + G_2 \hat{\mathbf{u}}) \\
&\quad + a_3 (\hat{\mathbf{L}} \mathbf{O}^2 \mathbf{D} \hat{\mathbf{u}} + G_2 \mathbf{D} \hat{\mathbf{u}}) + H_1 \hat{\mathbf{u}} \\
\hat{\mathbf{K}} \hat{\mathbf{u}}_M(x) &= a_0 \hat{\mathbf{K}} \mathbf{O}^2 \hat{\mathbf{u}} + a_0 G_1 \hat{\mathbf{u}} + a_1 \hat{\mathbf{K}} \mathbf{O}^2 \mathbf{D} \hat{\mathbf{u}} + a_1 G_1 \mathbf{D} \hat{\mathbf{u}} + a_2 \hat{\mathbf{L}} \mathbf{O}^2 \hat{\mathbf{u}} + a_2 G_2 \hat{\mathbf{u}} \\
&\quad + a_3 \hat{\mathbf{L}} \mathbf{O}^2 \mathbf{D} \hat{\mathbf{u}} + a_3 G_2 \mathbf{D} \hat{\mathbf{u}} + H_1 \hat{\mathbf{u}} \\
\hat{\mathbf{K}} \hat{\mathbf{u}}_M(x) &= \left(a_0 \hat{\mathbf{K}} \mathbf{O}^2 + a_0 G_1 + a_1 \hat{\mathbf{K}} \mathbf{O}^2 \mathbf{D} + a_1 G_1 \mathbf{D} + a_2 \hat{\mathbf{L}} \mathbf{O}^2 + a_2 G_2 \right. \\
&\quad \left. + a_3 \hat{\mathbf{L}} \mathbf{O}^2 \mathbf{D} + a_3 G_2 \mathbf{D} + H_1 \right) \hat{\mathbf{u}}
\end{aligned}$$

Cancelling the common terms

$$\hat{\mathbf{K}} = a_0 \hat{\mathbf{K}} \mathbf{O}^2 + a_0 G_1 + a_1 \hat{\mathbf{K}} \mathbf{O}^2 \mathbf{D} + a_1 G_1 \mathbf{D} + a_2 \hat{\mathbf{L}} \mathbf{O}^2 + a_2 G_2 + a_3 \hat{\mathbf{L}} \mathbf{O}^2 \mathbf{D} + a_3 G_2 \mathbf{D} + H_1 \tag{4.37}$$

$$\begin{aligned}
\hat{\mathbf{L}}\hat{\mathbf{M}}(x) &= b_0(\hat{\mathbf{K}}\mathbf{O}^2\hat{\mathbf{M}} + G_1\hat{\mathbf{M}}) + b_1(\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D}\hat{\mathbf{M}} + G_1\mathbf{D}\hat{\mathbf{M}}) + b_2(\hat{\mathbf{L}}\mathbf{O}^2\hat{\mathbf{M}} + G_2\hat{\mathbf{M}}) \\
&\quad + b_3(\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D}\hat{\mathbf{M}} + G_2\mathbf{D}\hat{\mathbf{M}}) + H_2\hat{\mathbf{M}} \\
\hat{\mathbf{L}}\hat{\mathbf{M}}(x) &= b_0\hat{\mathbf{K}}\mathbf{O}^2\hat{\mathbf{M}} + b_0G_1\hat{\mathbf{M}} + b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D}\hat{\mathbf{M}} + b_1G_1\mathbf{D}\hat{\mathbf{M}} + b_2\hat{\mathbf{L}}\mathbf{O}^2\hat{\mathbf{M}} + b_2G_2\hat{\mathbf{M}} \\
&\quad + b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D}\hat{\mathbf{M}} + b_3G_2\mathbf{D}\hat{\mathbf{M}} + H_2\hat{\mathbf{M}} \\
\hat{\mathbf{L}}\hat{\mathbf{M}}(x) &= \left(b_0\hat{\mathbf{K}}\mathbf{O}^2 + b_0G_1 + b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} + b_1G_1\mathbf{D} + b_2\hat{\mathbf{L}}\mathbf{O}^2 + b_2G_2 \right. \\
&\quad \left. + b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} + b_3G_2\mathbf{D} + H_2 \right) \hat{\mathbf{M}}
\end{aligned}$$

cancelling the common terms

$$\hat{\mathbf{L}} = b_0\hat{\mathbf{K}}\mathbf{O}^2 + b_0G_1 + b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} + b_1G_1\mathbf{D} + b_2\hat{\mathbf{L}}\mathbf{O}^2 + b_2G_2 + b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} + b_3G_2\mathbf{D} + H_2 \quad (4.38)$$

we express the (4.37) and (4.38) in matrix form as,

$$\begin{bmatrix} \hat{\mathbf{K}} \\ \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} a_0\hat{\mathbf{K}}\mathbf{O}^2 \\ b_2\hat{\mathbf{L}}\mathbf{O}^2 \end{bmatrix} + \begin{bmatrix} a_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} \\ b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} a_2\hat{\mathbf{L}}\mathbf{O}^2 \\ b_0\hat{\mathbf{K}}\mathbf{O}^2 \end{bmatrix} + \begin{bmatrix} a_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} \\ b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} a_0G_1 + a_1G_1\mathbf{D} + a_2G_2 + a_3G_2\mathbf{D} + H_1 \\ b_0G_1 + b_1G_1\mathbf{D} + b_2G_2 + b_3G_2\mathbf{D} + H_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} \\ \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} a_0\hat{\mathbf{K}}\mathbf{O}^2 \\ b_2\hat{\mathbf{L}}\mathbf{O}^2 \end{bmatrix} + \begin{bmatrix} a_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} \\ b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} a_2\hat{\mathbf{L}}\mathbf{O}^2 \\ b_0\hat{\mathbf{K}}\mathbf{O}^2 \end{bmatrix} + \begin{bmatrix} a_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} \\ b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} A \\ B \end{bmatrix}$$

Taking transpose we get,

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} a_0\hat{\mathbf{K}}\mathbf{O}^2 & b_2\hat{\mathbf{L}}\mathbf{O}^2 \end{bmatrix} + \begin{bmatrix} a_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} & b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} a_2\hat{\mathbf{L}}\mathbf{O}^2 & b_0\hat{\mathbf{K}}\mathbf{O}^2 \end{bmatrix} + \begin{bmatrix} a_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} & b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} A & B \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} a_0\hat{\mathbf{K}}\mathbf{O}^2 + a_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} & b_2\hat{\mathbf{L}}\mathbf{O}^2 + b_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} a_2\hat{\mathbf{L}}\mathbf{O}^2 + a_3\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} & b_0\hat{\mathbf{K}}\mathbf{O}^2 + b_1\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} A & B \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} a_0\mathbf{O}^2 + a_1\mathbf{O}^2\mathbf{D} & 0 \\ 0 & b_2\mathbf{O}^2 + b_3\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} 0 & b_0\mathbf{O}^2 + b_1\mathbf{O}^2\mathbf{D} \\ a_2\mathbf{O}^2 + a_3\mathbf{O}^2\mathbf{D} & 0 \end{bmatrix} + \begin{bmatrix} A & B \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \begin{bmatrix} a_0\mathbf{O}^2 + a_1\mathbf{O}^2\mathbf{D} & b_0\mathbf{O}^2 + b_1\mathbf{O}^2\mathbf{D} \\ a_2\mathbf{O}^2 + a_3\mathbf{O}^2\mathbf{D} & b_2\mathbf{O}^2 + b_3\mathbf{O}^2\mathbf{D} \end{bmatrix} + \begin{bmatrix} A & B \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \left(\begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix} - \begin{bmatrix} a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D} & b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D} \\ a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D} & b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D} \end{bmatrix} \right) = \begin{bmatrix} A & B \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} \left(\begin{bmatrix} I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D}) & -(b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D}) \\ -(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D}) & I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D}) \end{bmatrix} \right) = \begin{bmatrix} A & B \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D}) & -(b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D}) \\ -(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D}) & I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D}) \end{bmatrix}^{-1}$$

$$\blacksquare = \begin{bmatrix} I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D}) & -(b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D}) \\ -(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D}) & I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D}) \end{bmatrix}$$

$$(\blacksquare)^{-1} = \frac{1}{\det \blacksquare} \text{adj}(\blacksquare)$$

$$\text{adj}(\blacksquare) = \begin{bmatrix} I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D}) & b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D} \\ a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D} & I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D}) \end{bmatrix}$$

$$(\blacksquare)^{-1} = \begin{bmatrix} \frac{I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} & \frac{(b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} \\ \frac{(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} & \frac{I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} \frac{I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} & \frac{(b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} \\ \frac{(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} & \frac{I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} \frac{A(I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D})) + B(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare} & \frac{A(b_0 \mathbf{O}^2 + b_1 \mathbf{O}^2 \mathbf{D}) + B(I - (a_0 \mathbf{O}^2 + a_1 \mathbf{O}^2 \mathbf{D}))}{\det \blacksquare} \end{bmatrix}$$

$$\hat{\mathbf{K}} = \frac{A(I - (b_2 \mathbf{O}^2 + b_3 \mathbf{O}^2 \mathbf{D})) + B(a_2 \mathbf{O}^2 + a_3 \mathbf{O}^2 \mathbf{D})}{\det \blacksquare}$$

$$\hat{\mathbf{L}} = \frac{A(b_0\mathbf{O}^2 + b_1\mathbf{O}^2\mathbf{D}) + B(I - (a_0\mathbf{O}^2 + a_1\mathbf{O}^2\mathbf{D}))}{\det\mathbf{L}}$$

by putting the obtained results in the (4.33) and (4.34) we get the solution for the second order system of equations.

4.2.3 Third-order Coupled Systems

Consider the following system of equations of third order.

$$\frac{d^3}{dx^3}u(x) = a_0u(x) + a_1\frac{d}{dx}u(x) + a_2\frac{d^2}{dx^2}u(x) + a_3v(x) + a_4\frac{d}{dx}v(x) + a_5\frac{d^2}{dx^2}v(x) + f(x) \quad (4.39)$$

$$\frac{d^3}{dx^3}v(x) = a_0u(x) + a_1\frac{d}{dx}u(x) + a_2\frac{d^2}{dx^2}u(x) + a_3v(x) + a_4\frac{d}{dx}v(x) + a_5\frac{d^2}{dx^2}v(x) + g(x) \quad (4.40)$$

$$u(0) = u_0, \quad \frac{d}{dx}u(0) = u_1, \quad \frac{d^2}{dx^2}u(0) = u_2$$

and

$$v(0) = v_0, \quad \frac{d}{dx}v(0) = v_1, \quad \frac{d^2}{dx^2}v(0) = v_2$$

we express the highest order derivative in terms of Shifted Legendre Polynomials.

$$\frac{d^3}{dx^3}u(x) = \hat{\mathbf{K}}\mathbf{O} \quad (4.41)$$

$$\frac{d^3}{dx^3}v(x) = \hat{\mathbf{L}}\mathbf{O} \quad (4.42)$$

Integrating (4.41) from 0 to x we get

$$\begin{aligned} \int_0^x \frac{d^3}{dx^3}u(x)dx &= \hat{\mathbf{K}}\mathbf{O} \\ \frac{d^2}{dx^2}u(x)\Big|_0^x &= \hat{\mathbf{K}}\mathbf{O} \\ \frac{d^2}{dx^2}u(x) - \frac{d^2}{dx^2}u(0) &= \hat{\mathbf{K}}\mathbf{O} \\ \frac{d^2}{dx^2}u(x) - u_2 &= \hat{\mathbf{K}}\mathbf{O} \end{aligned}$$

Again integrating the above equation from 0 to x we get

$$\begin{aligned}
\int_0^x \left(\frac{d^2}{dx^2} u(x) - u_2 \right) dx &= \hat{\mathbf{K}} \mathbf{O}^2 \blacksquare \\
\left. \frac{d}{dx} u(x) \right|_0^x - u_2 x \Big|_0^x &= \hat{\mathbf{K}} \mathbf{O}^2 \blacksquare \\
\frac{d}{dx} u(x) - \frac{d}{dx} u(0) - u_2 x &= \hat{\mathbf{K}} \mathbf{O}^2 \blacksquare \\
\frac{d}{dx} u(x) - u_1 - u_2 x &= \hat{\mathbf{K}} \mathbf{O}^2 \blacksquare
\end{aligned}$$

Again integrating the above equation from 0 to x we get

$$\begin{aligned}
\int_0^x \left(\frac{d}{dx} u(x) - u_1 - u_2 x \right) dx &= \hat{\mathbf{K}} \mathbf{O}^3 \blacksquare \\
u(x) \Big|_0^x - u_1 x \Big|_0^x - u_2 \frac{x^2}{2} \Big|_0^x &= \hat{\mathbf{K}} \mathbf{O}^3 \blacksquare \\
u(x) - u(0) - u_1 x - u_2 \frac{x^2}{2} &= \hat{\mathbf{K}} \mathbf{O}^3 \blacksquare \\
u(x) - u_0 - u_1 x - u_2 \frac{x^2}{2} &= \hat{\mathbf{K}} \mathbf{O}^3 \blacksquare \\
u(x) &= \hat{\mathbf{K}} \mathbf{O}^3 \blacksquare + u_0 + u_1 x + u_2 \frac{x^2}{2}
\end{aligned}$$

$$u(x) = \hat{\mathbf{K}} \mathbf{O}^3 \blacksquare + G_1 \blacksquare \quad (4.43)$$

where $\hat{\mathbf{K}}$ is the unknown Shifted Legendre coefficient, $\mathbf{O}^3$ is the operational matrix of integration of third order, $\blacksquare$ is the function vector of Shifted Legendre Polynomials and G_1 is the shifted legendre coefficient for $u_0 + u_1 x + u_2 \frac{x^2}{2}$.

Now integrating (4.42) from 0 to x we get

$$\begin{aligned}
\int_0^x \frac{d^3}{dx^3} v(x) dx &= \hat{\mathbf{L}} \mathbf{O} \blacksquare \\
\left. \frac{d^2}{dx^2} v(x) \right|_0^x &= \hat{\mathbf{L}} \mathbf{O} \blacksquare \\
\frac{d^2}{dx^2} v(x) - \frac{d^2}{dx^2} v(0) &= \hat{\mathbf{L}} \mathbf{O} \blacksquare \\
\frac{d^2}{dx^2} v(x) - v_2 &= \hat{\mathbf{L}} \mathbf{O} \blacksquare
\end{aligned}$$

Again integrating the above equation from 0 to x we get

$$\begin{aligned}\int_0^x \left(\frac{d^2}{dx^2} v(x) - v_2 \right) dx &= \hat{\mathbf{L}} \mathbf{O}^2 \blacksquare \\ \frac{d}{dx} v(x) \Big|_0^x - v_2 x \Big|_0^x &= \hat{\mathbf{L}} \mathbf{O}^2 \blacksquare \\ \frac{d}{dx} v(x) - \frac{d}{dx} v(0) - v_2 x &= \hat{\mathbf{L}} \mathbf{O}^2 \blacksquare \\ \frac{d}{dx} v(x) - v_1 - v_2 x &= \hat{\mathbf{L}} \mathbf{O}^2 \blacksquare\end{aligned}$$

Again integrating the above equation from 0 to x we get

$$\begin{aligned}\int_0^x \left(\frac{d}{dx} v(x) - v_1 - v_2 x \right) dx &= \hat{\mathbf{L}} \mathbf{O}^3 \blacksquare \\ v(x) \Big|_0^x - v_1 x \Big|_0^x - v_2 \frac{x^2}{2} \Big|_0^x &= \hat{\mathbf{L}} \mathbf{O}^3 \blacksquare \\ v(x) - v(0) - v_1 x - v_2 \frac{x^2}{2} &= \hat{\mathbf{L}} \mathbf{O}^3 \blacksquare \\ v(x) - v_0 - v_1 x - v_2 \frac{x^2}{2} &= \hat{\mathbf{L}} \mathbf{O}^3 \blacksquare \\ v(x) &= \hat{\mathbf{L}} \mathbf{O}^3 \blacksquare + v_0 + v_1 x + v_2 \frac{x^2}{2}\end{aligned}$$

$$v(x) = \hat{\mathbf{L}} \mathbf{O}^3 \blacksquare + G_2 \blacksquare \quad (4.44)$$

where $\hat{\mathbf{L}}$ is the unknown Shifted Legendre coefficient and G_2 is the shifted legendre coefficient for $v_0 + v_1 x + v_2 \frac{x^2}{2}$. we approximate the substituting the obtained results in (4.39) we get

$$\begin{aligned}\hat{\mathbf{K}} \blacksquare &= a_0 (\hat{\mathbf{K}} \mathbf{O}^3 \blacksquare + \mathbf{G}_1 \blacksquare) + a_1 (\hat{\mathbf{K}} \mathbf{O}^3 \mathbf{D} \blacksquare + \mathbf{G}_1 \mathbf{D} \blacksquare) + a_2 (\hat{\mathbf{K}} \mathbf{O}^3 \mathbf{D}^2 \blacksquare + \mathbf{G}_1 \mathbf{D}^2 \blacksquare) \\ &+ a_3 (\hat{\mathbf{L}} \mathbf{O}^3 \blacksquare + \mathbf{G}_2 \blacksquare) + a_4 (\hat{\mathbf{L}} \mathbf{O}^3 \mathbf{D} \blacksquare + \mathbf{G}_2 \mathbf{D} \blacksquare) + a_5 (\hat{\mathbf{L}} \mathbf{O}^3 \mathbf{D}^2 \blacksquare + \mathbf{G}_2 \mathbf{D}^2 \blacksquare) \\ &+ \mathbf{H}_1 \blacksquare.\end{aligned}$$

$$\begin{aligned}\hat{\mathbf{K}} \hat{\blacksquare} &= \hat{\blacksquare} (a_0 (\hat{\mathbf{K}} \mathbf{O}^3 + \mathbf{G}_1) + a_1 (\hat{\mathbf{K}} \mathbf{O}^3 \mathbf{D} + \mathbf{G}_1 \mathbf{D}) + a_2 (\hat{\mathbf{K}} \mathbf{O}^3 \mathbf{D}^2 + \mathbf{G}_1 \mathbf{D}^2) + a_3 (\hat{\mathbf{L}} \mathbf{O}^3 + \mathbf{G}_2) \\ &+ a_4 (\hat{\mathbf{L}} \mathbf{O}^3 \mathbf{D} + \mathbf{G}_2 \mathbf{D}) + a_5 (\hat{\mathbf{L}} \mathbf{O}^3 \mathbf{D}^2 + \mathbf{G}_2 \mathbf{D}^2) + \mathbf{H}_1).\end{aligned}$$

cancelling common terms

$$\hat{\mathbf{K}} = (a_0(\hat{\mathbf{K}}\mathbf{O}^3 + \mathbf{G}_1) + a_1(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + \mathbf{G}_1\mathbf{D}) + a_2(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + \mathbf{G}_1\mathbf{D}^2) + a_3(\hat{\mathbf{L}}\mathbf{O}^3 + \mathbf{G}_2) + a_4(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + \mathbf{G}_2\mathbf{D}) + a_5(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + \mathbf{G}_2\mathbf{D}^2) + \mathbf{H}_1).$$

$$\hat{\mathbf{K}} = a_0\hat{\mathbf{K}}\mathbf{O}^3 + a_0\mathbf{G}_1 + a_1\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + a_1\mathbf{G}_1\mathbf{D} + a_2\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + a_2\mathbf{G}_1\mathbf{D}^2 + a_3\hat{\mathbf{L}}\mathbf{O}^3 + a_3\mathbf{G}_2 + a_4\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + a_4\mathbf{G}_2\mathbf{D} + a_5\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + a_5\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_1. \quad (4.45)$$

substituting the obtained results in (4.40) we get

$$\begin{aligned} \hat{\mathbf{L}}_{\blacksquare} &= b_0(\hat{\mathbf{K}}\mathbf{O}^3_{\blacksquare} + \mathbf{G}_{1\blacksquare}) + b_1(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}_{\blacksquare} + \mathbf{G}_1\mathbf{D}_{\blacksquare}) + b_2(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2_{\blacksquare} + \mathbf{G}_1\mathbf{D}^2_{\blacksquare}) \\ &\quad + b_3(\hat{\mathbf{L}}\mathbf{O}^3_{\blacksquare} + \mathbf{G}_{2\blacksquare}) + b_4(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}_{\blacksquare} + \mathbf{G}_2\mathbf{D}_{\blacksquare}) + b_5(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2_{\blacksquare} + \mathbf{G}_2\mathbf{D}^2_{\blacksquare}) \\ &\quad + \mathbf{H}_{2\blacksquare}. \end{aligned}$$

$$\hat{\mathbf{L}}_{\blacksquare} = \hat{\blacksquare}(b_0(\hat{\mathbf{K}}\mathbf{O}^3 + \mathbf{G}_1) + b_1(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + \mathbf{G}_1\mathbf{D}) + b_2(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + \mathbf{G}_1\mathbf{D}^2) + b_3(\hat{\mathbf{L}}\mathbf{O}^3 + \mathbf{G}_2) + b_4(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + \mathbf{G}_2\mathbf{D}) + b_5(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + \mathbf{G}_2\mathbf{D}^2) + \mathbf{H}_2).$$

$$\hat{\mathbf{L}} = (b_0(\hat{\mathbf{K}}\mathbf{O}^3 + \mathbf{G}_1) + b_1(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + \mathbf{G}_1\mathbf{D}) + b_2(\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + \mathbf{G}_1\mathbf{D}^2) + b_3(\hat{\mathbf{L}}\mathbf{O}^3 + \mathbf{G}_2) + b_4(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + \mathbf{G}_2\mathbf{D}) + b_5(\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + \mathbf{G}_2\mathbf{D}^2) + \mathbf{H}_2).$$

$$\begin{aligned} \hat{\mathbf{L}} &= b_0\hat{\mathbf{K}}\mathbf{O}^3 + b_0\mathbf{G}_1 + b_1\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + b_1\mathbf{G}_1\mathbf{D} + b_2\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + b_2\mathbf{G}_1\mathbf{D}^2 + b_3\hat{\mathbf{L}}\mathbf{O}^3 + b_3\mathbf{G}_2 \\ &\quad + b_4\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + b_4\mathbf{G}_2\mathbf{D} + b_5\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + b_5\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_2. \end{aligned} \quad (4.46)$$

we express (4.45) and (4.46) in matrix form as,

$$\begin{aligned} \begin{bmatrix} \hat{\mathbf{K}} \\ \hat{\mathbf{L}} \end{bmatrix} &= \begin{bmatrix} a_0\hat{\mathbf{K}}\mathbf{O}^3 + a_1\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + a_2\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 \\ b_3\hat{\mathbf{L}}\mathbf{O}^3 + b_4\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + b_5\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 \end{bmatrix} + \begin{bmatrix} a_3\hat{\mathbf{L}}\mathbf{O}^3 + a_4\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + a_5\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 \\ b_0\hat{\mathbf{K}}\mathbf{O}^3 + b_1\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + b_2\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 \end{bmatrix} \\ &\quad + \begin{bmatrix} a_0\mathbf{G}_1 + a_1\mathbf{G}_1\mathbf{D} + a_2\mathbf{G}_1\mathbf{D}^2 + a_3\mathbf{G}_2 + a_4\mathbf{G}_2\mathbf{D} + a_5\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_1 \\ b_0\mathbf{G}_1 + b_1\mathbf{G}_1\mathbf{D} + b_2\mathbf{G}_1\mathbf{D}^2 + b_3\mathbf{G}_2 + b_4\mathbf{G}_2\mathbf{D} + b_5\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_2 \end{bmatrix} \end{aligned}$$

$$A = a_0\mathbf{G}_1 + a_1\mathbf{G}_1\mathbf{D} + a_2\mathbf{G}_1\mathbf{D}^2 + a_3\mathbf{G}_2 + a_4\mathbf{G}_2\mathbf{D} + a_5\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_1$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} I - (a_0 \mathbf{O}^3 + a_1 \mathbf{O}^3 \mathbf{D} + a_2 \mathbf{O}^3 \mathbf{D}^2) & -(b_0 \mathbf{O}^3 + b_1 \mathbf{O}^3 \mathbf{D} + b_2 \mathbf{O}^3 \mathbf{D}^2) \\ -(a_3 \mathbf{O}^3 + a_4 \mathbf{O}^3 \mathbf{D} + a_5 \mathbf{O}^3 \mathbf{D}^2) & I - (b_3 \mathbf{O}^3 + b_4 \mathbf{O}^3 \mathbf{D} + b_5 \mathbf{O}^3 \mathbf{D}^2) \end{bmatrix}^{-1}$$

$$\blacksquare = \begin{bmatrix} I - (a_0 \mathbf{O}^3 + a_1 \mathbf{O}^3 \mathbf{D} + a_2 \mathbf{O}^3 \mathbf{D}^2) & -(b_0 \mathbf{O}^3 + b_1 \mathbf{O}^3 \mathbf{D} + b_2 \mathbf{O}^3 \mathbf{D}^2) \\ -(a_3 \mathbf{O}^3 + a_4 \mathbf{O}^3 \mathbf{D} + a_5 \mathbf{O}^3 \mathbf{D}^2) & I - (b_3 \mathbf{O}^3 + b_4 \mathbf{O}^3 \mathbf{D} + b_5 \mathbf{O}^3 \mathbf{D}^2) \end{bmatrix}^{-1}$$

$$(\blacksquare)^{-1} = \frac{1}{\det \blacksquare} \text{adj}(\blacksquare)$$

$$\text{adj}(\blacksquare) = \begin{bmatrix} I - (b_3 \mathbf{O}^3 + b_4 \mathbf{O}^3 \mathbf{D} + b_5 \mathbf{O}^3 \mathbf{D}^2) & (b_0 \mathbf{O}^3 + b_1 \mathbf{O}^3 \mathbf{D} + b_2 \mathbf{O}^3 \mathbf{D}^2) \\ (a_3 \mathbf{O}^3 + a_4 \mathbf{O}^3 \mathbf{D} + a_5 \mathbf{O}^3 \mathbf{D}^2) & I - (a_0 \mathbf{O}^3 + a_1 \mathbf{O}^3 \mathbf{D} + a_2 \mathbf{O}^3 \mathbf{D}^2) \end{bmatrix}$$

$$(\blacksquare)^{-1} = \begin{bmatrix} \frac{I - (b_3 \mathbf{O}^3 + b_4 \mathbf{O}^3 \mathbf{D} + b_5 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} & \frac{(b_0 \mathbf{O}^3 + b_1 \mathbf{O}^3 \mathbf{D} + b_2 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} \\ \frac{(a_3 \mathbf{O}^3 + a_4 \mathbf{O}^3 \mathbf{D} + a_5 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} & \frac{I - (a_0 \mathbf{O}^3 + a_1 \mathbf{O}^3 \mathbf{D} + a_2 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} \frac{I - (b_3 \mathbf{O}^3 + b_4 \mathbf{O}^3 \mathbf{D} + b_5 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} & \frac{(b_0 \mathbf{O}^3 + b_1 \mathbf{O}^3 \mathbf{D} + b_2 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} \\ \frac{(a_3 \mathbf{O}^3 + a_4 \mathbf{O}^3 \mathbf{D} + a_5 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} & \frac{I - (a_0 \mathbf{O}^3 + a_1 \mathbf{O}^3 \mathbf{D} + a_2 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare} \end{bmatrix}$$

After Multiplying the above matrices we get the following results.

$$\hat{\mathbf{K}} = \frac{A \left(I - (b_3 \mathbf{O}^3 + b_4 \mathbf{O}^3 \mathbf{D} + b_5 \mathbf{O}^3 \mathbf{D}^2) \right) + B(a_3 \mathbf{O}^3 + a_4 \mathbf{O}^3 \mathbf{D} + a_5 \mathbf{O}^3 \mathbf{D}^2)}{\det \blacksquare}$$

$$\hat{\mathbf{L}} = \frac{A(b_0 \mathbf{O}^3 + b_1 \mathbf{O}^3 \mathbf{D} + b_2 \mathbf{O}^3 \mathbf{D}^2) + B \left(I - (a_0 \mathbf{O}^3 + a_1 \mathbf{O}^3 \mathbf{D} + a_2 \mathbf{O}^3 \mathbf{D}^2) \right)}{\det \blacksquare}$$

4.3 Partial Differential Equations(PDEs)

4.3.1 Generalized Heat Conduction Equation

Consider the following generalized heat conduction equation

$$u_t = a_0 u_{xx} + a_1 u(x, t) + \phi(x, t) \quad (4.47)$$

$$u(0, t) = g_1(t),$$

$$u(1, t) = g_2(t),$$

$$u(x, 0) = f(x)$$

let $u_t = \hat{\mathbf{K}}\hat{\mathbf{u}}$ integrating w.r.t t ,

$$u - u(x, 0) = \hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{u}}$$

$$u = \hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{u}} + u(x, 0)$$

$$u = \hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{u}} + f(x)$$

we approximate $f(x)$,

$$f(x) = G_1\hat{\mathbf{u}}$$

$$\boxed{u = \hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{u}} + G_1\hat{\mathbf{u}}} \quad (4.48)$$

$$a_0 u_{xx} = u_t - a_1 u - \phi(x, t)$$

$$a_0 u_{xx} = \hat{\mathbf{K}}\hat{\mathbf{u}} - a_1 (\hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{u}} + G_1\hat{\mathbf{u}}) - G_2\hat{\mathbf{u}}$$

$$u_{xx} = \frac{1}{a_0} \left(\hat{\mathbf{K}}\hat{\mathbf{u}} - a_1 (\hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{u}} + G_1\hat{\mathbf{u}}) - G_2\hat{\mathbf{u}} \right)$$

$$u_{xx} = \frac{1}{a_0} \left(\hat{\mathbf{K}} - a_1 \hat{\mathbf{K}}\mathbf{O}_t - a_1 G_1 - G_2 \right) \hat{\mathbf{u}}$$

let

$$W = \frac{1}{a_0} \left(\hat{\mathbf{K}} - a_1 \hat{\mathbf{K}}\mathbf{O}_t - a_1 G_1 - G_2 \right),$$

$$u_{xx} = W\hat{\mathbf{u}}$$

double integrating above equation w.r.t x on both sides we get,

$$u = W\mathbf{O}_x^2\hat{\mathbf{u}} + d_0 + xd_1 \quad (4.49)$$

using first boundary condition.

$$\begin{aligned} u(0, t) &= W\mathbf{O}_x^2\hat{\mathbf{n}}(0, t) + d_0 \\ d_0 &= u(0, t) \\ d_0 &= g_1(t) \end{aligned}$$

putting d_0 in (4.49)

$$u = W\mathbf{O}_x^2\hat{\mathbf{n}} + g_1(t) + xd_1$$

using second boundary condition,

$$\begin{aligned} u(1, t) &= W\mathbf{O}_x^2\hat{\mathbf{n}}(1, t) + g_1(t) + d_1 \\ g_2(t) &= W\mathbf{O}_x^2\hat{\mathbf{n}}(1, t) + g_1(t) + d_1 \\ d_1 &= g_2(t) - W\mathbf{O}_x^2\hat{\mathbf{n}}(1, t) - g_1(t) \end{aligned}$$

putting d_1 in (4.49) we get,

$$u = W\mathbf{O}_x^2\hat{\mathbf{n}} + g_1(t) + xg_2(t) - xg_1(t) - W\mathbf{O}_x^2\hat{\mathbf{n}}(1, t)$$

approximating $g_1(t) + xg_2(t) - xg_1(t)$,

$$g_1(t) + xg_2(t) - xg_1(t) = H_1\hat{\mathbf{n}}$$

$$u = W\mathbf{O}_x^2\hat{\mathbf{n}} + H_1\hat{\mathbf{n}} - W\mathbf{O}_x^2x\hat{\mathbf{n}}(1, t) \quad (4.50)$$

Now by comparing (4.48) and (4.50)

$$\begin{aligned} \hat{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{n}} + G_1\hat{\mathbf{n}} &= W\mathbf{O}_x^2\hat{\mathbf{n}} + H_1\hat{\mathbf{n}} - W\mathbf{O}_x^2x\hat{\mathbf{n}}(1, t) \\ \hat{\mathbf{n}}(\hat{\mathbf{K}}\mathbf{O}_t + G_1) &= W\mathbf{O}_x^2(\hat{\mathbf{n}} - x\hat{\mathbf{n}}(1, t)) + H_1\hat{\mathbf{n}} \\ \hat{\mathbf{n}}(\hat{\mathbf{K}}\mathbf{O}_t + G_1) &= W\mathbf{O}_x^2(\hat{\mathbf{n}} - A^{(1,1,1)}\hat{\mathbf{n}}) + H_1\hat{\mathbf{n}} \\ \hat{\mathbf{K}}\mathbf{O}_t + G_1 &= W\mathbf{O}_x^2(I - A^{(1,1,1)}) + H_1 \end{aligned}$$

let $\mathbf{Q} = \mathbf{O}_x^2(I - A^{(1,1,1)})$

$$\begin{aligned}\hat{\mathbf{K}}\mathbf{O}_t + G_1 &= \frac{1}{a_0} \left(\hat{\mathbf{K}} - a_1 \hat{\mathbf{K}}\mathbf{O}_t - a_1 G_1 - G_2 \right) \mathbf{Q} + H_1 \\ \hat{\mathbf{K}}\mathbf{O}_t + G_1 &= \frac{1}{a_0} \left(\hat{\mathbf{K}}(I - a_1 \mathbf{O}_t) - a_1 G_1 - G_2 \right) \mathbf{Q} + H_1 \\ \hat{\mathbf{K}}\mathbf{O}_t + G_1 &= \frac{1}{a_0} \left(\hat{\mathbf{K}}(I - a_1 \mathbf{O}_t) - a_1 G_1 - G_2 \right) \mathbf{Q} + H_1 \\ \hat{\mathbf{K}}\mathbf{O}_t + G_1 &= \frac{1}{a_0} \left(\hat{\mathbf{K}}\mathbf{Q}(I - a_1 \mathbf{O}_t) - \mathbf{Q}(a_1 G_1 + G_2) \right) + H_1\end{aligned}$$

$$\hat{\mathbf{K}}\mathbf{O}_t - \frac{1}{a_0} \left(\hat{\mathbf{K}}\mathbf{Q}(I - a_1 \mathbf{O}_t) \right) = -\frac{1}{a_0} \left(\mathbf{Q}(a_1 G_1 + G_2) \right) + H_1 - G_1$$

4.3.2 Coupled System of Generalized Heat Conduction Model

Consider the following coupled system of heat conduction equation,

$$\begin{aligned}u_t &= \alpha_1 u_{xx} + \alpha_2 v_{xx} + a_0 u + a_1 v + F(x, t) \\ v_t &= \alpha_3 u_{xx} + \alpha_4 v_{xx} + b_0 u + b_1 v + G(x, t)\end{aligned}\tag{4.51}$$

Initial and boundary conditions are given by,

$$u(x, 0) = g_1(x) \quad v(x, 0) = g_2(x)$$

,

$$u(0, t) = f_1(t) \quad v(0, t) = f_3(t)$$

,

$$u(1, t) = f_2(t) \quad v(1, t) = f_4(t)$$

let $u_t = \tilde{\mathbf{K}}\hat{\mathbf{\Pi}}$ and $v_t = \tilde{\mathbf{L}}\hat{\mathbf{\Pi}}$

after integrating the above expressions w.r.t t we get,

$$u = \tilde{\mathbf{K}}\mathbf{O}_t\hat{\mathbf{\Pi}} + G_1\hat{\mathbf{\Pi}}\tag{4.52}$$

taking double derivative of the (4.52),

$$u_{xx} = \tilde{\mathbf{K}}\mathbf{O}_t\mathbf{D}^2\hat{\mathbf{\Pi}} + G_1\mathbf{D}^2\hat{\mathbf{\Pi}}$$

$$v = \tilde{\mathbf{L}}\mathbf{O}\hat{\mathbf{\Pi}} + G_2\hat{\mathbf{\Pi}}\tag{4.53}$$

After taking double derivative of the (4.52) we get,

$$v_{xx} = \bar{\mathbf{L}}\mathbf{O}\mathbf{D}^2\hat{\mathbf{n}} + G_2\mathbf{D}^2\hat{\mathbf{n}}$$

from (4.51),

$$\alpha_1 u_{xx} = u_t - \alpha_2 v_{xx} - a_0 u - a_1 v - F(x, t)$$

$$u_{xx} = \frac{1}{\alpha_1} (u_t - \alpha_2 v_{xx} - a_0 u - a_1 v - F(x, t))$$

$$\alpha_4 v_{xx} = v_t - \alpha_3 u_{xx} - b_0 u - b_1 v - G(x, t)$$

$$v_{xx} = \frac{1}{\alpha_4} (v_t - \alpha_3 u_{xx} - b_0 u - b_1 v - G(x, t))$$

$$u_{xx} = \frac{1}{\alpha_1} (\bar{\mathbf{K}}\hat{\mathbf{n}} - \alpha_2 \bar{\mathbf{L}}\mathbf{O}\mathbf{D}^2\hat{\mathbf{n}} - \alpha_2 G_2 \mathbf{D}^2\hat{\mathbf{n}} - a_0 \bar{\mathbf{K}}\mathbf{O}\hat{\mathbf{n}} - a_0 G_1 \hat{\mathbf{n}} - a_1 \bar{\mathbf{L}}\mathbf{O}\hat{\mathbf{n}} - a_1 G_2 \hat{\mathbf{n}} - G_3 \hat{\mathbf{n}})$$

$$v_{xx} = \frac{1}{\alpha_4} (\bar{\mathbf{L}}\hat{\mathbf{n}} - \alpha_3 \bar{\mathbf{K}}\mathbf{O}\mathbf{D}^2\hat{\mathbf{n}} - \alpha_3 G_1 \mathbf{D}^2\hat{\mathbf{n}} - b_0 \bar{\mathbf{K}}\mathbf{O}\hat{\mathbf{n}} - b_0 G_1 \hat{\mathbf{n}} - b_1 \bar{\mathbf{L}}\mathbf{O}\hat{\mathbf{n}} - b_1 G_2 \hat{\mathbf{n}} - G_4 \hat{\mathbf{n}})$$

$$u_{xx} = \frac{1}{\alpha_1} (\bar{\mathbf{K}} - \alpha_2 \bar{\mathbf{L}}\mathbf{O}\mathbf{D}^2 - \alpha_2 G_2 \mathbf{D}^2 - a_0 \bar{\mathbf{K}}\mathbf{O} - a_0 G_1 - a_1 \bar{\mathbf{L}}\mathbf{O} - a_1 G_2 - G_3) \hat{\mathbf{n}}$$

$$v_{xx} = \frac{1}{\alpha_4} (\bar{\mathbf{L}} - b_1 \bar{\mathbf{L}}\mathbf{O} - \alpha_3 \bar{\mathbf{K}}\mathbf{O}\mathbf{D}^2 - b_0 \bar{\mathbf{K}}\mathbf{O} - \alpha_3 G_1 \mathbf{D}^2 - b_0 G_1 - b_1 G_2 - G_4) \hat{\mathbf{n}}$$

$$u_{xx} = \frac{1}{\alpha_1} (\bar{\mathbf{K}} - \alpha_2 (\bar{\mathbf{L}}\mathbf{O}\mathbf{D}^2 + G_2 \mathbf{D}^2) - a_0 (\bar{\mathbf{K}}\mathbf{O} + G_1) - a_1 (\bar{\mathbf{L}}\mathbf{O} + G_2) - G_3) \hat{\mathbf{n}}$$

$$v_{xx} = \frac{1}{\alpha_4} (\bar{\mathbf{L}}(I - b_1 \mathbf{O}) - \bar{\mathbf{K}}(\alpha_3 \mathbf{O}\mathbf{D}^2 - b_0 \mathbf{O}) - \alpha_3 G_1 \mathbf{D}^2 - b_0 G_1 - b_1 G_2 - G_4) \hat{\mathbf{n}}$$

$$W = \frac{1}{\alpha_1} (\bar{\mathbf{K}} - \alpha_2 (\bar{\mathbf{L}}\mathbf{O}\mathbf{D}^2 + G_2 \mathbf{D}^2) - a_0 (\bar{\mathbf{K}}\mathbf{O} + G_1) - a_1 (\bar{\mathbf{L}}\mathbf{O} + G_2) - G_3)$$

$$W_1 = \frac{1}{\alpha_4} (\bar{\mathbf{L}}(I - b_1 \mathbf{O}) - \bar{\mathbf{K}}(\alpha_3 \mathbf{O}\mathbf{D}^2 - b_0 \mathbf{O}) - \alpha_3 G_1 \mathbf{D}^2 - b_0 G_1 - b_1 G_2 - G_4)$$

$$u_{xx} = W\hat{\mathbf{n}}$$

$$v_{xx} = W_1\hat{\mathbf{n}}$$

double integrating the above equations we get the following expression,

$$\begin{aligned} u &= W\mathbf{O}_x^2\hat{\mathbf{n}} + d_0 + xd_1 \\ v &= W_1\mathbf{O}_x^2\hat{\mathbf{n}} + e_0 + xe_1 \end{aligned} \tag{4.54}$$

using the boundary conditions $u(0,t) = f_1(t)$ and $v(0,t) = f_3(t)$

$$\begin{aligned} u(0,t) &= W\mathbf{O}_x^2\hat{\mathbf{n}}(0,t) + d_0 + 0 \\ d_0 &= f_1(t) \end{aligned}$$

$$\begin{aligned} v(0,t) &= W_1\mathbf{O}_x^2\hat{\mathbf{n}}(0,t) + e_0 + 0 \\ e_0 &= f_3(t) \end{aligned}$$

Putting d_0 and e_0 in (4.54)

$$\begin{aligned} u &= W\mathbf{O}_x^2\hat{\mathbf{n}} + f_1(t) + xd_1 \\ v &= W_1\mathbf{O}_x^2\hat{\mathbf{n}} + f_3(t) + xe_1 \end{aligned}$$

Now we use the boundary conditions $u(1,t) = f_2(t)$ and $v(1,t) = f_4(t)$

$$\begin{aligned} u(1,t) &= W\mathbf{O}_x^2\hat{\mathbf{n}}(1,t) + f_1(t) + d_1 \\ v(1,t) &= W_1\mathbf{O}_x^2\hat{\mathbf{n}}(1,t) + f_3(t) + e_1 \\ d_1 &= f_2(t) - f_1(t) - W\mathbf{O}_x^2\hat{\mathbf{n}}(1,t) \\ e_1 &= f_4(t) - f_3(t) - W_1\mathbf{O}_x^2\hat{\mathbf{n}}(1,t) \end{aligned}$$

$$\begin{aligned} u &= W\mathbf{O}_x^2\hat{\mathbf{n}} + f_1(t) + xf_2(t) - xf_1(t) - W\mathbf{O}_x^2x\hat{\mathbf{n}}(1,t) \\ v &= W_1\mathbf{O}_x^2\hat{\mathbf{n}} + f_3(t) + xf_4(t) - xf_3(t) - W_1\mathbf{O}_x^2x\hat{\mathbf{n}}(1,t) \end{aligned}$$

we approximate $f_1(t) + xf_2(t) - xf_1(t) = H_1\hat{\mathbf{n}}$, and $f_3(t) + xf_4(t) - xf_3(t) = H_2\hat{\mathbf{n}}$

$$\begin{aligned} u &= W\mathbf{O}_x^2\hat{\mathbf{n}} + H_1\hat{\mathbf{n}} - W\mathbf{O}_x^2\hat{\mathbf{n}}(1,t) \\ v &= W_1\mathbf{O}_x^2\hat{\mathbf{n}} + H_2\hat{\mathbf{n}} - W_1\mathbf{O}_x^2\hat{\mathbf{n}}(1,t) \end{aligned} \tag{4.55}$$

Comparing the (4.55) and (4.52)

$$\begin{aligned} \bar{\mathbf{K}}\mathbf{O}\hat{\mathbf{n}} + G_1\hat{\mathbf{n}} &= W\mathbf{O}_x^2(\hat{\mathbf{n}} - x\hat{\mathbf{n}}(1,t)) + H_1\hat{\mathbf{n}} \\ \bar{\mathbf{L}}\mathbf{O}\hat{\mathbf{n}} + G_2\hat{\mathbf{n}} &= W_1\mathbf{O}_x^2(\hat{\mathbf{n}} - x\hat{\mathbf{n}}(1,t)) + H_2\hat{\mathbf{n}} \end{aligned}$$

cancelling the common terms,

$$\begin{aligned}\bar{\mathbf{K}}\mathbf{O} + G_1 &= W\mathbf{O}_x^2(I - A^{(1,1,1)}) + H_1 \\ \bar{\mathbf{L}}\mathbf{O} + G_2 &= W_1\mathbf{O}_x^2(I - A^{(1,1,1)}) + H_2\end{aligned}$$

let $\mathbf{O}_x^2(I - A^{(1,1,1)}) = Q$ then the above equation becomes,

$$\begin{aligned}\bar{\mathbf{K}}\mathbf{O} + G_1 &= (WQ) + H_1 \\ \bar{\mathbf{L}}\mathbf{O} + G_2 &= (W_1Q) + H_2\end{aligned}$$

$$\bar{\mathbf{K}}\mathbf{O} = \frac{1}{\alpha_1}(\bar{\mathbf{K}}(I - a_0\mathbf{O}) - \bar{\mathbf{L}}(\alpha_2\mathbf{O}\mathbf{D}^2 + a_1\mathbf{O}) - a_0G_1 - \alpha_2G_2\mathbf{D}^2 - a_1G_2 - G_3)Q + H_1 - G_1$$

$$\bar{\mathbf{L}}\mathbf{O} = \frac{1}{\alpha_4}(\bar{\mathbf{L}}(I - b_1\mathbf{O}) - \bar{\mathbf{K}}(\alpha_3\mathbf{O}\mathbf{D}^2 + b_0\mathbf{O}) - b_0G_1 - \alpha_3G_1\mathbf{D}^2 - b_1G_2 - G_4)Q + H_2 - G_2$$

5 RESULTS AND DISCUSSION

In this chapter, we will test the scheme developed in the previous chapter for the solution of the ordinary differential equations and their coupled systems. We will start from testing the single linear differential equations of first order to check the validity of the proposed method developed for the generalized linear differential equations.

5.1 Test Problems

To check the validity and the convergence of the proposed method, this section presents the different test problems whose exact solutions $u(x)$ are known. To show the convergence experimentally, the absolute error between the exact solution $u(x)$ and the proposed method solution $\bar{u}(x)$ is found and shown graphically. We can observe from the graphical results the proposed method works well for the wide variety of the problems. Also, to check that how the truncation error in the proposed method solution decreases, we find and show the solution graphically at different scale levels M and compare it with the known exact solution $u(x)$ using the absolute error and Mean Absolute Error. From the average absolute error graphs of almost all the problems, we can observe our suggested method converges linearly by upscaling M .

We use the following formula for finding the Absolute Error(AE)

$$AE_u = |u(x) - \bar{u}(x)|,$$

and to find the Mean Absolute Error

$$MAE_u = \frac{1}{N} \sum_{i=1}^N |u_i - \bar{u}_i|,$$

We will use the notation $u(x)$ for the exact solution and $\bar{u}(x)$ for the proposed method solution throughout the discussion of the results and in the graphs.

5.1.1 Ordinary Differential Equations

Example 1.

Solve the following First Order Differential equation.

$$\frac{d}{dx}u(x) = 5u(x) + f(x) \tag{5.1}$$

subject to

$$u(0) = 0$$

exact solution $u(x)$,

$$u(x) = \sin(\pi x) + x$$

source term $f(x)$ is,

$$f(x) = \pi \cos(\pi x) - 5 \sin(\pi x) - 5x + 1$$

Solution:

To explain the procedure that how the proposed method works for this problem, we solve the problem using the proposed method step by step by fixing the scale level $M = 5$. The problem is given by

$$\frac{d}{dx}u(x) = 5u(x) + f(x)$$

Analogous to (4.2), we express the $\frac{d}{dx}u(x)$ in terms of the Shifted Legendre Polynomials.

$$\frac{d}{dx}u(x) = \mathbf{V}_5 \hat{\mathbf{u}}_5(x) \quad (5.2)$$

where $\mathbf{V}_5$ is the row vector of unknown Shifted Legendre coefficients and $\hat{\mathbf{u}}_5(x)$ is the column vector of Shifted Legendre Polynomials After Integrating (5.2) from 0 to x we get.

$$\begin{aligned} \int_0^x \frac{d}{dx}u(x) dx &= \int_0^x \mathbf{V}_5 \hat{\mathbf{u}}_5(x) dx \\ u(x) &= \mathbf{V}_5 \mathbf{O}_{5 \times 5} \hat{\mathbf{u}}_5(x) + u_0 \end{aligned}$$

$$u(x) = \mathbf{V}_5 \mathbf{O}_{5 \times 5} \hat{\mathbf{u}}_5(x) + G_1 \hat{\mathbf{u}}_5(x) \quad (5.3)$$

where G_1 is the Shifted Legendre co-efficient vector for u_0 and $\mathbf{O}_{5 \times 5}$ is the operational matrix of integration of first order. Now we have to just determine the co-efficient vector $\mathbf{V}_5$. We know from (4.5) by setting $a_0 = 5$

$$\mathbf{V}_5 = (5\mathbf{G}_1 + \mathbf{G}_2)(\mathbf{I} - 5\mathbf{O}_{5 \times 5})^{-1} \quad (5.4)$$

Where $\mathbf{G}_2$ is Shifted Legendre co-efficient vector of the source term $f(x)$. Now we define the known integration matrix $\mathbf{O}_{5 \times 5}$ of first order, and legendre co-efficient vectors $\mathbf{G}_1$ and $\mathbf{G}_2$.

$$\mathbf{O}_{5 \times 5} = \begin{bmatrix} 0.5000 & 0.5000 & 0 & 0 & 0 \\ -0.1667 & 0 & 0.1667 & 0 & 0 \\ 0 & -0.1000 & 0 & 0.1000 & 0 \\ 0 & 0 & -0.0714 & 0 & 0.0714 \\ 0 & 0 & 0 & -0.0556 & 0 \end{bmatrix}$$

$$\mathbf{G}_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{G}_2 = \begin{bmatrix} -4.6831 & -6.3197 & 3.4354 & 0.7065 & -0.2589 \end{bmatrix}$$

$$5\mathbf{G}_1 + \mathbf{G}_2 = 5 \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} -4.6831 & -6.3197 & 3.4354 & 0.7065 & -0.2589 \end{bmatrix}$$

$$5\mathbf{G}_1 + \mathbf{G}_2 = \begin{bmatrix} -4.6831 & -6.3197 & 3.4354 & 0.7065 & -0.2589 \end{bmatrix} \quad (5.5)$$

$$(\mathbf{I} - 5\mathbf{O}_{5 \times 5})^{-1} = \left(\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} - 5 \begin{bmatrix} 0.5000 & 0.5000 & 0 & 0 & 0 \\ -0.1667 & 0 & 0.1667 & 0 & 0 \\ 0 & -0.1000 & 0 & 0.1000 & 0 \\ 0 & 0 & -0.0714 & 0 & 0.0714 \\ 0 & 0 & 0 & -0.0556 & 0 \end{bmatrix} \right)^{-1}$$

$$(\mathbf{I} - 5\mathbf{O}_{5 \times 5})^{-1} = \begin{bmatrix} 29.7394 & 54.7309 & 39.2351 & 17.8470 & 6.3739 \\ -18.2436 & -32.8385 & -23.5411 & -10.7082 & -3.8244 \\ 7.8470 & 14.1246 & 10.9858 & 4.9972 & 1.7847 \\ -2.5496 & -4.5892 & -3.5694 & -0.7139 & -0.2550 \\ 0.7082 & 1.2748 & 0.9915 & 0.1983 & 1.0708 \end{bmatrix} \quad (5.6)$$

By putting the results from (5.5) and (5.6) in (5.4) we get the value of unknown coeffi-

cient vector $\mathbf{V}_M$ and is given by,

$$\mathbf{V}_5 = \begin{bmatrix} -4.6831 & -6.3197 & 3.4354 & 0.7065 & -0.2589 \end{bmatrix} \times \begin{bmatrix} 29.7394 & 54.7309 & 39.2351 & 17.8470 & 6.3739 \\ -18.2436 & -32.8385 & -23.5411 & -10.7082 & -3.8244 \\ 7.8470 & 14.1246 & 10.9858 & 4.9972 & 1.7847 \\ -2.5496 & -4.5892 & -3.5694 & -0.7139 & -0.2550 \\ 0.7082 & 1.2748 & 0.9915 & 0.1983 & 1.0708 \end{bmatrix}$$

$$\mathbf{V}_5 = \begin{bmatrix} 1.000 & -3.8197 & 0 & 0.7065 & 0 \end{bmatrix}$$

Now we have found the required unknown coefficient vector $\mathbf{V}_M$, we will put this in (5.3) to get solution $u(x)$ for our proposed method.

$$\boxed{u(x) = 3.526x^4 - 7.057x^3 + 0.411x^2 + 4.116x - 0.0003907} \quad (5.7)$$

Since for the large values of M , there is a large calculation. Therefore, we are not solving for the higher scale levels. Although we represent the results in Figure to check the convergence of proposed method. The examination focuses on comparing the true analytical solution $u(x)$ with its numerical counterpart $\bar{u}(x)$, utilizing our developed framework based on operational matrices. These matrices for both integration and differentiation exhibit improved precision as the resolution level increases. We approximated the function $u(x) = \sin(\pi x) + x$. Figure 5.1 (a) shows the graphical results of the comparison between the exact solution $u(x)$ and its approximation $\bar{u}(x)$ at scale levels $M = 5, 7, 9$ and 11 and (b) visualize absolute error at $M = 5, 7, 9$ and 11. In 5.1 (a) the exact solution is represented by red dots and coloured lines denote the approximations at scale level $M = 5, 7, 9$ and 11 respectively. The graphical results demonstrate progressive improvement in solution quality by increasing M . Figure 5.1 (b) shows the absolute error $|u(x) - \bar{u}(x)|$ at $M = 5, 7, 9$ and 11 represented by solid coloured lines. The error clearly decreases as we increase M , the error decreases from 10^{-2} to 10^{-12} . The graphical results show that our proposed method is highly accurate and converges quickly as scale level M is raised up.

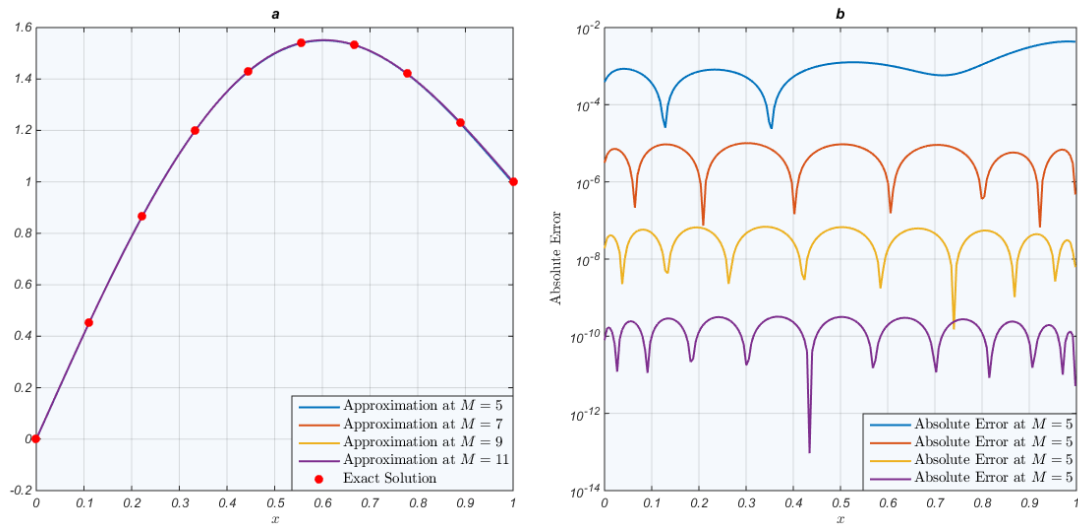


Figure 5.1: (a) Comparison of Exact and Approximations at $M = 5, 7, 9$ and 11 . (b) Absolute Error at $M = 5, 7, 9$ and 11 .

The Figure 5.2 shows the mean absolute error $\text{mean}|u(x) - \bar{u}(x)|$ at scale levels $M = 5, 7, 9$ and 11 . We can observe that as we increase the scale level M mean absolute error declines rapidly the error points at scale level are represented by red dots. The mean absolute error decreases from 10^{-3} to 10^{-10} . This shows that the proposed scheme works with accuracy and is reliable.

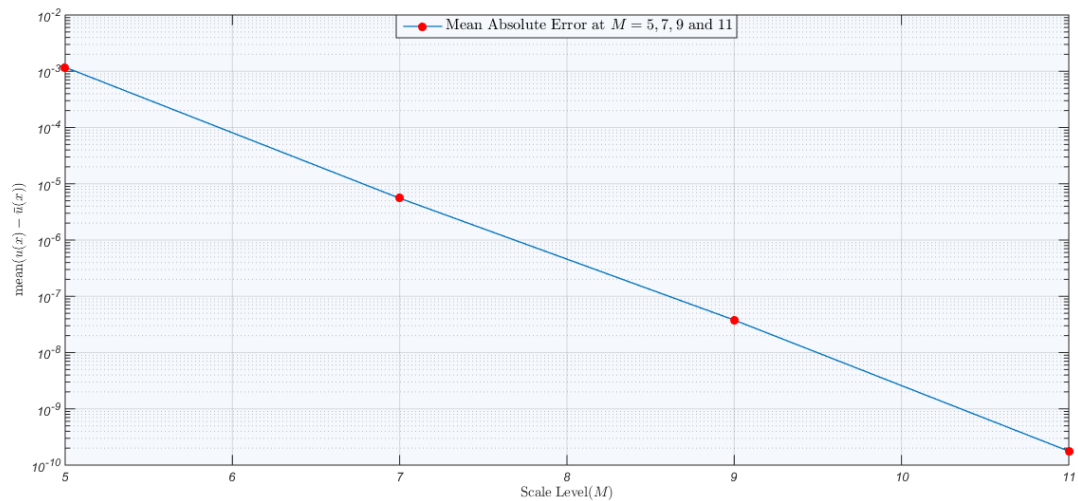


Figure 5.2: Mean Absolute Error at $M = 5, 7, 9$ and 11

Example 2.

Solve the following Second Order Differential Equation.

$$\frac{d^2}{dx^2}u(x) = 5\frac{d}{dx}u(x) + 2u(x) + f(x) \quad (5.8)$$

the exact solution $u(x)$ for the problem is given by,

$$u(x) = \sin(\pi x)$$

The initial condition is given by,

$$u(0) = u_0 = 0$$

and

$$u_1(x) = 3.1416$$

$f(x)$ is the source term and is given below,

$$f(x) = -2 \sin(\pi x) - 5 \pi \cos(\pi x) - \pi^2 \sin(\pi x)$$

Solution:

To explain the procedure that how the proposed method works for this problem, we solve the problem using the proposed method step by step by fixing the scale level $M = 6$. The problem is given by

$$\frac{d^2}{dx^2}u(x) = 5 \frac{d}{dx}u(x) + 2u(x) + f(x)$$

Analogous to (4.7), we express the $\frac{d^2}{dx^2}u(x)$ in terms of the Shifted Legendre Polynomials.

$$\frac{d^2}{dx^2}u(x) = \mathbf{V}_6 \hat{\mathbf{L}}_6(x) \quad (5.9)$$

where $\mathbf{V}_6$ is the unknown row vector of Shifted Legendre Polynomials, and $\hat{\mathbf{L}}_6(x)$ is the function vector of Shifted Legendre Polynomials.

After integrating (5.9) from 0 to x we get the following expression,

$$\int_0^x \frac{d^2}{dx^2}u(x)dx = \int_0^x \mathbf{V}_6 \hat{\mathbf{L}}_6(x)dx$$

$$\frac{d}{dx}u(x) - \frac{d}{dx}u_0 = \mathbf{V}_6 \mathbf{O}_{6 \times 6} \hat{\mathbf{L}}_6(x)$$

Again integrating from 0 to x we get $u(x)$,

$$\begin{aligned}
 \int_0^x \frac{d}{dx} u(x) - \int_0^x \frac{d}{dx} u_0 &= \mathbf{V}_6 \mathbf{O}_{6 \times 6}^2 \hat{\mathbf{u}}_6(x) \\
 u(x) \Big|_0^x - u_1 x \Big|_0^x &= \mathbf{V}_6 \mathbf{O}_{6 \times 6}^2 \hat{\mathbf{u}}_6(x) \\
 u(x) - u_0 - u_1 x &= \mathbf{V}_6 \mathbf{O}_{6 \times 6}^2 \hat{\mathbf{u}}_6(x) \\
 u(x) &= \mathbf{V}_6 \mathbf{O}_{6 \times 6}^2 \hat{\mathbf{u}}_6(x) + (u_1(x) + u_0)
 \end{aligned} \tag{5.10}$$

$$u(x) = \mathbf{V}_6 \mathbf{O}_{6 \times 6}^2 \hat{\mathbf{u}}_6(x) + G_1 \hat{\mathbf{u}}_6(x) \tag{5.11}$$

where G_1 is the Shifted Legendre co-efficient vector for $(u_1(x) + u_0)$ and $\mathbf{O}_{6 \times 6}^2$ is the operational matrix of integration of second order.

We approximate the source term $f(x)$,

$$f(x) = \mathbf{G}_2 \hat{\mathbf{u}}_6(x)$$

where G_2 is the Shifted Legendre co-efficient vector of the source term $f(x)$. Now we have to determine the coefficient vector $\mathbf{V}_M$, we know from (4.18) we set $a_0 = 5$ and $a_1 = 2$,

$$\mathbf{V}_6 = \{I - (5\mathbf{O}_{6 \times 6}^2 \mathbf{D}_{6 \times 6} + 2\mathbf{O}_{6 \times 6}^2)\}^{-1} (5G_1 \mathbf{D}_{6 \times 6} + 2G_1 + G_2) \tag{5.12}$$

where $\mathbf{O}_{6 \times 6}^2$ is the operational matrix of integration of second order, $\mathbf{D}_{6 \times 6}$ is the operational matrix derivatives of first order. Now we define the known integration matrix of second order $\mathbf{O}_{6 \times 6}^2$, known operational matrix of derivatives of first order $\mathbf{D}_{6 \times 6}$, and Shifted Legendre coefficients G_1 and G_2 .

$$\mathbf{O}_{6 \times 6}^2 = \begin{bmatrix} 0.1667 & 0.25 & 0.08333 & 0 & 0 & 0 \\ -0.08333 & -0.1 & 0 & 0.01667 & 0 & 0 \\ 0.01667 & 0 & -0.02381 & 0 & 0.007143 & 0 \\ 0 & 0.007143 & 0 & -0.01111 & 0 & 0.003968 \\ 0 & 0 & 0.003968 & 0 & -0.006494 & 0 \\ 0 & 0 & 0 & 0.002525 & 0 & -0.004274 \end{bmatrix}$$

$$\mathbf{D}_{6 \times 6} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 & 0 \\ 2 & 0 & 10 & 0 & 18 & 0 \end{bmatrix}$$

$$G_1 = \begin{bmatrix} 1.5708 & 1.5708 & 0 & 0 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} -7.5564 & 19.0986 & 8.1554 & -3.5326 & -0.6146 \end{bmatrix}$$

$$\begin{aligned} (5G_1\mathbf{D}_{6 \times 6} + 2G_1 + G_2) &= 5 \begin{bmatrix} 1.5708 & 1.5708 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 & 0 \\ 2 & 0 & 10 & 0 & 18 & 0 \end{bmatrix} \\ &+ 2 \begin{bmatrix} 1.5708 & 1.5708 & 0 & 0 & 0 \end{bmatrix} \\ &+ \begin{bmatrix} -7.5564 & 19.0986 & 8.1554 & -3.5326 & -0.6146 \end{bmatrix} \\ (5G_1\mathbf{D}_{6 \times 6} + 2G_1 + G_2) &= \begin{bmatrix} 11.29 & 22.24 & 8.155 & -3.533 & -0.6146 & 0.1445 \end{bmatrix} \end{aligned} \quad (5.13)$$

$$\begin{aligned} (I - (5\mathbf{O}_{6 \times 6}^2 \mathbf{D}_{6 \times 6} + 2\mathbf{O}_{6 \times 6}^2))^{-1} &= \begin{bmatrix} 98.77 & 186.8 & 142.4 & 64.77 & 24.65 & 0.5097 \\ -62.31 & -117.2 & -89.37 & -40.66 & -15.47 & -0.32 \\ 29.12 & 54.81 & 42.63 & 19.36 & 7.369 & 0.1524 \\ -8.122 & -15.29 & -11.91 & -4.561 & -1.762 & -0.03589 \\ 8.533 & 16.05 & 12.36 & 5.239 & 2.993 & 0.04123 \\ -7.531 & -14.19 & -10.94 & -4.822 & -2.216 & 0.9536 \end{bmatrix} \end{aligned} \quad (5.14)$$

now after putting the results from (5.13) and (5.14) values in (5.12) we get the value of $\mathbf{V}_6$,

$$\mathbf{V}_6 = \begin{bmatrix} 11.29 & 22.24 & 8.155 & -3.533 & -0.6146 & 0.1445 \end{bmatrix} \times \begin{bmatrix} 98.77 & 186.8 & 142.4 & 64.77 & 24.65 & 0.5097 \\ -62.31 & -117.2 & -89.37 & -40.66 & -15.47 & -0.32 \\ 29.12 & 54.81 & 42.63 & 19.36 & 7.369 & 0.1524 \\ -8.122 & -15.29 & -11.91 & -4.561 & -1.762 & -0.03589 \\ 8.533 & 16.05 & 12.36 & 5.239 & 2.993 & 0.04123 \\ -7.531 & -14.19 & -10.94 & -4.822 & -2.216 & 0.9536 \end{bmatrix}$$

$$\mathbf{V}_6 = \begin{bmatrix} -10.52 & -7.969 & 0.6649 & -2.683 & -1.59 & 0.1221 \end{bmatrix}$$

Now we have found the required unknown coefficient vector $\mathbf{V}_6$, we will put this in (5.11) to get solution $u(x)$ for our proposed method.

$$u(x) = -2.815x^5 + 8.091x^4 - 10.42x^3 + 1.391x^2 + 2.961x + 0.004228 \quad (5.15)$$

Since for the large values of M , there is a large calculation. Therefore, we are not solving for the higher scale levels. Although we respresent the results in Figure 5.3 to check the convergence of proposed method as Parameter M increases.

Figure 5.3 (a) shows the comparative behaviour between the exact solution $u(x)$ and the approximations $\bar{u}(x)$ at $M = 5, 7, 9$ and 11 . Red dots represent the exact solution $u(x)$ and the solid coloured lines (blue, orange, yellow and purple) represent the approximate solution at $M = 5, 7, 9$ and 11 respectively. We can observe that as we increase the scale level M the computed approximations $\bar{u}(x)$ improves and becomes close to $u(x)$. Figure 5.3 (b) displays the absolute error between exact solution $u(x)$ and approximate solution $\bar{u}(x)$ at scale levels $M = 5, 7, 9$ and 11 . We can see clearly that the absolute error decreases by elevating Parameter M . Error drops from 10^{-1} to 10^{-8} by increasing Parameter from $M = 5$ to 11 . Convergence of approximations $\bar{u}(x)$ towards real solution $u(x)$ clearly show that our proposed scheme works efficiently and is valid, and the decrease in error by upscaling M confirms authenticity.

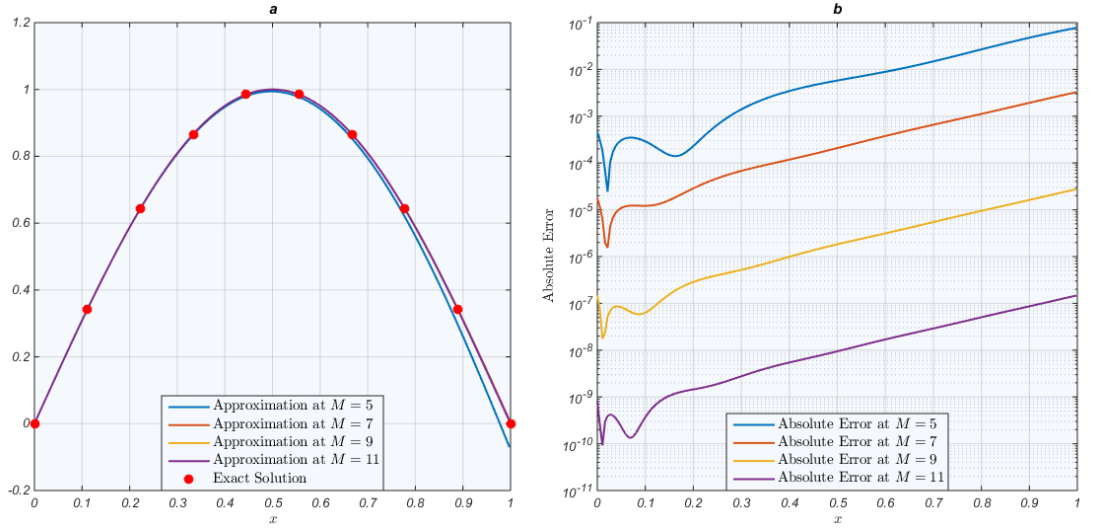


Figure 5.3: (a) Comparison of the exact solution with the approximate solution at $M = 5, 7, 9$ and 11 . Absolute Error at $M = 5, 7, 9$ and 11

In Figure 5.4 the mean absolute error $mean|u(x) - \bar{u}(x)|$ between the exact solution $u(x)$ and the approximate solution $\bar{u}(x)$ is shown graphically. The Figure shows error lowers down as we upscale M . Error is large 10^{-2} at $M = 5$ and error reduces to 10^{-8} at $M = 11$. The graphical results confirms and validates the accuracy of our proposed method of solution.

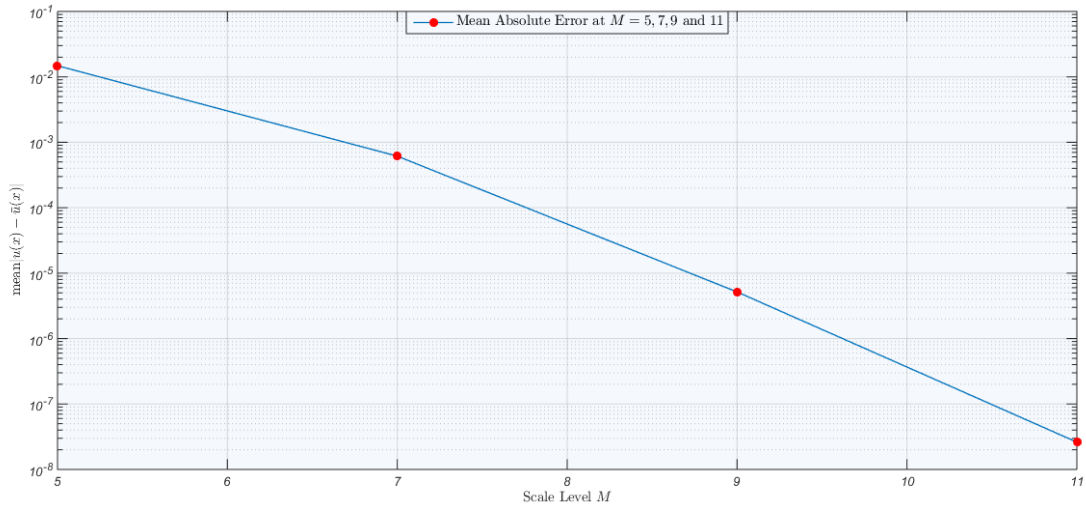


Figure 5.4: Mean Absolute Error

Example 3.

Solve the following third order Differential Equation.

$$\frac{d^3}{dx^3}u(x) = 5\frac{d^2}{dx^2}u(x) + 2\frac{d}{dx}u(x) + 3u(x) + f(x) \quad (5.16)$$

The exact solution is given by,

$$u(x) = \cos(\pi x)$$

Initial conditions for the problem are given by,

$$u(0) = u_0 = 1$$

$$u_1 = 0$$

and

$$u_2 = -9.8696$$

$f(x)$ is the source term and is defined as,

$$f(x) = 46.35 \cos(3.142x) + 37.29 \sin(3.142x) \quad (5.17)$$

Solution:

To explain the procedure that how the proposed method works for this problem, we solve the problem using the proposed method step by step by fixing the scale level

$M = 5$. The problem is given by

$$\frac{d^3}{dx^3}u(x) = 5\frac{d^2}{dx^2}u(x) + 2\frac{d}{dx}u(x) + 3u(x) + f(x)$$

Analogous to (4.12), we express the $\frac{d^3}{dx^3}u(x)$ in terms of the Shifted Legendre Polynomials.

$$\frac{d^3}{dx^3}u(x) = \mathbf{V}_5 \hat{\mathbf{L}}_5(x) \quad (5.18)$$

where $\mathbf{V}_5$ is the unknown row vector of Shifted Legendre Polynomials, and $\hat{\mathbf{L}}_5(x)$ is the function vector of Shifted Legendre Polynomials.

Integrating (5.18) from 0 to x we get the following expression,

$$\begin{aligned} \int_0^x \frac{d^3}{dx^3}u(x) &= \mathbf{V}_5 \mathbf{O}_{5 \times 5} \hat{\mathbf{L}}_5(x) \\ \frac{d^2}{dx^2}u(x) \Big|_0^x &= \mathbf{V}_5 \mathbf{O}_{5 \times 5} \hat{\mathbf{L}}_5(x) \\ \frac{d^2}{dx^2}u(x) - \frac{d^2}{dx^2}u(0) &= \mathbf{V}_5 \mathbf{O}_{5 \times 5} \hat{\mathbf{L}}_5(x) \\ \frac{d^2}{dx^2}u(x) - u_2 &= \mathbf{V}_5 \mathbf{O}_{5 \times 5} \hat{\mathbf{L}}_5(x) \end{aligned}$$

Again integrating from 0 to x ,

$$\begin{aligned} \int_0^x \frac{d^2}{dx^2}u(x) - \int_0^x u_2 &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^2 \hat{\mathbf{L}}_5(x) \\ \frac{d}{dx}u(x) \Big|_0^x - u_2x \Big|_0^x &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^2 \hat{\mathbf{L}}_5(x) \\ \frac{d}{dx}u(x) - \frac{d}{dx}u(0) - u_2x - 0 &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^2 \hat{\mathbf{L}}_5(x) \\ \frac{d}{dx}u(x) - u_1 - u_2x &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^2 \hat{\mathbf{L}}_5(x) \end{aligned}$$

Again Integrating from 0 to x we get the required expression,

$$\begin{aligned} \int_0^x \frac{d}{dx}u(x)dx - \int_0^x u_1dx - \int_0^x u_2xdx &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^3 \hat{\mathbf{L}}_5(x) \\ u(x) \Big|_0^x - u_1x \Big|_0^x - u_2\frac{x^2}{2} \Big|_0^x &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^3 \hat{\mathbf{L}}_5(x) \\ u(x) - u_0 - u_1x - u_2\frac{x^2}{2} &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^3 \hat{\mathbf{L}}_5(x) \\ u(x) &= \mathbf{V}_5 \mathbf{O}_{5 \times 5}^3 \hat{\mathbf{L}}_5(x) + u_0 + u_1x + u_2\frac{x^2}{2} \end{aligned}$$

$$u(x) = \mathbf{V}_5 \mathbf{O}_{5 \times 5}^3 \hat{\mathbf{u}}_5(x) + G_1 \hat{\mathbf{u}}_5(x) \quad (5.19)$$

where G_1 is the Shifted Legendre co-efficient vector for $u_0 + u_1x + u_2\frac{x^2}{2}$ and $\mathbf{O}_{5 \times 5}^3$ is the operational matrix of integration of third order.

we approximate the source term $f(x)$,

$$f(x) = G_2 \hat{\mathbf{u}}_5(x)$$

where G_2 is the Shifted Legendre coefficient for source term $f(x)$. Now we have to determine the coefficient vector, we know from (4.18) we set $a_0 = 5$, $a_1 = 2$ and $a_2 = 3$

$$\mathbf{V}_5 = \left\{ I - (5\mathbf{O}_{5 \times 5}^3 \mathbf{D}_{5 \times 5}^2 + 2\mathbf{O}_{5 \times 5}^3 \mathbf{D}_{5 \times 5} + 3\mathbf{O}_{5 \times 5}^3) \right\}^{-1} (5G_1 \mathbf{D}_{5 \times 5}^2 + 2G_1 \mathbf{D}_{5 \times 5} + 3G_1 + G_2) \quad (5.20)$$

where $\mathbf{D}_{5 \times 5}^2$ is the operational matrix of derivatives of second order, $\mathbf{D}_{5 \times 5}$ is the operational matrix of derivatives of first order. Now we define the known operational matrix of integration of third order $\mathbf{O}_{5 \times 5}^3$, operational matrix of derivatives of second order $\mathbf{D}_{5 \times 5}^2$, operational matrix of derivatives of first order $\mathbf{D}_{5 \times 5}$ and Shifted Legendre coefficients G_1 and G_2 .

$$\mathbf{O}_{5 \times 5}^3 = \begin{bmatrix} 0.0417 & 0.0750 & 0.0417 & 0.0083 & 0 \\ -0.0250 & -0.0417 & -0.0179 & 0 & 0.0012 \\ 0.0083 & 0.0107 & 0 & -0.0028 & 0 \\ -0.0012 & 0 & 0.0020 & 0 & -0.0010 \\ 0 & -0.0004 & 0 & 0.0008 & 0 \end{bmatrix}$$

$$\mathbf{D}_{5 \times 5}^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 12 & 0 & 0 & 0 & 0 \\ 0 & 60 & 0 & 0 & 0 \\ 40 & 0 & 140 & 0 & 0 \end{bmatrix}$$

$$\mathbf{D}_{5 \times 5} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 \end{bmatrix}$$

$$G_1 = \begin{bmatrix} -0.6449 & -2.4674 & -0.8225 & 0 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 23.7392 & -56.3524 & -25.6210 & 10.4233 & 1.9308 \end{bmatrix}$$

$$\begin{aligned} (5G_1\mathbf{D}_{5 \times 5}^2 + 2G_1\mathbf{D}_{5 \times 5} + 3G_1 + G_2) &= 5 \begin{bmatrix} -0.6449 & -2.4674 & -0.8225 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 12 & 0 & 0 & 0 & 0 \\ 0 & 60 & 0 & 0 & 0 \\ 40 & 0 & 140 & 0 & 0 \end{bmatrix} \\ &+ 2 \begin{bmatrix} -0.6449 & -2.4674 & -0.8225 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 \end{bmatrix} \\ &+ 3 \begin{bmatrix} -0.6449 & -2.4674 & -0.8225 & 0 & 0 \end{bmatrix} \\ &+ \begin{bmatrix} 23.7392 & -56.3524 & -25.6210 & 10.4233 & 1.9308 \end{bmatrix} \\ (5G_1\mathbf{D}_{5 \times 5}^2 + 2G_1\mathbf{D}_{5 \times 5} + 3G_1 + G_2) &= \begin{bmatrix} -37.41 & -73.62 & -28.09 & 10.42 & 1.931 \end{bmatrix} \\ (I - (5\mathbf{O}_{5 \times 5}^3\mathbf{D}_{5 \times 5}^2 + 2\mathbf{O}_{5 \times 5}^3\mathbf{D}_{5 \times 5} + 3\mathbf{O}_{5 \times 5}^3))^{-1} &= \begin{bmatrix} -3.0401 & -4.8191 & -4.2749 & -0.1957 & -0.0166 \\ 1.6560 & 3.1508 & 2.7081 & 0.1206 & 0.0109 \\ -1.4298 & -2.6677 & -1.3478 & -0.1104 & -0.0092 \\ 1.1940 & 2.1625 & 1.2472 & 1.0626 & 0.0046 \\ 0.3511 & 0.6699 & 0.5885 & 0.0277 & 1.0023 \end{bmatrix} \end{aligned}$$

Now we put the calculated results in the (5.20) to get $\mathbf{V}_5$,

$$\mathbf{V}_5 = \begin{bmatrix} -37.41 & -73.62 & -28.09 & 10.42 & 1.931 \end{bmatrix} \begin{bmatrix} -3.0401 & -4.8191 & -4.2749 & -0.1957 & -0.0166 \\ 1.6560 & 3.1508 & 2.7081 & 0.1206 & 0.0109 \\ -1.4298 & -2.6677 & -1.3478 & -0.1104 & -0.0092 \\ 1.1940 & 2.1625 & 1.2472 & 1.0626 & 0.0046 \\ 0.3511 & 0.6699 & 0.5885 & 0.0277 & 1.0023 \end{bmatrix}$$

$$\mathbf{V}_5 = \begin{bmatrix} 45.1003 & 47.0863 & 12.5496 & 12.6748 & 2.0619 \end{bmatrix} \quad (5.21)$$

Now we have found the required unknown coefficient vector $\mathbf{V}_5$, we will put this in (5.19) to get solution $u(x)$ for our proposed method.

$$u(x) = 3.06x^4 + 0.7314x^3 - 4.896x^2 - 0.03627x + 1.002 \quad (5.22)$$

To check the validity and accuracy of our developed scheme of operational matrices, outcomes are shown graphically in Figure 5.5. In Figure 5.5 (a) exact solution $u(x)$ is represented by red dots and the approximations $\bar{u}(x)$ is shown by the solid lines on scale levels $M = 5, 7, 9$ and 11. The obtained results show that the approximate solution $\bar{u}(x)$ converges towards the analytical solution $u(x)$ as we increase the scale level M . Figure 5.5 (b) represents absolute error among real solution $u(x)$ and the approximations $\bar{u}(x)$ at $M = 5, 7, 9$ and 11. The graph shows error declines when we rise M . The error reduces quickly from 10^{-2} to 10^{-10} by increasing M from 5 to 11.

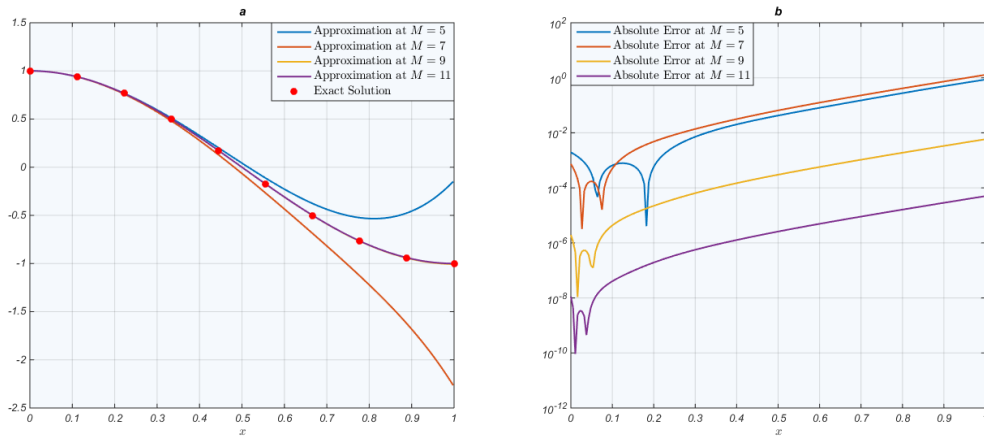


Figure 5.5: (a) Comparison of the exact solution to the approximate solution at $M = 5, 7, 9$ and 11. (b) Absolute Error at $M = 5, 7, 9$ and 11

The Figure 5.6 displays the mean absolute error among exact solution $u(x)$ and approximations $\bar{u}(x)$ at scale levels $M = 5, 7, 9$ and 11. The result show that there is a decrease in the error as the Parameter M increases. Error at $M = 5$ is 10^{-1} and the error reduces to 10^{-5} at $M = 11$. This shows the accuracy of our proposed scheme.

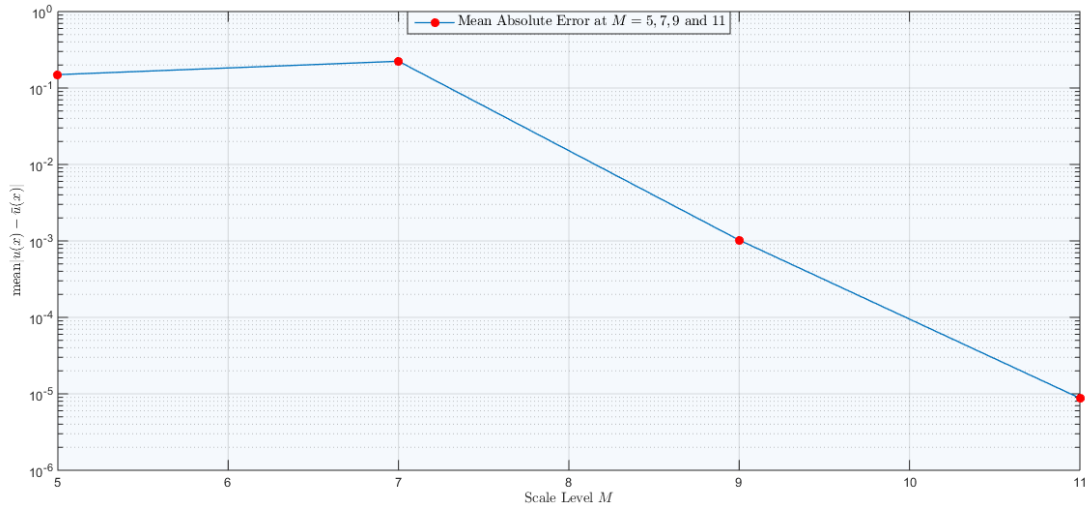


Figure 5.6: Mean Absolute Error at $M = 5, 7, 9$ and 11

5.1.2 Coupled System Of Equations

Example 1.

Consider the following system of equations of first order.

$$\frac{d}{dx}u(x) = 6u(x) + 5v(x) + f(x) \quad (5.23)$$

$$\frac{d}{dx}v(x) = 10u(x) + 3v(x) + g(x) \quad (5.24)$$

the exact solution for $u(x)$ and $v(x)$ is given by,

$$u(x) = \sin(x)$$

and

$$v(x) = \cos(x)$$

the initial conditions are given by,

$$u(0) = u_0 = 0$$

and

$$v(0) = v_0 = 1$$

where $f(x)$ and $g(x)$ are the source terms

$$f(x) = -4 \cos(x) - 6 \sin(x)$$

$$g(x) = -3 \cos(x) - 11 \sin(x)$$

Solution:

To explain the procedure that how the proposed method works for this problem, we solve the problem using the proposed method step by step by fixing the scale level $M = 6$. The problem is given by

$$\frac{d}{dx}u(x) = 6u(x) + 5v(x) + f(x)$$

$$\frac{d}{dx}v(x) = 10u(x) + 3v(x) + g(x)$$

Analogous to (4.21) and (4.22) we express the highest order terms in terms of Shifted Legendre Polynomials.

$$\frac{d}{dx}u(x) = \hat{\mathbf{K}}\hat{\mathbf{u}}(x) \quad (5.25)$$

$$\frac{d}{dx}v(x) = \hat{\mathbf{L}}\hat{\mathbf{u}}(x) \quad (5.26)$$

where $\tilde{\mathbf{K}}$ and $\tilde{\mathbf{L}}$ are the unknown Shifted Legendre coefficient vectors, and $\hat{\mathbf{u}}(x)$ is the function vector of Shifted Legendre Polynomials.

Integrating (5.25) from 0 to x we get the following expression

$$\begin{aligned} \int_0^x \frac{d}{dx}u(x)dx &= \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}} \\ u(x)\Big|_0^x &= \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_5(x) \\ u(x) - u_0 &= \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_5(x) \\ u(x) &= \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_5(x) + u_0 \\ u(x) &= \hat{\mathbf{K}}\mathbf{O}\hat{\mathbf{u}}_5(x) + G_1\hat{\mathbf{u}}_5(x) \end{aligned} \quad (5.27)$$

where G_1 is the Shifted Legendre coefficient vector for u_0

Integrating (5.26) from 0 to x we get the following expression

$$\begin{aligned} \int_0^x \frac{d}{dx}v(x)dx &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{u}}(x) \\ v(x)\Big|_0^x &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{u}} \\ v(x) - v_0 &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{u}} \\ u(x) &= \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{u}} + v_0 \end{aligned}$$

$$v(x) = \hat{\mathbf{L}}\mathbf{O}\hat{\mathbf{u}} + G_2\hat{\mathbf{u}} \quad (5.28)$$

we approximate the source terms $f(x)$ and $g(x)$ as

$$f(x) = H_1\hat{\mathbf{u}}$$

$$g(x) = H_2\hat{\mathbf{u}}$$

where H_1 and H_2 are the Shifted Legendre coefficient vectors for $f(x)$ and $g(x)$ respectively. Now we have to determine the coefficient vectors $\hat{\mathbf{K}}$ and $\hat{\mathbf{L}}$. we know from (4.28), setting $a_0 = 6$, $a_1 = 6$, $b_0 = 10$ and $b_1 = 3$ according to our problem.

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} \frac{I - 3\mathbf{O}}{\det \mathbf{A}} & \frac{10\mathbf{O}}{\det \mathbf{A}} \\ \frac{5\mathbf{O}}{\det \mathbf{A}} & \frac{I - 6\mathbf{O}}{\det \mathbf{A}} \end{bmatrix}$$

After Multiplying the matrices we get

$$\hat{\mathbf{K}} = \frac{A(I - 3\mathbf{O}) + B(5\mathbf{O})}{\det \mathbf{A}}$$

$$\hat{\mathbf{L}} = \frac{A(10\mathbf{O}) + B(I - 6\mathbf{O})}{\det \mathbf{A}}$$

$$A = a_0G_1 + a_1G_2 + H_1$$

and

$$B = b_0G_1 + b_1G_2 + H_2$$

Now we define the known operational matrix of integration of first order $\mathbf{O}$, the Shifted Legendre coefficient vectors G_1 , G_2 , H_1 , H_2

$$\mathbf{O} = \begin{bmatrix} 0.5000 & 0.5000 & 0 & 0 & 0 & 0 \\ -0.1667 & 0 & 0.1667 & 0 & 0 & 0 \\ 0 & -0.1000 & 0 & 0.1000 & 0 & 0 \\ 0 & 0 & -0.0714 & 0 & 0.0714 & 0 \\ 0 & 0 & 0 & -0.0556 & 0 & 0.0556 \\ 0 & 0 & 0 & 0 & -0.0455 & 0 \end{bmatrix}$$

$$G_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$H_1 = \begin{bmatrix} -6.1241 & -1.6324 & 0.5228 & 0.0275 & -0.0038 & -0.0001 \end{bmatrix}$$

$$H_2 = \begin{bmatrix} -7.5811 & -4.0058 & 0.6472 & 0.0675 & -0.0047 & -0.0003 \end{bmatrix}$$

we put all the calculated results to get the unknown Shifted Legendre coefficient vector $\hat{\mathbf{K}}$ and $\hat{\mathbf{L}}$

$$\hat{\mathbf{K}} = \begin{bmatrix} 0.8415 & -0.2338 & -0.0718 & 0.0039 & 0.0005 & -0.0000 \end{bmatrix}$$

$$\hat{\mathbf{L}} = \begin{bmatrix} -0.4597 & -0.4279 & 0.0392 & 0.0072 & -0.0003 & -0.0000 \end{bmatrix}$$

we have determined the unknown Shifted Legendre coefficients, now we obtained $u(x)$ and $v(x)$ by our proposed method.

$$u(x) = 0.007261x^5 + 0.0016104x^4 - 0.16763x^3 + 0.00025605x^2 + 0.99997x + 6.57 \times 10^{-7}$$

$$v(x) = -3.9155 \times 10^{-3}x^5 + 4.5956 \times 10^{-2}x^4 - 2.2291 \times 10^{-3}x^3 - 4.9945 \times 10^{-1}x^2 - 5.3552 \times 10^{-5}x -$$

In Figure 5.7(a) Comparability among exact solution $u(x)$ and approximations $\bar{u}(x)$ is shown at different scale levels $M = 5, 7, 9$ and 11 . Exact solution $u(x)$ is shown by red dotted lines and approximations $\bar{u}(x)$ is represented by solid lines. Approximate solution is almost similar to real solution $u(x)$ on the whole domain as M is increased. For smaller values of M we can see there is slight difference but as we increase M the approximations reaches analytical solution. Similarly, In Figure 5.7 (b) exact solution $v(x)$ and approximations $\bar{v}(x)$ are compared. Similar trend is observed, as we elevate M the approximations matches true solution. This shows that increase the scale level gives us more accurate solution.

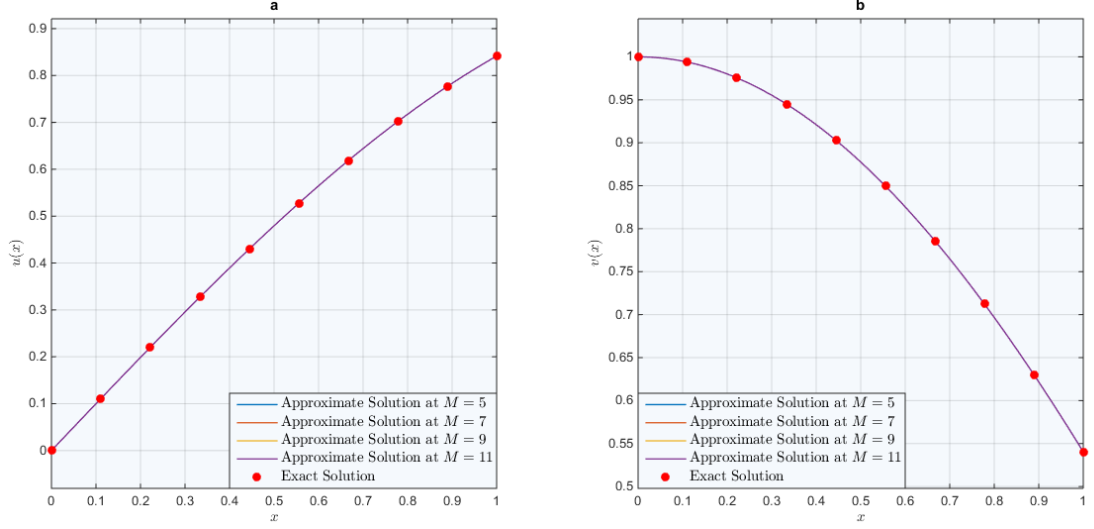


Figure 5.7: (a) Exact Solution and Approximate Solution for $u(x)$ (b) Exact Solution and Approximate Solution for $v(x)$

In Figure 5.8 (a) Absolute error between exact solution $u(x)$ and approximate solution $\bar{u}(x)$ is represented at different scale levels. The solid lines represent the error at $M = 5, 7, 9$ and 11 . We noticed that at $M = 7$ the error is 10^{-4} and as we increase the M the error is reduced to 10^{-14} at $M = 11$. In Figure 5.8 (b) Absolute error between exact solution $v(x)$ and approximate solution $\bar{v}(x)$ is shown at different scale levels of M . The graph follows the similar trend, as we increase the scale there is clear reduction in the error. At $M = 5$ the error is 10^{-4} and it becomes 10^{-14} at $M = 11$. The big reduction in error shows that our proposed method is highly accurate.

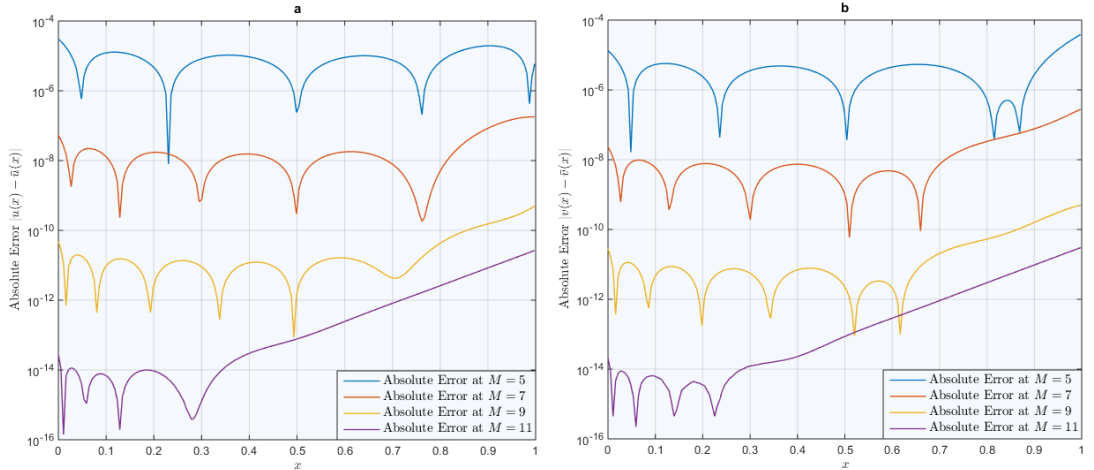


Figure 5.8: (a) absolute error $u(x) - \bar{u}(x)$ (b) absolute error $v(x) - \bar{v}(x)$

In Figure 5.9 (a) and (b) the mean absolute error for $u(x)$ and $v(x)$ is presented for $M = 5, 7, 9$ and 11 . Results obtained from figure (a) shows that there is significant decrease in error as M is increased. At $M = 5$ the error is 10^{-5} and as we increase

the scale to 11 error reduces to 10^{-11} . Similarly we observe in Figure (b) the same decreasing trend is observed. At $M = 5$ we see that the error is 10^{-5} and by increasing $M = 11$ the error rapidly drops to 10^{-11} . The obtained results gives validity for the accuracy and effectiveness of our proposed numerical method.

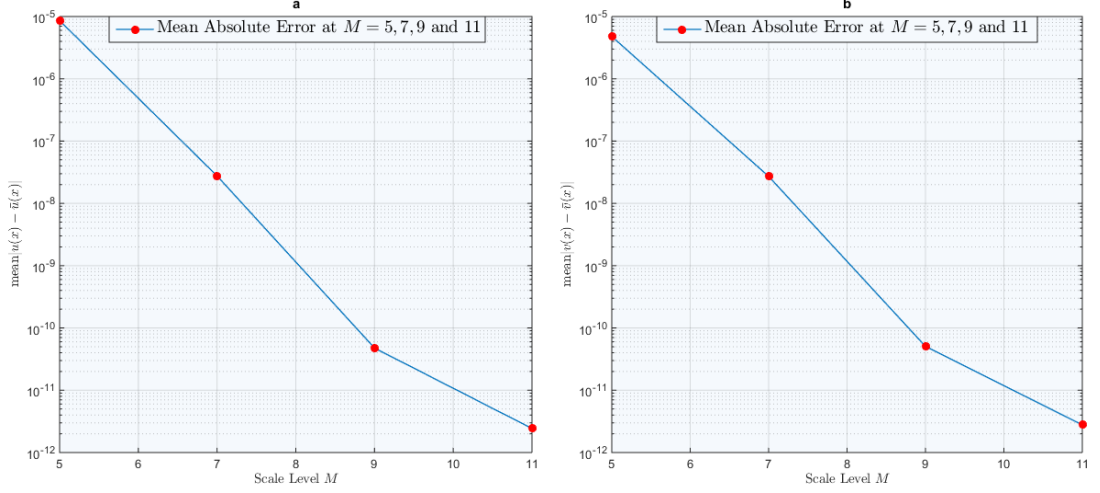


Figure 5.9: (a) Mean Absolute Error $\text{mean}|u(x) - \bar{u}(x)|$ at $M = 5, 7, 9, 11$ (b) Mean Absolute Error $\text{mean}|v(x) - \bar{v}(x)|$ at $M = 5, 7, 9, 11$

Example 2.

Consider the following system of equations of second order

$$\frac{d^2}{dx^2}u(x) = 6u(x) + 5\frac{d}{dx}u(x) + 2v(x) + \frac{d}{dx}v(x) + f(x) \quad (5.29)$$

$$\frac{d^2}{dx^2}v(x) = 10u(x) + 3\frac{d}{dx}u(x) + 4v(x) + 2\frac{d}{dx}v(x) + g(x) \quad (5.30)$$

The exact solution is given by,

$$u(x) = x(x - 0.3)(x - 0.5)(x - 0.7)(x - 1)$$

and

$$v(x) = \sin(\pi x)$$

Initial conditions for the problem are given by

$$u(0) = u_0 = 0, \quad \frac{d}{dx}u(0) = u_1 = 3.1416$$

and

$$v(0) = v_0 = 1, \quad \frac{d}{dx}v(0) = v_1 = 0$$

Solution:

To explain the procedure that how the proposed method works for this problem, we solve the problem using the proposed method step by step by fixing the scale level $M = 5$. Analogous to (4.31) we express the highest order term in (5.23) in Shifted Legendre Polynomials as

$$\frac{d^2}{dx^2}u(x) = \hat{\mathbf{K}}\hat{\mathbf{u}}(x) \quad (5.31)$$

where $\hat{\mathbf{K}}$ is the unknown shifted Legendre coefficient to be determined and $\hat{\mathbf{u}}(x)$ is the function vector of Shifted Legendre Polynomials.

As we know from (4.33) two times Integrating (5.31) from 0 to x we get the following expression

$$u(x) = \hat{\mathbf{K}}\mathbf{O}^2\hat{\mathbf{u}}_M(x) + G_1\hat{\mathbf{u}}_M(x) \quad (5.32)$$

where $\mathbf{O}^2$ is the operational matrix of integration of second order. G_1 is Shifted Legendre coefficient vector.

Analogous to (4.31) we express the highest order term in (5.30) in Shifted Legendre Polynomials.

$$\frac{d^2}{dx^2}v(x) = \hat{\mathbf{L}}\hat{\mathbf{v}}(x) \quad (5.33)$$

where $\hat{\mathbf{L}}$ is the unknown Shifted Legendre coefficient vector.

similarly two times integrating (5.33), we know from (4.34) we get the following expression

$$v(x) = \hat{\mathbf{L}}\mathbf{O}^2\hat{\mathbf{v}}_M(x) + G_2\hat{\mathbf{v}}_M(x) \quad (5.34)$$

where G_2 is the Shifted Legendre coefficient vector. Taking derivative of (5.32)

$$\frac{d}{dx}u(x) = \hat{\mathbf{K}}\mathbf{O}^2\mathbf{D}\hat{\mathbf{u}}_M(x) + G_1\mathbf{D}\hat{\mathbf{u}}_M(x) \quad (5.35)$$

Taking derivative of (5.33)

$$\frac{d}{dx}v(x) = \hat{\mathbf{L}}\mathbf{O}^2\mathbf{D}\hat{\mathbf{v}}_M(x) + G_2\mathbf{D}\hat{\mathbf{v}}_M(x) \quad (5.36)$$

we approximate the source term $f(x)$ in (5.29) and $g(x)$ in (5.30),

$$f(x) = H_1\hat{\mathbf{u}}_M(x) \quad (5.37)$$

where H_1 is the Shifted Legendre coefficient vector for $f(x)$.

$$g(x) = H_2 \hat{\mathbf{M}}_M(x) \quad (5.38)$$

where H_2 is the Shifted Legendre coefficient vector for $g(x)$.

substituting all the results from (5.31),(5.32),(5.33),(5.34),(5.35),(5.36),(5.37) and (5.38) in (5.29) and (5.30) we get the following expressions.

$$\hat{\mathbf{K}} = 6\hat{\mathbf{K}}\mathbf{O}^2 + 6G_1 + 5\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} + 5G_1\mathbf{D} + 2\hat{\mathbf{L}}\mathbf{O}^2 + 2G_2 + \hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} + G_2\mathbf{D} + H_1$$

$$\hat{\mathbf{L}} = 10\hat{\mathbf{K}}\mathbf{O}^2 + 10G_1 + 3\hat{\mathbf{K}}\mathbf{O}^2\mathbf{D} + 3G_1\mathbf{D} + 4\hat{\mathbf{L}}\mathbf{O}^2 + 4G_2 + 2\hat{\mathbf{L}}\mathbf{O}^2\mathbf{D} + 2G_2\mathbf{D} + H_2$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} \frac{I - (4\mathbf{O}^2 + 2\mathbf{O}^2\mathbf{D})}{\det \blacksquare} & \frac{(10\mathbf{O}^2 + 3\mathbf{O}^2\mathbf{D})}{\det \blacksquare} \\ \frac{(2\mathbf{O}^2 + 1\mathbf{O}^2\mathbf{D})}{\det \blacksquare} & \frac{I - (6\mathbf{O}^2 + 5\mathbf{O}^2\mathbf{D})}{\det \blacksquare} \end{bmatrix}$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \frac{1}{\det \blacksquare} \begin{bmatrix} A(I - 4\mathbf{O}^2 - 2\mathbf{O}^2\mathbf{D}) + B(2\mathbf{O}^2 + \mathbf{O}^2\mathbf{D}) & A(10\mathbf{O}^2 + 3\mathbf{O}^2\mathbf{D}) + B(I - 6\mathbf{O}^2 - 5\mathbf{O}^2\mathbf{D}) \end{bmatrix}$$

$$\hat{\mathbf{K}} = \frac{A(I - 4\mathbf{O}^2 - 2\mathbf{O}^2\mathbf{D}) + B(2\mathbf{O}^2 + \mathbf{O}^2\mathbf{D})}{\det \blacksquare} \quad (5.39)$$

$$\hat{\mathbf{L}} = \frac{A(10\mathbf{O}^2 + 3\mathbf{O}^2\mathbf{D}) + B(I - 6\mathbf{O}^2 - 5\mathbf{O}^2\mathbf{D})}{\det \blacksquare}$$

Now we define known operational matrix of integration of second order $\mathbf{O}^2$, operational matrix of derivatives of first order $\mathbf{D}$, G_1 , G_2 , H_1 and H_2 .

$$\mathbf{O}^2 = \begin{bmatrix} 0.1667 & 0.2500 & 0.0833 & 0 & 0 \\ -0.0833 & -0.1000 & 0 & 0.0167 & 0 \\ 0.0167 & 0 & -0.0238 & 0 & 0.0071 \\ 0 & 0.0071 & 0 & -0.0111 & 0 \\ 0 & 0 & 0.0040 & 0 & -0.0065 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 \end{bmatrix}$$

$$G_1 = \begin{bmatrix} 1.5708 & 1.5708 & 0 & 0 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$H_1 = \begin{bmatrix} -8.1029 & 21.5303 & 8.7452 & -3.9824 & -0.6590 \end{bmatrix}$$

$$H_2 = \begin{bmatrix} -2.3662 & 28.3226 & 2.5538 & -5.2387 & -0.1925 \end{bmatrix}$$

by substituting the results in the we get $u(x)$ and $v(x)$ through our proposed method.

$$\hat{\mathbf{K}} = \begin{bmatrix} -6.0112 & 0.6178 & 7.3353 & 0.3665 & -0.3273 \end{bmatrix}$$

$$\hat{\mathbf{L}} = \begin{bmatrix} 0.3270 & 12.6191 & 0.6521 & -1.8964 & 0.3620 \end{bmatrix}$$

$$u(x) = 3.8164x^4 - 7.5084x^3 + 0.65883x^2 + 3.0632x + 0.0022809$$

$$v(x) = 0.16149x^4 + 4.3048x^3 - 6.6551x^2 + 0.26416x + 0.99156$$

Figure 5.10 demonstrates how the approximations converges towards the exact solution for $u(x)$ and $v(x)$ in Figure5.10 (a) and (b) respectively. In (a) the approximate solutions for $u(x)$ are plotted at scale levels $M = 5, 7, 9$ and 11. Exact solution is denoted by red dotted lines and solid lines show the approximate solution. We notice that the lines are closer to exact solution. As we increase the scale M the lines become coincident to the exact solution. Figure (b) shows the same behaviour for $v(x)$. There are slight deviations in the lines at small scale level. As we increase the level the lines match the exact solution. The results in the graph proves the correctness of our proposed method.

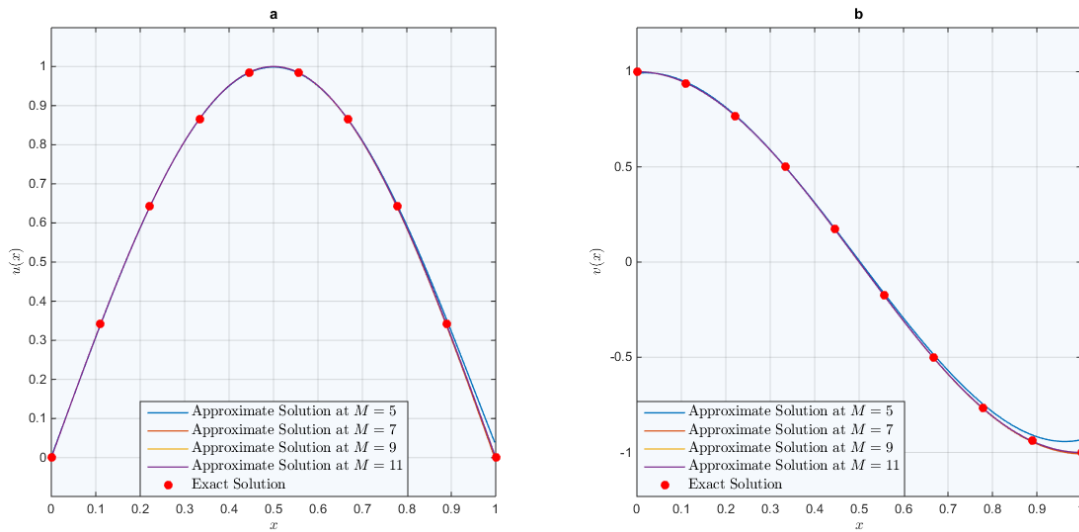


Figure 5.10: (a) approximate vs exact solution for $u(x)$ (b) approximate vs exact solution for $v(x)$

Figure 5.11 visualizes the absolute error $|u(x) - \bar{u}(x)|$ between the exact and the approximate solution for $u(x)$ and $v(x)$ in Figure 5.11 (a) and (b) for various scales of M ranging from 5 to 11. In Figure 5.11 (a) and (b) almost same pattern is portrayed. In Figure (a) the error at $M=5$ (blue line) we see the error is almost 10^{-2} and as we increase scale level to 11 (purple line) we clearly notice a considerable reduction in error and the error reaches to 10^{-10} . In (b) the error at $M=5$ (blue line) is 10^{-2} and as we raise scale level the error becomes 10^{-8} at $M=11$.

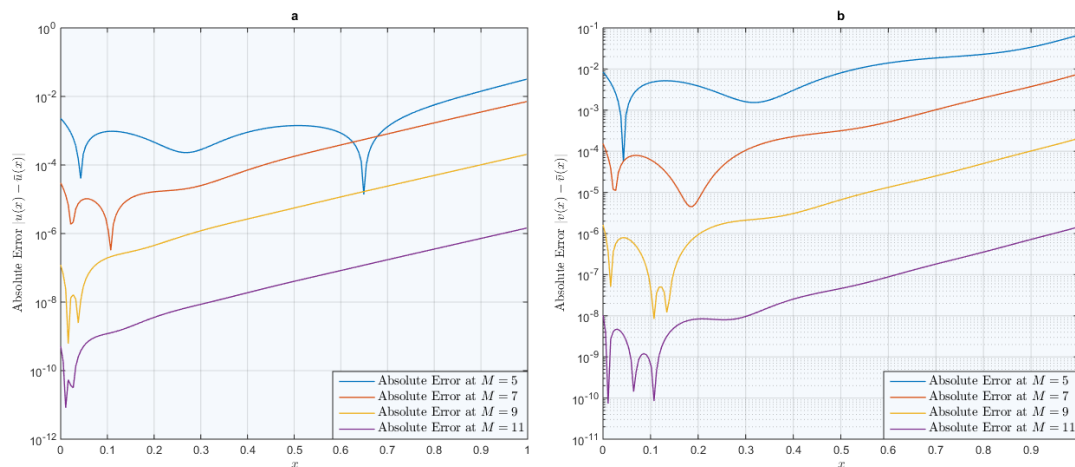


Figure 5.11: (a) absolute error for $u(x)$ (b) absolute error for $v(x)$

The figure 5.12 shows the mean absolute error for $u(x)$ and $v(x)$ in (a) and (b) respectively. In figure (a) the mean absolute error is denoted by red dotted lines for $u(x)$ at $M=5, 7, 9$ and 11. The graphical study demonstrates that there is gradual decrease in error as rise in Parameter M . At $M=5$ graph shows the error is between 10^{-2} and

10^{-3} and it decreases to 10^{-7} at $M = 11$. We observe the same trend for $v(x)$ in Figure (b). In (b) the error decreases by increasing the scale level M . Error at $M = 5$ is 10^{-2} and decreases to 10^{-6} at $M = 11$.

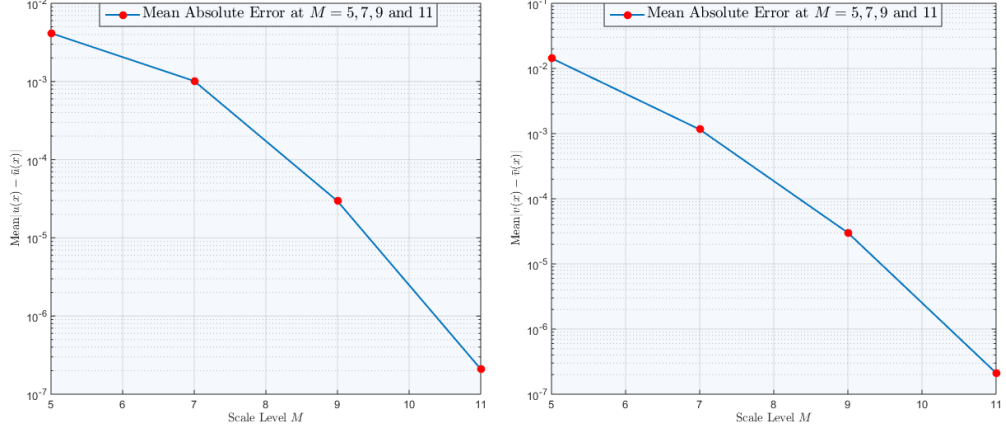


Figure 5.12: Mean Absolute Error (a) $u(x)$ (b) $v(x)$

Example 3.

$$\frac{d^3}{dx^3}u(x) = 6u(x) + 5\frac{d}{dx}u(x) + 2\frac{d^2}{dx^2}u(x) + 1v(x) + 5\frac{d}{dx}v(x) + 4\frac{d^2}{dx^2}v(x) + f(x) \quad (5.40)$$

$$\frac{d^3}{dx^3}v(x) = 10u(x) + 3\frac{d}{dx}u(x) + 4\frac{d^2}{dx^2}u(x) + 2v(x) + 7\frac{d}{dx}v(x) + 1\frac{d^2}{dx^2}v(x) + g(x) \quad (5.41)$$

the exact solution $u(x)$ and $v(x)$ are given by

$$u(x) = x(x - 0.3)(x - 0.5)(x - 0.8)$$

and

$$v(x) = x(x - 0.1)(x - 0.4)(x - 0.9)$$

the initial conditions are given by,

$$u(0) = u_0 = 0, \quad u_1 = -0.1200 \quad u_2 = 1.5800$$

and

$$v(0) = v_0 = 0, \quad v_1 = -0.0360 \quad v_2 = 0.9800$$

Solution:

To explain the procedure that how the proposed method works for this problem, we solve the problem using the proposed method step by step by fixing the scale level $M = 7$.

we express the highest order derivative in (5.40) in terms of Shifted Legendre Polynomials.

$$\frac{d^3}{dx^3}u(x) = \hat{\mathbf{K}}\mathbf{\Phi} \quad (5.42)$$

where $\hat{\mathbf{K}}$ is the unknown Shifted Legendre coefficient vector and $\mathbf{\Phi}$ is the function vector of Shifted Legendre Polynomials.

three times integrating (5.42) from 0 to x , as we know from (4.43) we get the following expression

$$u(x) = \hat{\mathbf{K}}\mathbf{O}^3\mathbf{\Phi} + G_1\mathbf{\Phi} \quad (5.43)$$

where $\mathbf{O}^3$ is the operational matrix of integration of third order. G_1 is Shifted Legendre coefficient vector.

we express the highest order term in (5.41) in terms of Shifted Legendre Polynomials.

$$\frac{d^3}{dx^3}v(x) = \hat{\mathbf{L}}\mathbf{\Phi} \quad (5.44)$$

three times integrating (5.44) from 0 to x , as we know from (4.44) we get the following expression

$$v(x) = \hat{\mathbf{L}}\mathbf{O}^3\mathbf{\Phi} + G_2\mathbf{\Phi} \quad (5.45)$$

where $\hat{\mathbf{L}}$ is the unknown Shifted Legendre coefficient and G_2 is the Shifted Legendre coefficient vector.

Taking derivative of (5.43) we get

$$\frac{d}{dx}u(x) = \hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}\mathbf{\Phi} + G_1\mathbf{D}\mathbf{\Phi} \quad (5.46)$$

where $\mathbf{D}$ is the operational matrix of derivative of first order. again taking derivative of the above equation

$$\frac{d^2}{dx^2}u(x) = \hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2\mathbf{\Phi} + G_1\mathbf{D}^2\mathbf{\Phi} \quad (5.47)$$

where $\mathbf{D}^2$ is the operational matrix of derivative of second order.

Taking derivative of (5.45) we get

$$\frac{d}{dx}v(x) = \hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}\blacksquare + G_2\mathbf{D}\blacksquare \quad (5.48)$$

Again taking derivative of the above equation we get the following expression

$$\frac{d^2}{dx^2}v(x) = \hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2\blacksquare + G_2\mathbf{D}^2\blacksquare \quad (5.49)$$

we approximate the source term $f(x)$ in (5.40) as

$$f(x) = H_1\blacksquare \quad (5.50)$$

where H_1 is the Shifted Legendre coefficient vector for the source term $f(x)$

we approximate the source term $g(x)$ in (5.41) as

$$g(x) = H_2\blacksquare \quad (5.51)$$

where H_2 is the Shifted Legendre coefficient vector for the source term $g(x)$.

Now we substitute the obtained results in (5.40) and we get the following expression

$$\begin{aligned} \hat{\mathbf{K}} = & 6\hat{\mathbf{K}}\mathbf{O}^3 + 6\mathbf{G}_1 + 5\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + 5\mathbf{G}_1\mathbf{D} + 2\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + 2\mathbf{G}_1\mathbf{D}^2 + 1\hat{\mathbf{L}}\mathbf{O}^3 + 1\mathbf{G}_2 \\ & + 5\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + 5\mathbf{G}_2\mathbf{D} + 4\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + 4\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_1. \end{aligned} \quad (5.52)$$

Similarly substituting the obtained results in (5.40) we get the following expression

$$\begin{aligned} \hat{\mathbf{L}} = & 10\hat{\mathbf{K}}\mathbf{O}^3 + 10\mathbf{G}_1 + 3\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D} + 3\mathbf{G}_1\mathbf{D} + 4\hat{\mathbf{K}}\mathbf{O}^3\mathbf{D}^2 + 4\mathbf{G}_1\mathbf{D}^2 + 2\hat{\mathbf{L}}\mathbf{O}^3 + 2\mathbf{G}_2 \\ & + 7\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D} + 7\mathbf{G}_2\mathbf{D} + 1\hat{\mathbf{L}}\mathbf{O}^3\mathbf{D}^2 + 1\mathbf{G}_2\mathbf{D}^2 + \mathbf{H}_2. \end{aligned} \quad (5.53)$$

$$\begin{bmatrix} \hat{\mathbf{K}} & \hat{\mathbf{L}} \end{bmatrix} = \begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} \frac{I - (2\mathbf{O}^3 + 7\mathbf{O}^3\mathbf{D} + 1\mathbf{O}^3\mathbf{D}^2)}{\det\blacksquare} & \frac{(10\mathbf{O}^3 + 3\mathbf{O}^3\mathbf{D} + 4\mathbf{O}^3\mathbf{D}^2)}{\det\blacksquare} \\ \frac{(1\mathbf{O}^3 + 5\mathbf{O}^3\mathbf{D} + 4\mathbf{O}^3\mathbf{D}^2)}{\det\blacksquare} & \frac{I - (6\mathbf{O}^3 + 5\mathbf{O}^3\mathbf{D} + 2\mathbf{O}^3\mathbf{D}^2)}{\det\blacksquare} \end{bmatrix}$$

After Multiplying the above matrices we get

$$\hat{\mathbf{K}} = \frac{A[I - (2\mathbf{O}^3 + 7\mathbf{O}^3\mathbf{D} + 1\mathbf{O}^3\mathbf{D}^2)] + B[1\mathbf{O}^3 + 5\mathbf{O}^3\mathbf{D} + 4\mathbf{O}^3\mathbf{D}^2]}{\det\blacksquare} \quad (5.54)$$

$$\hat{\mathbf{L}} = \frac{A [10\mathbf{O}^3 + 3\mathbf{O}^3\mathbf{D} + 4\mathbf{O}^3\mathbf{D}^2] + B [I - (6\mathbf{O}^3 + 5\mathbf{O}^3\mathbf{D} + 2\mathbf{O}^3\mathbf{D}^2)]}{\det \mathbf{L}} \quad (5.55)$$

Now we define the known operational matrix of integration of third order $\mathbf{O}^3$, operational matrix of derivatives of first order $\mathbf{D}$, operational matrix of derivatives of second order $\mathbf{D}^2$, G_1, G_2, H_1 and H_2

$$\mathbf{O}^3 = \begin{bmatrix} 0.0417 & 0.0750 & 0.0417 & 0.0083 & 0 & 0 & 0 \\ -0.0250 & -0.0417 & -0.0179 & 0 & 0.0012 & 0 & 0 \\ 0.0083 & 0.0107 & 0 & -0.0028 & 0 & 0.0004 & 0 \\ -0.0012 & 0 & 0.0020 & 0 & -0.0010 & 0 & 0.0002 \\ 0 & -0.0004 & 0 & 0.0008 & 0 & -0.0005 & 0 \\ 0 & 0 & -0.0002 & 0 & 0.0004 & 0 & -0.0003 \\ 0 & 0 & 0 & -0.0001 & 0 & 0.0002 & 0 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 & 0 & 0 \\ 2 & 0 & 10 & 0 & 18 & 0 & 0 \\ 0 & 6 & 0 & 14 & 0 & 22 & 0 \end{bmatrix}$$

$$\mathbf{D}^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 12 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 60 & 0 & 0 & 0 & 0 & 0 \\ 40 & 0 & 140 & 0 & 0 & 0 & 0 \\ 0 & 168 & 0 & 252 & 0 & 0 & 0 \\ 84 & 0 & 360 & 0 & 396 & 0 & 0 \end{bmatrix}$$

$$G_1 = \begin{bmatrix} 0.2033 & 0.3350 & 0.1317 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$G_2 = \begin{bmatrix} 0.1453 & 0.2270 & 0.0817 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$H_1 = \begin{bmatrix} -2.9153 & 0.4130 & -14.6217 & -2.1500 & -0.1000 & 0 & 0 \end{bmatrix}$$

$$H_2 = \begin{bmatrix} -0.9120 & 3.3560 & -12.9086 & -2.2600 & -0.1714 & 0 & 0 \end{bmatrix}$$

Now we substitute the results in (5.54) we get the unknown Shifted Legendre coefficient $\hat{\mathbf{K}}$

$$\hat{\mathbf{K}} = \begin{bmatrix} 2.4000 & 12.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & -0.0000 \end{bmatrix}$$

Now we substitute the results in (5.55) we get the unknown Shifted Legendre coefficient $\hat{\mathbf{L}}$

$$\hat{\mathbf{L}} = \begin{bmatrix} 3.6000 & 12.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0000 \end{bmatrix}$$

Now we have all the required results to find the solution $u(x)$ and $v(x)$ through our proposed method we substitute all the obtained results in the (5.43) and (5.45) we get

$$u(x) = 1.4017 \times 10^{-14} x^6 - 2.8023 \times 10^{-14} x^5 + 1.0x^4 - 1.6x^3 + 0.79x^2 - 0.12x + 2.9987 \times 10^{-18}$$

$$v(x) = 1.3545 \times 10^{-14} x^6 - 2.7361 \times 10^{-14} x^5 + 1.0x^4 - 1.4x^3 + 4.9 \times 10^{-1} x^2 - 3.6 \times 10^{-2} x + 2.7336 \times 10^{-18}$$

Figure 5.13 shows a co-relation between the approximations and exact solution for $u(x)$ and $v(x)$ in (a) and (b) respectively. In figure (a) the co-relation among approximations $\bar{u}(x)$ and exact solution $u(x)$ is shown. The rounded red dots represent the exact solution and the solid lines (blue, orange, yellow and purple) represent the approximate solutions at scale level $M = 5, 7, 9$ and 11 respectively. We noticed as we raise M approximations $\bar{u}(x)$ approaches the exact solution $u(x)$. Figure 5.13 (b) shows comparability among approximations $\bar{v}(x)$ and exact solution $v(x)$ at $M = 5, 7, 9$ and 11 . The exact solution $u(x)$ is expressed by red dots and the approximate solution $\bar{u}(x)$ is represented by the solid lines. Results in (b) show that by increasing the scale level M the approximations $\bar{u}(x)$ reaches exact solution $u(x)$.

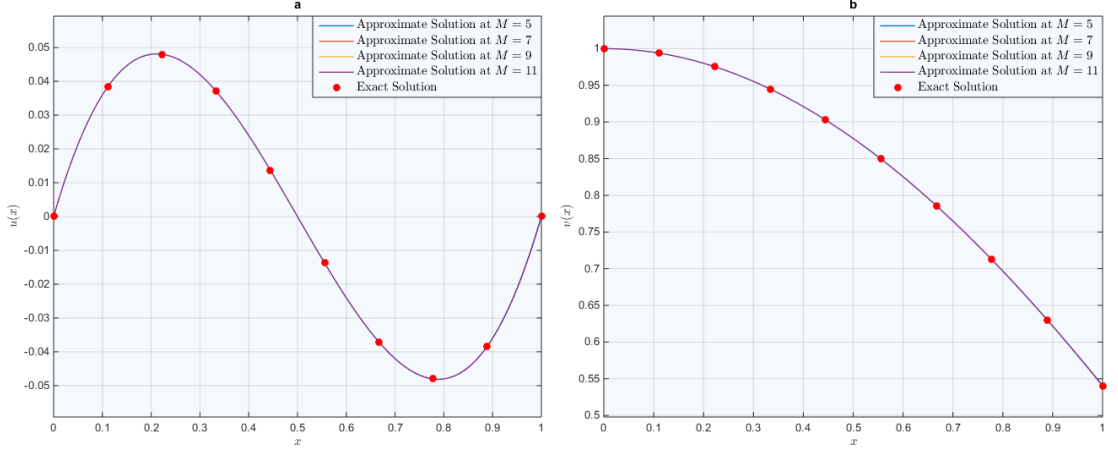


Figure 5.13: (a) Comparison of Exact solution and approximations for $u(x)$ (b) Comparison of Exact and approximations for $v(x)$

Figure 5.14 shows the absolute error for functions $u(x)$ and $v(x)$ in (a) and (b). In figure (a) the absolute error $|u(x) - \bar{u}(x)|$ for $u(x)$ at different scale levels $M = 5, 7, 9$ and 11 is shown. It can be noticed that for smaller scale levels the error is large, for $M = 5$ the error is 10^{-5} . After Upscale in level $M = 11$ error drops to 10^{-15} . Figure (b) represents the absolute error $|v(x) - \bar{v}(x)|$ for $v(x)$ at scale level $M = 5, 7, 9$ and 11. It is observed that the error at small scale levels M . At $M = 5$ the error is 10^{-2} and the error declines to 10^{-16} at $M = 11$. The graphical results show that our proposed method gives accurate results.

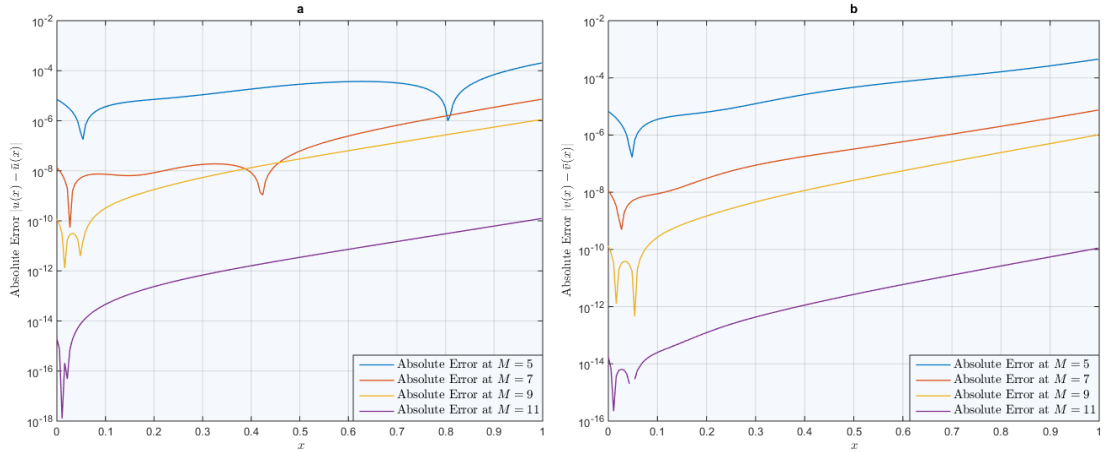


Figure 5.14: (a) Absolute Error $u(x) - \bar{u}(x)$ (b) Absolute Error $v(x) - \bar{v}(x)$

Figure 5.15 shows the mean absolute error for $u(x)$ and $v(x)$ in (a) and (b) respectively. In figure (a) Red dots represent the mean absolute error for $u(x)$ at $M = 5, 7, 9$ and 11 respectively. The graphical outcomes express that there is gradual decrease in error after raising M . At $M = 5$ graph shows the error is between 10^{-4} and 10^{-5} and it decreases to 10^{-10} at $M = 11$. We observe that same trend is followed for $v(x)$ in

Figure (b). In (b) error decreases by increasing scale level M . Error at $M = 5$ is 10^{-4} and decreases to 10^{-11} at $M = 11$.

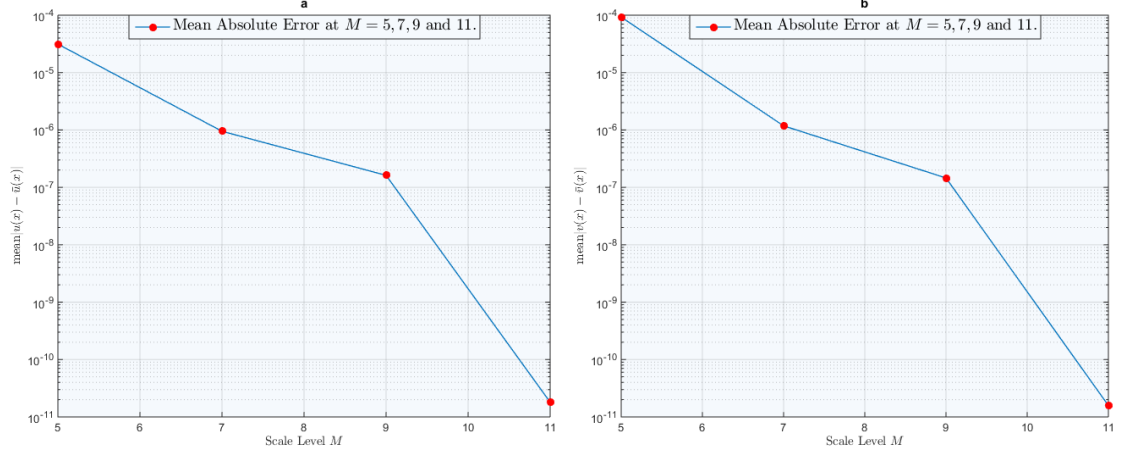


Figure 5.15: (a) Mean Absolute Error at $M = 5, 7, 9$ and 11 . (b) Mean Absolute Error at $M = 5, 7, 9$ and 11 .

5.1.3 Partial Differential Equations(PDEs)

Generalized Heat Conduction Equation

Test problem 1.

$$u_t = 0.1u_{xx} + 2u(x, t) - x \exp(t) \quad (5.56)$$

$$u(0, t) = 0,$$

$$u(1, t) = \exp(t),$$

$$u(x, 0) = x$$

The exact solution is

$$u(x, t) = x \exp(t)$$

To check validity and accuracy of our proposed method of operational matrices for the Solution of Generalized Heat Conduction Model the results are shown graphically. In Figure 5.16 the left graph indicates the comparison between exact solution $u(x, t)$ and approximate solution $\bar{u}(x, t)$ at scale different scale levels M . The red rounded dots represents the exact solution $u(x, t)$ and coloured surfaces represent the approximate solution $\bar{u}(x, t)$ at $M = 5, 7, 9$ and 11 . It can be observed that as we increase the scale level M approximate solution $\bar{u}(x, t)$ converges rapidly towards the exact solution $u(x, t)$. On the right side the graphical results for the absolute error for $u(x, t)$ are represented at $M = 5, 7, 9$ and 11 . The right graph clearly shows that there is sizeable decline in error as we elevate M . At $M = 5$ error is 10^{-5} and reduces to 10^{-15}

at $M = 11$. The graphical results show that our proposed method is valid and gives accurate results for higher scale levels M .

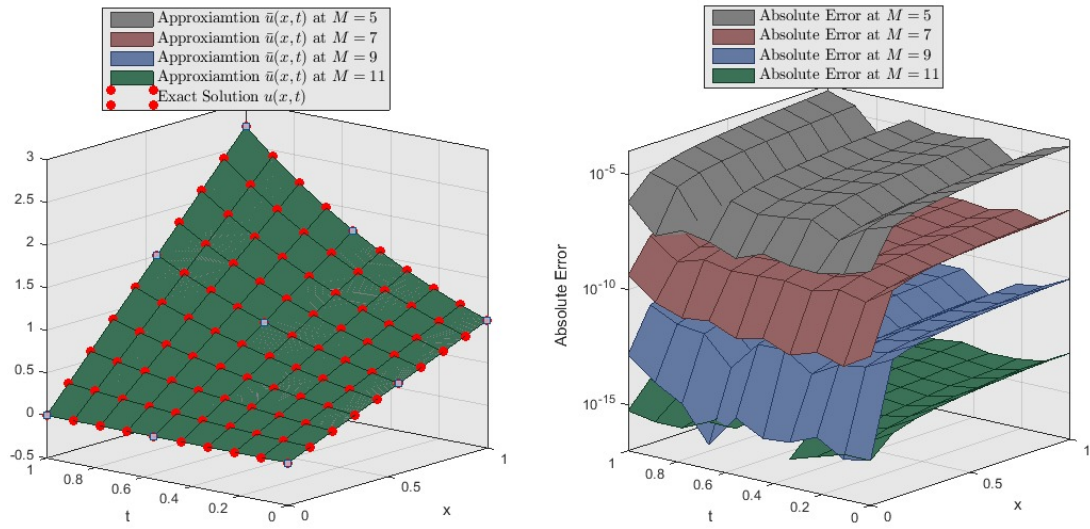


Figure 5.16: Comparison between Exact solution $u(x,t)$ and approximate solution $\bar{u}(x,t)$ (left graph) with absolute error $|u(x,t) - \bar{u}(x,t)|$ (right graph)

The Figure 5.17 shows the maximum absolute error for $u(x)$ at scale levels $M = 3, 5, 7, 9$ and 11 . The results indicate that there is reduction in error as the scale level is raised. The maximum absolute error at $M = 3$ is 10^{-2} and there is a significant decline in error at $M = 11$ error is almost 10^{-14} . The graph shows that the results obtained by using our proposed method are reliable and efficient.

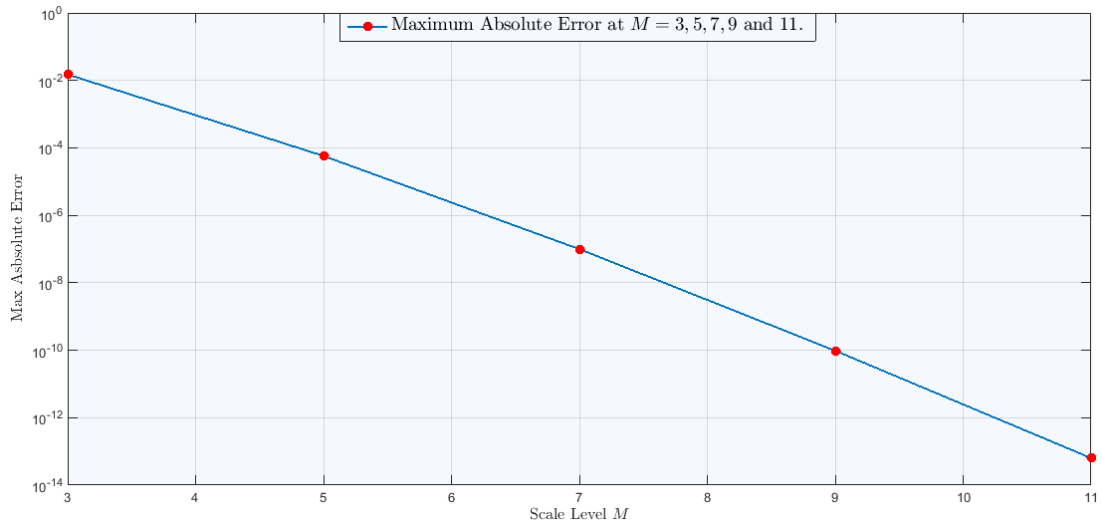


Figure 5.17: Maximum Absolute Error

In the figure 5.18 the RMS(Root Mean Square) Error is shown graphically. RMS Error at various scale levels = 3, 5, 7, 9 and 11 is represented. We noticed that there is

large decrease in error as we raise Parameter M . RMS Error at $M = 3$ is large almost 10^{-3} and as we increase the scale level $M = 11$ the RMS Error reduces to 10^{-15} . This shows that there is a large decrease in RMS Error by elevating scale level M .

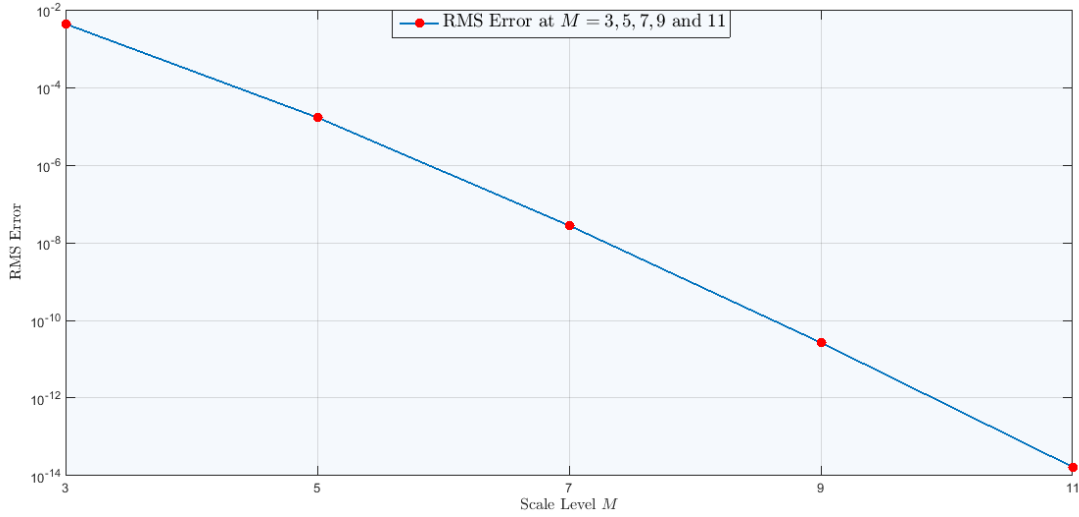


Figure 5.18: Root Mean Square Error at $M = 3, 5, 7, 9$ and 11

Coupled System of Generalized Heat Conduction Model

Test problem 2. Solve the following coupled system of Heat Conduction Model

$$u_t = 0.7u_{xx} + 0.5v_{xx} + 1u + 2v + x \cos(t) - \frac{3t \cos(x)}{2} - x \sin(t) \quad (5.57)$$

$$v_t = 0.3u_{xx} + 0.1v_{xx} + 3u + 5v + \cos(x) - \frac{49t \cos(x)}{10} - 3x \sin(t). \quad (5.58)$$

Initial and boundary conditions are given by,

$$u(x, 0) = 0 \quad v(x, 0) = 0$$

,

$$u(0, t) = 0 \quad v(0, t) = t$$

,

$$u(1, t) = \sin(t) \quad v(1, t) = t \cos(1)$$

The Exact solutions are given by

$$u(x, t) = x \sin(t)$$

$$v(x, t) = t \cos(x)$$

To check the validity and accuracy of our proposed method of operational matrices for the Solution of Coupled System of Generalized Heat Conduction Model the results are shown graphically. In Figure 5.19 the graph(left side) indicates the comparability among exact solution $u(x,t)$ and approximations $\bar{u}(x,t)$ at scale different scale levels M . The red rounded dots represents exact solution $u(x,t)$ and coloured surfaces represent the approximations $\bar{u}(x,t)$ at $M = 4, 6, 8, 10$ and 12 . It can be observed that as we increase the scale level M approximate solution $\bar{u}(x,t)$ converges rapidly towards the exact solution $u(x,t)$. Similar flow is observed for $v(x,t)$ in graph(right side). Graph(left side) indicates the comparison between exact solution $v(x,t)$ and approximate solution $\bar{v}(x,t)$ at scale different scale levels M . The red stars symbolize the exact solution $v(x,t)$ and the approximate solution $\bar{v}(x,t)$ is expressed by coloured surfaces at $M = 4, 6, 8, 10$ and 12 . It can be observed that as we increase the scale level M approximate solution $\bar{v}(x,t)$ converges rapidly towards the exact solution $v(x,t)$.

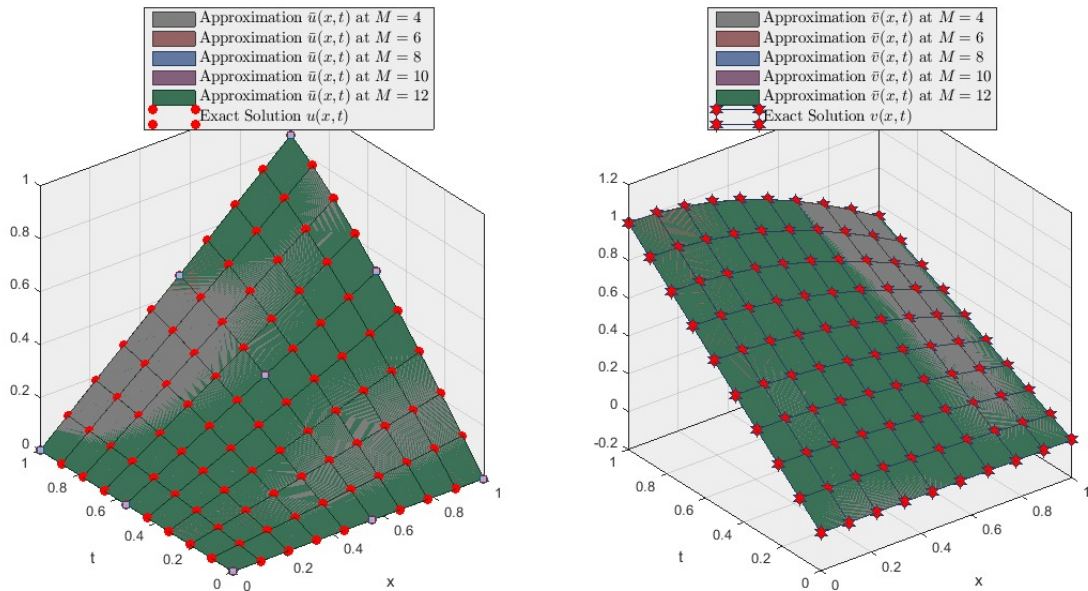


Figure 5.19: Comparison between exact solution $u(x,t)$ (left side), $v(x,t)$ (right side) and approximate solution $\bar{u}(x,t)$ (left side), $\bar{v}(x,t)$ (right side)

Figure 5.20 shows the absolute error $|u(x,t) - \bar{u}(x,t)|$ (left side) for $u(x,t)$ and absolute error $|v(x,t) - \bar{v}(x,t)|$ (right side) for $v(x,t)$. The absolute error is shown for scale level $M = 4, 6, 8, 10$ and 12 . The graphical results for $u(x,t)$ show that the error decreases as we raise the scale M . We notice that error at $M = 4$ is 10^{-4} which is high and as there is rise in scale level at $M = 12$ the error lowers down to 10^{-20} . Similarly, in the graph on the right side the coloured surfaces represent the absolute error for $v(x,t)$ at $M = 4, 6, 8, 10$ and 12 . The graph shows that there is a notable decrease in the error after upscaling scale level M . Error at $M = 4$ is 10^{-5} and reduces rapidly to 10^{-20} at $M = 12$. The graph shows that our proposed method gives precise results as we increase

the scale level.

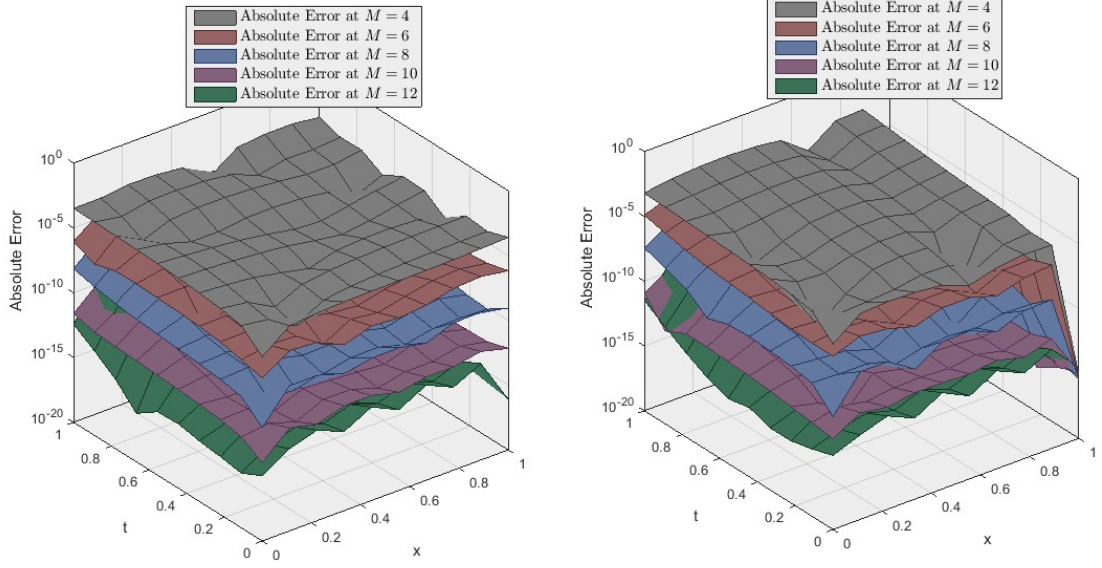


Figure 5.20: Absolute Error $|u(x,t) - \bar{u}(x,t)|$ (left side) Absolute Error $|v(x,t) - \bar{v}(x,t)|$ (right side)

The results for the maximum error for $u(x,t)$ (left side) and $v(x,t)$ (right side) for scale level $M = 4, 6, 8, 10$ and 12 are shown graphically in Figure 5.21. The coloured surfaces represent the error at scale levels $M = 4, 6, 8, 10$ and 12 . The results in the right graph of the 5.21 reflect that there is a sharp decrease in the error as we rise scale level M . Figure shows that the error decrease from 10^{-3} to 10^{-9} as there is elevation in scale level from 4 to 12 . Figure 5.21(right side) indicates the error for $v(x,t)$. Coloured surfaces represent the error at $M = 4, 6, 8, 10$ and 12 . We noticed that there is a reduction in error by upscaling the level from 4 to 12 . The right side graph shows that the error at $M = 4$ is 10^{-3} and reduces to 10^{-10} at $M = 12$.

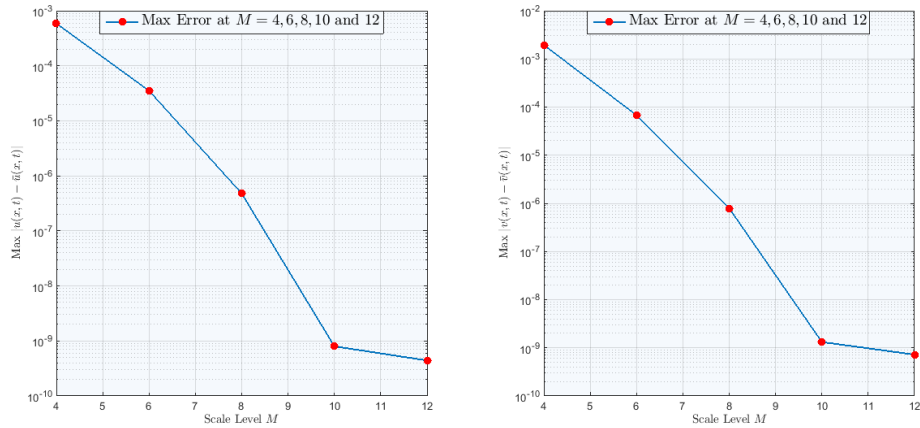


Figure 5.21: Maximum Error for $u(x,t)$ (left side) and $v(x,t)$ (right side)

Figure 5.22 represents the graphical results for the RMS(Root Mean Square) Error for $u(x,t)$ (left side) and $v(x,t)$ (right side) for different scale levels M . The graph on the left side shows the RMS Error for scale levels $M = 4, 6, 8, 10$ and 12 . The RMS Error is denoted by coloured surfaces. We observed that as we rise the level of M there is rapid decrease in RMS Error. Error at $M = 4$ is 10^{-4} and reduces to 10^{-10} at $M = 12$. Similarly on the right side the RMS Error for $v(x,t)$ is represented by coloured surfaces for $M = 4, 6, 8, 10$ and 12 . The figure 5.22 (right side) shows that the error at $M = 4$ is 10^{-3} and error declines to 10^{-10} at $M = 12$.

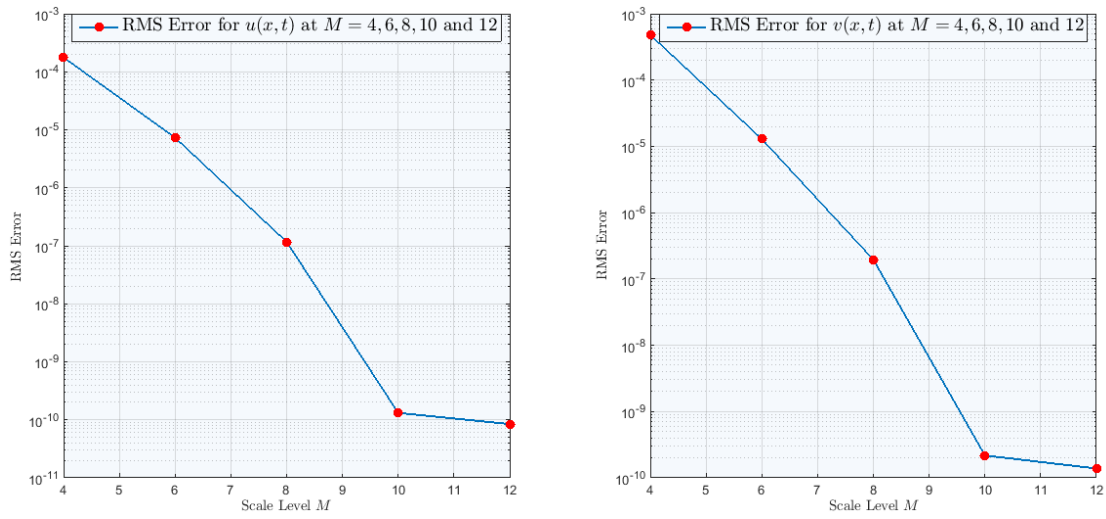


Figure 5.22: RMS(Root Mean Square) Error for $u(x,t)$ (left side) and $v(x,t)$ (right side)

6 CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis studied an integer-order coupled heat-type system related to generalized heat conduction under boundary constraints. The main aim was to compute accurate numerical approximations for the two unknown functions, while keeping the method systematic and not too hard to implement in MATLAB. To do this, shifted Legendre polynomials were used as the approximation basis on a finite interval, and operational matrices were applied so that differential operations could be handled in a matrix form.

A key point of this work is that the coupled PDE system can be converted into an algebraic system for the unknown coefficients of the shifted Legendre expansions. This step makes the computational procedure clean. Instead of doing repeated symbolic differentiation and integration, the operational matrices handle these operators directly, and the final problem becomes a linear algebra problem. This type of idea is well known in spectral and operational-matrix based computation, and it is considered helpful when the solution is smooth and the domain is regular (Abd-Elhameed et al., 2015; Khalil & Khan, 2014; Trefethen, 2000).

The MATLAB implementation was completed for the coupled model cases considered in this thesis. The approximate solutions were compared with the exact solutions for test problems, and different error measures were computed. The overall behaviour from the numerical results was consistent: when the scale level M was increased, the error reduced strongly. In other words, the method showed fast convergence and high accuracy for the smooth benchmark problems. This is an expected benefit of polynomial spectral-type approximations, but it was also clearly visible in the computed tables and figures in this work (Trefethen, 2000).

Another useful observation is that the method remained stable for the tested parameter settings and it produced smooth approximations without unwanted oscillations in the reported examples. Since the model is coupled, this stability is important. The approach also remained practical from a coding point of view because most steps were based on matrix operations and solving an algebraic system, which is convenient in MATLAB.

So, based on the results and the analysis carried out in this thesis, it can be said that the shifted Legendre operational matrix technique is a reliable and accurate tool for solving the classical coupled generalized heat conduction type models considered here. It is especially suitable when the computational domain is simple and when the exact or

expected solution is smooth, because the polynomial approximation can then give very small error with moderate scale levels.

6.2 Limitations

Although the method performed well, some limitations should be mentioned honestly. First, polynomial spectral methods usually work best for smooth solutions. If a problem has sharp thermal fronts, discontinuities, or very strong boundary layers, then the approximation may require a larger scale level, or it may require special treatment such as domain decomposition or filtering (Trefethen, 2000). Second, as M increases, the size of the algebraic system also increases, so computation time and memory usage can grow. This is not a serious issue for moderate M , but it becomes noticeable if very large expansions are used.

Another limitation is related to modelling scope. This thesis mainly focused on a one-dimensional spatial variable setting (with time dependence) and linear coupled equations with standard boundary constraints. More complex geometries and strongly nonlinear couplings were not the main target here. For those cases, other numerical approaches or hybrid strategies may be more suitable.

6.3 Future work

There are several clear directions for extending this work, and they can be done step by step.

One direction is to extend the present approach to more complicated boundary conditions. In many real heat transfer problems, boundary conditions are not only Dirichlet type. Neumann or Robin boundary conditions may appear, and sometimes mixed boundary conditions are used. It will be useful to develop a systematic way to impose such boundary conditions within the operational-matrix framework, so that the method remains stable and accurate.

Another direction is to study higher-dimensional problems. Many practical heat conduction systems depend on two or three spatial variables. Shifted Legendre bases can be extended in a tensor-product form, but then the algebraic system becomes larger. So, efficient storage and solver strategies will be needed. In that case, iterative solvers and preconditioning ideas can be investigated to reduce computation cost.

It will also be interesting to apply the method to nonlinear coupled systems. In real materials and thermal processes, coefficients can depend on temperature, and source terms can be nonlinear. In such cases, the algebraic system will become nonlinear, and a suitable iterative method (for example Newton-type iteration) will be re-

quired. The operational matrix structure can still help, but careful convergence control will be important.

A practical improvement can be made by using an adaptive strategy for the scale level. Instead of fixing a single M from the start, the method can begin with a small M and then increase it until a required accuracy is reached. This will help in saving computation time when high M is not needed.

Another future direction is validation with more realistic data or experimental settings. Generalized heat conduction models are often motivated by experimental evidence and special materials. So, fitting the model parameters (such as relaxation or lag parameters) from experimental temperature data and then solving the coupled system using the proposed method can make the work more application-oriented. It will also help in understanding how sensitive the solution is to parameter changes.

Finally, comparative studies can be expanded. In this thesis, the method was tested mainly with benchmark problems and exact solutions. In future, comparisons can be done against finite difference methods and finite element methods for the same coupled problems. This will provide a clearer picture of accuracy versus computational cost, and it will also show where the operational matrix method gives the most advantage.

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A APPENDIX

A.1 MATLAB CODES

This appendix contains the MATLAB codes used for generating shifted Legendre polynomials, forming two-dimensional basis functions, computing Legendre coefficients, constructing operational matrices, and implementing the numerical solution for both the single heat equation and the coupled system. All codes are included in the same form used during implementation so that the results can be reproduced.

A.2 Shifted Legendre polynomials (1D)

```

1 function P=leg_poly(m,x)
2 syms k ;
3 for i=0:m-1;
4 p1=(-1)^(i+k)*factorial(i+k);
5 p2=factorial(i-k)*(factorial(k))^2;
6 p3=(p1/p2)*(x^k);
7 p4=symsum(p3,k,0,i);
8 P(i+1,1)=p4;
9 end

```

Listing A.1: Shifted Legendre polynomials (1D): leg_poly(m,x)

A.3 Two-dimensional shifted Legendre basis (2D)

```

1 function PPP=l_poly(m,x,t)
2 P=leg_poly(m,x);
3 P1=leg_poly(m,t);
4 for r=1:m^2;
5 [i,j]=index_file(m,r);
6 PP(r)=P(i+1)*P1(j+1);
7 PPP(r,1)=PP(r);
8 end

```

```
9     end
```

Listing A.2: Two-dimensional basis: `l_poly(m,x,t)`

A.4 Legendre coefficients for one-dimensional approximation

```
1 function C=leg_coe(f,m,x)
2 P=leg_poly(m,x);
3 Pf=f*P;
4 PFI=int(Pf,x,0,1);
5 for i=1:m;
6 PFIC(i)=double(PFI(i)*(2*i-1));
7 end
8 C=(PFIC);
```

Listing A.3: 1D Legendre coefficients: `leg_coe(f,m,x)`

A.5 Legendre coefficients for two-dimensional approximation

```
1 function K=l_coe2(f,m)
2 J=m;
3 syms x y t;
4 a=0;
5 b=0;
6 eta=1;
7 %A=funvector(m,a,b,eta);
8 %b=(m^2)*(m-1)+m*(m-1)+m;
9 Aa=leg_poly(m,x);
10 Bb=subs(Aa,x,t);%jacob_fun_vector(m,y,a,b,eta);
11 for i=1:m;
12     Bx=ort(i);
13     for j=1:m;
14         Bt=ort(j);
15         constant=1/(Bx*Bt);
16         FU= f*Aa(i)*Bb(j);
```

```

17 n=J*(i-1)+(j);
18 As(n)=constant*(int(int(FU,x,0,eta),t,0,eta));
19     end
20 end
21
22 K=double(As);

```

Listing A.4: 2D coefficients: `l_coe2(f,m)`

A.6 Operational matrix for time derivative

```

1 function Q=L2_tder_mat(o,m);
2 a=0;
3 J=m-1;syms k i j l;size=(m^2)
4 A=creat_index(m);
5 Ab=Lsin_der(o,m)
6 for nn=1:size
7     a=A(nn,1)
8     b=A(nn,2)
9
10 for mm=1:size
11     d=A(mm,1)
12     e=A(mm,2)
13
14
15
16     if b<ceil(o)
17         Qa(nn,mm)=0
18     else
19         if d==a;
20             Qa(nn,mm)=Ab(b,e)
21         else
22             Qa(nn,mm)=0
23         end
24     end
25 end
26 end
27 Q=Qa

```

Listing A.5: Time derivative operational matrix: `L2_tder_mat(o,m)`

A.7 Operational matrix for time integration

```

1 function Q=L2_tint_mat(alpha,m)
2 J=m-1;syms k i j l;size=(m^2);
3 A=creat_index(m);
4 Ab=Lsin_int(alpha,m);
5 for nn=1:size;
6     a=A(nn,1);
7     b=A(nn,2);
8
9 for mm=1:size;
10     d=A(mm,1);
11     e=A(mm,2);
12
13
14
15         if d==a;
16             Qa(nn,mm)=Ab(b,e);
17         else
18             Qa(nn,mm)=0;
19         end
20     end
21 end
22 Q=double(Qa);

```

Listing A.6: Time integration operational matrix: L2_tint_mat(alpha,m)

A.8 Operational matrix for space integration

```

1 function Q=L2_xint_mat(alpha,m)
2 J=m-1;syms k i j l;size=(m^2);
3 A=creat_index(m);o
4 Ab=Lsin_int(alpha,m);
5 for nn=1:size;
6     a=A(nn,1);
7     b=A(nn,2);
8
9 for mm=1:size;

```

```

10     d=A(mm,1);
11     e=A(mm,2);
12     if b==e;
13         Qa(nn,mm)=Ab(a,d);
14     else
15         Qa(nn,mm)=0;
16     end
17 end
18 end
19
20 Q=double(Qa);

```

Listing A.7: Space integration operational matrix: `L2_xint_mat(alpha,m)`

A.9 Operational matrix for space derivative

```

1 function Q=L2_xder_mat(alpha,m);
2 J=m-1;syms k i j l;size=(m^2)
3 A=creat_index(m);
4 Ab=Lsin_der(alpha,m);
5 for nn=1:size;
6     a=A(nn,1);
7     b=A(nn,2);
8
9     for mm=1:size;
10        d=A(mm,1);
11        e=A(mm,2);
12        if b==e;
13            Qa(nn,mm)=Ab(a,d);
14        else
15            Qa(nn,mm)=0;
16        end
17    end
18 end
19
20 Q=double(Qa);

```

Listing A.8: Space derivative operational matrix: `L2_xder_mat(alpha,m)`

A.10 Single heat equation: approximation and absolute error plots

```

1  % Plotting approximations + absolute errors using MESH (grid
    discretization)
2
3  clear all
4  clc
5  syms x t
6
7  m_values = 3:2:11;
8
9  alpha = 1;
10 beta = 2;
11 lambda = 0.1;
12 a1 = 2;
13
14 % Exact solution (symbolic)
15 U = x*exp(t);
16
17 % Initial/boundary-type terms (same as your code)
18 f1 = subs(U,t,0);
19 f3 = f1;
20
21 g1 = subs(U,x,0);
22 g2 = subs(U,x,1);
23 g3 = g1 + x*g2 - x*g1;
24
25 % PDE forcing term
26 f = diff(U,t,alpha) - lambda*diff(U,x,beta) - a1*U;
27
28 % ----- Discretization grid for mesh -----
29 N = 10; % change grid density if you
    want
30 dom = linspace(0,1,N);
31 [Xg,Tg] = meshgrid(dom,dom);
32
33 % Exact solution on grid (numeric)
34 U_exact = Xg.*exp(Tg);

```

```

35
36 % Store errors (optional)
37 maxErr = zeros(length(m_values),1);
38 rmsErr = zeros(length(m_values),1);
39
40 for i = 1:length(m_values)
41
42     m = m_values(i);
43
44     OME = l_poly(m,x,t);
45     G1   = l_coe2(f3,m);
46     H1   = l_coe2(g3,m);
47     G2   = l_coe2(f,m);
48     Pt   = L2_tint_mat(alpha,m);
49     Px   = L2_xint_mat(beta,m);
50     A111 = var_subs2(1,1,x,m);
51     II   = eye(m^2);
52
53     Q = Px*(II - A111);
54
55     A = 0;
56     B = double( Pt - (1/lambda)*(II - a1*Pt)*Q );
57     C = -double( -(1/lambda)*(a1*G1 + G2)*Q + H1 - G1 );
58
59     XXX = lyap(A,B,C);
60
61     % Symbolic approximation
62     Uapp_sym = (XXX*Pt + G1)*OME;
63
64     % Convert to numeric function and evaluate on the grid
65     Uapp_fun = matlabFunction(Uapp_sym,'Vars',[x,t]);
66     U_app     = Uapp_fun(Xg,Tg);
67
68     % Absolute error on grid
69     Err = abs(U_exact - U_app);
70
71     % Optional error measures
72     maxErr(i) = max(Err(:));
73     rmsErr(i) = sqrt(mean(Err(:).^2));
74

```

```

75 % ----- Mesh plots -----
76 figure(1);
77 subplot(1,2,1)
78 mesh(Xg,Tg,U_app)
79 title(['Approximation (m = ',num2str(m),'')'])
80 xlabel('x'); ylabel('t'); zlabel('U_{app}')
81 hold on
82
83 subplot(1,2,2)
84 mesh(Xg,Tg,Err)
85 title('|U - U_{app}| (Absolute Error)')
86 xlabel('x'); ylabel('t'); zlabel('Error')
87 hold on
88
89 drawnow
90 % pause(1) % (optional) animate slowly
91 end
92
93 figure(1)
94 subplot(1,2,1)
95 hold on
96
97 mesh(Xg,Tg,U_exact,...
98 'EdgeLighting','flat',...
99 'MarkerFaceColor',[1 0 0],...
100 'MarkerEdgeColor',[0.152941182255745 0.227450981736183
101 0.372549027204514],...
102 'MarkerSize',10,...
103 'Marker','hexagram',...
104 'FaceColor','none',...
105 'EdgeColor',[0.152941182255745 0.227450981736183
106 0.372549027204514])
107 title '(a)'
108
109 % ----- Error vs m (optional) -----
110 figure(2);
111 plot(m_values,maxErr,'-o'); grid on
112 xlabel('m'); ylabel('Max abs error on grid')
113 title('Max Error vs m')
114 hold on

```



```

113
114 figure(3);
115 plot(m_values,rmsErr,'-o'); grid on
116 xlabel('m'); ylabel('RMS error on grid')
117 title('RMS Error vs m')
118 hold on

```

Listing A.9: Single heat equation (MESH): approximation and absolute error

A.11 Coupled system: approximation and absolute error plots

```

1  % Plot approximations + absolute errors for the coupled
   % system using MESH
2  % (Uapp approximates X, Vapp approximates Y) on a
   % discretized grid.
3
4  clc
5  clear all
6  syms x t
7
8  alpha = 1;
9  beta  = 2;
10
11 a0 = 1;  lambda1 = 0.7;
12 a1 = 2;  lambda2 = 0.5;
13 a2 = 3;  lambda3 = 0.3;
14 a3 = 5;  lambda4 = 0.1;
15
16 % Exact solutions (symbolic)
17 X = x*sin(t);
18 Y = t*cos(x);
19
20 % IC/BC-type terms
21 f1 = subs(X,t,0);
22 f2 = subs(Y,t,0);
23
24 g1 = subs(X,x,0);
25 g2 = subs(X,x,1);

```

```

26 g3 = subs(Y,x,0);
27 g4 = subs(Y,x,1);
28
29 h1 = g1 + x*g2 - x*g1;
30 h2 = g3 + x*g4 - x*g3;
31
32 % PDE forcing terms
33 f = diff(X,t,alpha) - lambda1*diff(X,x,beta) - lambda2*diff(
    Y,x,beta) - a0*X - a1*Y;
34 g = diff(Y,t,alpha) - lambda3*diff(X,x,beta) - lambda4*diff(
    Y,x,beta) - a2*X - a3*Y;
35 f = simplify(f);
36 g = simplify(g);
37
38 % ----- Discretization grid for mesh -----
39 N = 10; % grid density (increase/
    decrease as needed)
40 dom = linspace(0,1,N);
41 [Xg,Tg] = meshgrid(dom,dom);
42
43 % Exact values on the grid (numeric)
44 X_exact = Xg.*sin(Tg);
45 Y_exact = Tg.*cos(Xg);
46
47 % (optional) error summaries vs m
48 m_list = 4:2:12;
49 maxErrU = zeros(length(m_list),1);
50 rmsErrU = zeros(length(m_list),1);
51 maxErrV = zeros(length(m_list),1);
52 rmsErrV = zeros(length(m_list),1);
53
54 for ii = 1:length(m_list)
55
56     m = m_list(ii);
57
58     OME = l_poly(m,x,t);
59
60     F1 = l_coe2(f1,m);
61     G1 = l_coe2(f2,m);
62     H1 = l_coe2(h1,m);

```

```

63     H2 = l_coe2(h2,m);
64
65     F2 = l_coe2(f,m);
66     G2 = l_coe2(g,m);
67
68     Pt = L2_tint_mat(alpha,m);
69     Px = L2_xint_mat(beta,m);
70     A11 = var_subs2(1,1,x,m);
71     Dx = L2_xder_mat(beta,m);
72
73     II = eye(m^2);
74     III = zeros(m^2);
75
76     M = Px*(II - A11);
77
78     SS1 = -(1/lambda1)*(F2*M);
79     SS2 = -(1/lambda4)*(G2*M);
80
81     B1 = (1/lambda1)*(II - a0*Pt)*M;
82     B2 = (1/lambda1)*(-lambda2*Pt*Dx - a1*Pt)*M;
83     B3 = (1/lambda1)*(-a0)*M;
84     B4 = (1/lambda1)*(-lambda2*Dx - a1*II)*M;
85
86     Z2 = (1/lambda4)*(-lambda3*Pt*Dx - a2*Pt)*M;
87     Z1 = (1/lambda4)*(II - a3*Pt)*M;
88     Z3 = (1/lambda4)*(-lambda3*Dx - a2*II)*M;
89     Z4 = (1/lambda4)*(-a3)*M;
90
91     A = 0;
92     B = double([B1 Z2; B2 Z1] - [Pt III; III Pt]);
93     C = double([F1 G1]*[B3 Z3; B4 Z4] + [H1 H2] - [F1 G1] +
94               [SS1 SS2]);
95
96     XX = lyap(A,B,C);
97
98     % Split solution vector into UU and VV
99     UU = zeros(1,m^2);
100    VV = zeros(1,m^2);
101    for j = 1:m^2
        UU(j) = XX(j);
    end

```

```

102     VV(j) = XX(m^2 + j);
103 end
104
105 % Symbolic approximations
106 Uapp_sym = (UU*Pt + F1)*OME; % approximates X
107 Vapp_sym = (VV*Pt + G1)*OME; % approximates Y
108
109 % Evaluate on the grid
110 Uapp_fun = matlabFunction(Uapp_sym, 'Vars', [x,t]);
111 Vapp_fun = matlabFunction(Vapp_sym, 'Vars', [x,t]);
112
113 U_app = Uapp_fun(Xg,Tg);
114 V_app = Vapp_fun(Xg,Tg);
115
116 % Absolute errors on grid
117 ErrU = abs(X_exact - U_app);
118 ErrV = abs(Y_exact - V_app);
119
120 % Optional error measures
121 maxErrU(ii) = max(ErrU(:));
122 rmsErrU(ii) = sqrt(mean(ErrU(:).^2));
123 maxErrV(ii) = max(ErrV(:));
124 rmsErrV(ii) = sqrt(mean(ErrV(:).^2));
125
126 % ----- MESH plots (Exact vs Approx vs Error)
127 % -----
128 figure(1);
129 subplot(1,2,1)
130 mesh(Xg,Tg,U_app)
131 title(['U_{app} (m = ', num2str(m), ')'])
132 xlabel('x'); ylabel('t'); zlabel('U_{app}')
133 hold on
134 subplot(1,2,2)
135 mesh(Xg,Tg,V_app)
136 title(['V_{app} (m = ', num2str(m), ')'])
137 xlabel('x'); ylabel('t'); zlabel('V_{app}')
138 hold on
139 figure(2)
140 subplot(1,2,1)
141 mesh(Xg,Tg,ErrU)

```

```

141     title('|X - U_{app}|')
142     xlabel('x'); ylabel('t'); zlabel('Error')
143 hold on
144     subplot(1,2,2)
145     mesh(Xg,Tg,ErrV)
146     title('|Y - V_{app}|')
147     xlabel('x'); ylabel('t'); zlabel('Error')
148 hold on
149     drawnow
150     % pause(1) % optional animation delay
151 end
152
153 figure(1)
154 subplot(1,2,1)
155 hold on
156 mesh(Xg,Tg,X_exact,...
157 'EdgeLighting','flat',...
158     'MarkerFaceColor',[1 0 0],...
159     'MarkerEdgeColor',[0.152941182255745 0.227450981736183
160         0.372549027204514],...
161     'MarkerSize',10,...
162     'Marker','hexagram',...
163     'FaceColor','none',...
164     'EdgeColor',[0.152941182255745 0.227450981736183
165         0.372549027204514])
166 title '(a)'
167
168 subplot(1,2,2)
169 mesh(Xg,Tg,Y_exact,...
170 'EdgeLighting','flat',...
171     'MarkerFaceColor',[1 0 0],...
172     'MarkerEdgeColor',[0.152941182255745 0.227450981736183
173         0.372549027204514],...
174     'MarkerSize',10,...
175     'Marker','hexagram',...
176     'FaceColor','none',...
177     'EdgeColor',[0.152941182255745 0.227450981736183
178         0.372549027204514])
179 title '(a)'

```

```

177 % ----- Error vs m (optional) -----
178 figure(3);
179 subplot(1,2,1)
180 plot(m_list,maxErrU,'-o'); grid on
181 xlabel('m'); ylabel('Max |X - U_{app}|')
182 title('Max Error (U) vs m')
183
184 subplot(1,2,2)
185 plot(m_list,maxErrV,'-o'); grid on
186 xlabel('m'); ylabel('Max |Y - V_{app}|')
187 title('Max Error (V) vs m')
188
189 figure(4)
190 subplot(1,2,1)
191 plot(m_list,rmsErrU,'-o'); grid on
192 xlabel('m'); ylabel('RMS Error (U)')
193 title('RMS Error (U) vs m')
194
195 subplot(1,2,2)
196 plot(m_list,rmsErrV,'-o'); grid on
197 xlabel('m'); ylabel('RMS Error (V)')
198 title('RMS Error (V) vs m')

```

Listing A.10: Coupled system (MESH): approximations and absolute errors